Space Emanation Theory

Field capacity, source organization and gravitational observables

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Abstract

Space Emanation Theory begins with a physical question: how do gravity and motion affect the processes recorded by clocks? I propose that gravitational burden, translation and local event progression share a limited field interaction capacity. The source law assigns a source quantity to a specified mass and volume; its field response determines the burden entering that capacity budget. This paper develops the connection from sources to physical readings and then to observations.

For an isolated spherical source, the common-energy construction makes material motion, internal clock progression and the complete source response part of one calculation. Within the stated local static class, equality between complete source energy and its exterior reading selects the burden profile and radial response. The resulting probe dynamics gives escape speed and the perihelion equation, including a Mercury benchmark of approximately 42.98 arcseconds per century. The paper also follows a specified rotating extension into horizon and shadow calculations and summarizes the causal field completion and quadrupole radiation normalization.

For collective baryonic sources, an effective-area prescription relates source organization to boundary burden. The original twenty single-radius galaxy examples have a median absolute central speed discrepancy of 4.79%; among nineteen matched cases, the SET and Newtonian baryonic medians are 5.13% and 40.67%. Sixty-three additional worked examples extend the collection to 83, with a combined median absolute central speed discrepancy of 5.76%. Coupled reported-input propagation for the earlier 62 examples overlaps their individual observational ranges for 58 and 60 cases at the conditional 68% and 95% levels. A homogeneous cosmological source-clock prescription then calculates relative Type Ia supernova brightness. With the same fixed source inputs across two compilations, all fifteen grouped discrepancies decrease; the catalogue medians fall from 38.53% to 2.09% and from 36.39% to 2.96%. The comparison baselines, source prescriptions, measured inputs and uncertainty treatments are given with each calculation. These applications investigate whether effects attributed to dark matter and dark energy can arise through the collective response of the field to baryonic sources. The unchanged luminosity prescription also gives a full-covariance χ^2 of 255.61 for 260 additional target-minus-reference contrasts, compared with 322.10 for the baseline with a constant clock factor; no SET parameter or target intercept is fitted in that extension.

Reading guide

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Appendices A–J preserve the earlier source-family controls, alternative orbital forms, free-fall coordinates, galaxy inputs, collective energy and numerical checks. Appendix K gives the exact complete-energy selection proof; L develops the source and tidal invariant; M records causal evolution and radiation normalization; N connects the cosmological domain to common energy and clocks.

1 Why field interaction capacity

I began SET with a physical question: what happens to the processes of a clock when it records fewer ticks per reference second? A clock has to do something to produce each tick. The observations used to compare clocks also require physical interactions. I wanted to follow those processes through their interaction with the field and ask how the metric records the result.

“There are no motionless events” expresses my starting intuition: an observed event involves a physical occurrence or change. A body can be at rest as a whole while its internal processes continue. Observing that body also requires an interaction. Here, physical evolution includes interactions and recorded changes; it does not require a hidden classical trajectory within every quantum state. A stationary energy eigenstate need not produce distinguishable ticks, and a global phase change alone need not be observable. The hypothesis concerns the processes through which matter realizes events and observations.

My proposal is that the field has a limited interaction capacity shared by gravitational burden, translation and local event progression. A held clock has less capacity available for event progression where gravitational burden is greater. A moving probe shares the remaining capacity between translation and its local processes. These comparisons use a specified reference time.

A massive probe can approach the local speed c, while the energy law obtained within the common clock-and-motion construction below makes the required energy unbounded as that limit is approached. I interpret the accompanying reduction in event progression per reference second as increased participation of translation in the shared capacity budget.

In this interpretation, proper time records the progression of those physical processes. Fewer calibrated events per reference second means less accumulated proper time. I am not adding a separate time process that first slows and then acts on matter. Clocks following different paths can accumulate different event counts between the same endpoint events.

The observer’s clock participates in the field it is used to measure. This matters because the instrument and the phenomenon are subject to the same local conditions. In the stationary construction developed below, light’s propagation expressed as local distance per reference second and the held clock’s progression share the same factor. That factor cancels when propagation is timed by the local clock. Comparisons between clocks at different locations, or along different paths, and signal travel times referred to a specified clock reveal the gravitational effect. Section 4.4 distinguishes this local measurement from the radial coordinate speed.

Why introduce field interaction capacity when Newtonian gravity and general relativity already provide equations of motion?

Newtonian gravity supplies the successful weak-field mechanics used in many gravitational calculations. It can be expressed through a potential field, and it already accounts for the spatial distribution of extended sources, including flattened disks. Its limitation for the present purpose is that it does not itself supply the relativistic relations among clock progression, light propagation and motion. Those relations require a more complete description.

General relativity supplies that description through a dynamical metric coupled to energy and momentum. Its relativistic effects cannot be reduced to a delay added to Newton’s force law: even a stationary gravitational field changes the relations among measured distance, clock readings and orbital motion. SET retains these connections in the spherical construction presented here.

My incentive for introducing capacity is to investigate a physical account of those relations. I ask whether the metric can be understood through the participation of matter, clocks and signals in a limited field interaction

capacity. The proposed order of explanation begins with sources and their field response, follows how that response affects physical processes, and expresses the resulting clock and propagation relations geometrically.

Curvature therefore remains part of the description. With the source equation, propagation coupling and action specified in this paper, the capacity formulation recovers the Schwarzschild exterior and its test-particle dynamics. That recovery establishes the connection for the spherical problem considered here. It also identifies which physical relations the capacity interpretation must reproduce.

I denote gravitational burden by S, expressed in speed units. Its squared magnitude enters the capacity budget. Through the common-energy coupling developed below, it determines the escape-energy scale; its spatial variation determines the weak-field acceleration. The familiar spherical expression $S = \sqrt{(2GM/r)}$ is one solution for this quantity. The burden is established by the sources and their response, and the probe's escape speed and orbital motion follow through the stated dynamical relations.

This viewpoint leads to the collective question that motivates the second part of the paper. Can the source–boundary relation used for an individual body also apply to a system of disjoint masses, with their organization entering through an effective boundary area?

At Earth's approximately spherical surface, the spherical source normalization and surface area give the boundary burden. For a collective system, I apply SET's proposed source–boundary relation to the organization of its baryonic sources. At fixed baryonic mass and reference radius, the reference normalization is retained and the boundary burden follows from $Q_{ref} = A_{eff} S_{eff}$. A smaller effective area therefore gives a larger boundary burden without changing Q_{ref} or introducing a separate enhancement factor. This is why source organization is a natural route to investigate within SET: the proposed change in gravitational response enters through the relation between source normalization and boundary area, without adding unseen gravitating matter. The familiar spherical expression $S = \sqrt{(2GM/R)}$ is the spherical case of that reading; it is not held fixed while the effective area changes.

An effective boundary must describe the physical response. Rearranging the sources can change their collective field; merely redrawing a surface around an unchanged system cannot change that field at a fixed location. The area used in the calculation must consequently be connected to the source arrangement. In the galaxy examples, I estimate it with an oblate surface constrained by imaging and inclination.

The galaxy calculation examines whether the proposed capacity and boundary relation allows the adopted baryonic inventory to account for the measured orbital speed. Where matching models are available, I compare its result with spatially resolved Newtonian gas, disk and bulge contributions at the same radius.

The same question extends to the light received from distant sources. In galaxies, orbital motion probes a spatial response. In the supernova calculation, the advancing source domain includes more existing matter, while expansion spreads the matter already included over a larger region. These competing effects change the collective burden. Within the stated homogeneous realization, that burden history enters clock progression, light propagation and received brightness. I want to know whether the source–capacity relationship can account for both kinds of observation without assigning them to separate invisible components. The calculations below make that question concrete in each setting.

I begin with the source law and invariant, derive the spherical response and orbital motion, then examine strong fields and rotation. I next develop the effective-area calculation and its galaxy comparisons. The final application follows a collective source history into the observed supernova brightness trend. Detailed controls, alternative representations and numerical checks are retained in the appendices so that the main argument can be followed without losing the steps needed to reproduce it.

2 Capacity clocks and source quantities

2.1 The invariant and recorded events

For the isolated spherical problem, take a static reference time T normalized at infinity. A probe records proper time τ through its calibrated event accumulation. Let dℓ be the displacement norm in the static spatial description, and define β = dτ/dT and Vrel = dℓ/dT. A nearby held observer uses a different clock rate; its measured speed is derived below.

$$S^2 + V_{rel}^2 + c^2\beta^2 = c^2. \quad (1)$$

Each term has units m^2/s^2. S is the local gravitational burden expressed in speed units; S^2 is its contribution to the capacity budget. In SET's interpretation, sources establish the burden, and the energy coupling determines the escape speed associated with it, as derived in Section 4.4. Translation contributes $Vrel^2$, while $c^2\beta^2$ represents event progression. These are scalar contributions to a shared budget.

The sources and their spatial response determine S. The spherical expression derived below is one solution for that field. Section 9.1 shows the source contribution inside the invariant itself.

Place the source burden on the right. The left then shows how the probe shares the remaining capacity between translation and event progression:

$$V_{rel}^2 + c^2\beta^2 = c^2 - S^2. \quad (2)$$

$$u = \frac{S^2}{c^2}, \alpha^2 = 1 - u, \beta^2 = \alpha^2 - \frac{V_{rel}^2}{c^2}. \quad (3)$$

For a held probe, Vrel = 0 and β = α. Thus α is the held-clock rate relative to T. A local static observer uses dtlocal = α dT, so the local translation speed is vlocal = dℓ/dtlocal = Vrel/α. Substitution gives:

$$\beta^2 = \alpha^2\left(1 - \frac{v_{local}^2}{c^2}\right), \beta = \frac{\alpha}{\gamma_{local}}. \quad (4)$$

where γlocal = $1/\sqrt{(1 - vlocal^2/c^2)}$. Neither expression permits a massive probe with a real, positive proper-time increment to have vlocal ≥ c.

In SET, the field has a physical interaction limit. It does not need to classify an interaction as gravity, translation or event progression. These terms describe how the available capacity is shared. Different combinations of gravitational burden and translation can leave the same capacity for local events, compared using the same reference time. The resulting event-rate factor can therefore be the same even when the surrounding gravitational fields differ. Equal remaining capacity does not imply identical physical circumstances.

To give event progression an operational meaning, let N count the cycles of an ideal clock and let ν_0 be its calibrated number of cycles per unit local clock reading. Then $d\tau = dN/\nu_0$ and $dN = \nu_0 d\tau$. The reading records the events, so:

$$\frac{dN}{dT} = \nu_0\beta. \quad (5)$$

The clock completes $\nu_0\beta$ events per reference second. I take the common factor β for ideal clocks as part of the universal coupling to capacity. Real clock comparisons must also control temperature, acceleration-induced defects and other calibration changes. The scale ν_0 belongs to the clock: c alone specifies no

universal tick count. At u = 0 and Vrel = 0, β = 1 and this calibrated process has its maximal rate relative to the unloaded reference.

For example, take u = 0.36 and vlocal = 0.6c. Then α = 0.8, Vrel = 0.48c and β = 0.64. Burden, translation and event progression occupy 0.36, 0.2304 and 0.4096 of c^2, respectively. Their sum is one.

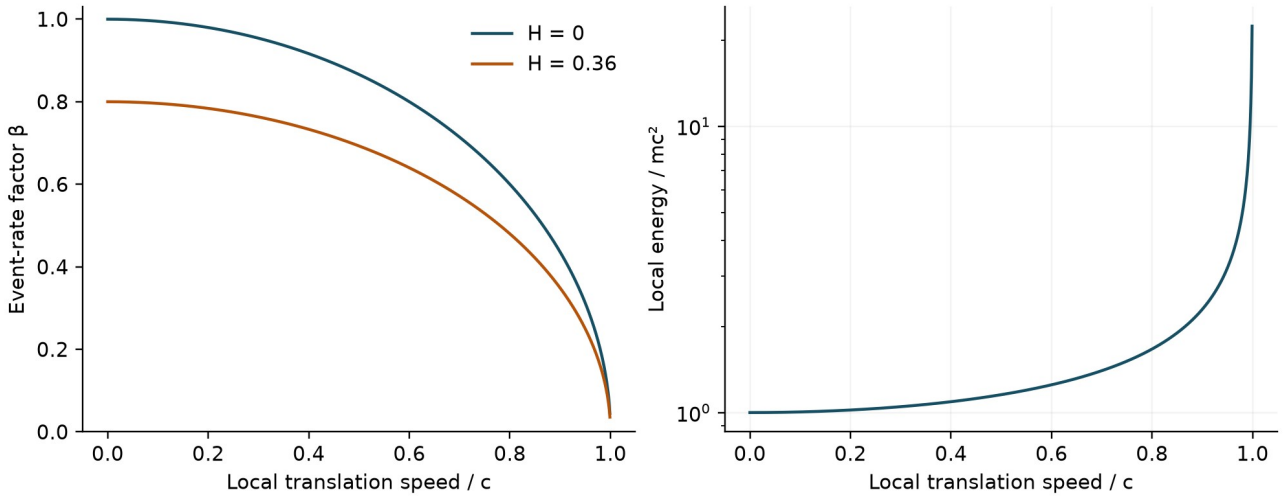


Figure 1. Left: event-rate factor β as local translation increases, for two fixed occupancies. Right: the local massive-probe energy obtained from the action in Section 3.6. The divergence is a limit as vlocal approaches c from below. In this retained plot, H denotes the occupancy u used in the text.

2.2 Displacement and proper velocity

Multiplying equation (2) by dT^2 turns the translation rate into a displacement:

$$d\ell^2 + c^2 d\tau^2 = (c^2 - S^2) dT^2. \quad (6)$$

$$c^2 d\tau^2 = \alpha^2 c^2 dT^2 - d\ell^2. \quad (7)$$

This gives the time–motion partition of the interval. To calculate a path, write $d\ell^2 = q_{ij} dx^i dx^j$, where q_{ij} is a positive spatial quadratic form in the static region and repeated spatial indices are summed. Section 3 determines its spherical radial response through the source and physical-current construction.

Now introduce proper velocity $u_p^i = dx^i/d\tau$ and its norm $u_p^2 = q_{ij} u_p^i u_p^j$. Since $dx^i/dT = \beta u_p^i$, $Vrel^2 = \beta^2 u_p^2$. Equation (2) becomes:

$$\beta^2 (c^2 + u_p^2) = c^2 \alpha^2. \quad (8)$$

$$\beta = \frac{\alpha}{\sqrt{1 + u_p^2/c^2}}, u_p = \gamma_{local} v_{local}. \quad (9)$$

Here u_p measures displacement per unit of the moving probe's own event accumulation. The local static observer measures vlocal using its clock. Equation (9) relates the two rates; the local causal limit applies to vlocal.

Equations (2)–(9) follow by rearranging the invariant with these definitions. To proceed, I specify how mass establishes S, how burden enters the spatial response, and which action determines a free path.

2.3 The source law and its domain

Begin with a specified source domain Ω containing mass M in volume VΩ. I use the SET source law:

$$Q_\Omega^2 = 24\pi GM V_\Omega, \mu_Q = \frac{Q_\Omega^2}{24\pi V_\Omega} = GM. \quad (10)$$

Here QΩ has units m³/s and μQ has units m³/s². With a spherical source-domain radius RΩ, VΩ = 4πRΩ³/3 and:

$$Q_\Omega = 4\pi\sqrt{2GM R_\Omega^3}. \quad (11)$$

The density form follows by substituting VΩ = M/ρ̄:

$$Q_\Omega = M\sqrt{\frac{24\pi G}{\bar{\rho}}}. \quad (12)$$

Keep the source domain attached to QΩ. For several sources, the volume-normalized sum is $\Sigma Q_a^2/V_a = 24\pi G\Sigma M_a$. This remains unchanged if we refine a numerical description of the same mass. For example, divide a uniform domain into N equal cells. Then Qcell = QΩ/N and $\Sigma Qcell^2 = QΩ^2/N$, while $\Sigma Qcell^2/Vcell = QΩ^2/VΩ$. It is the normalized quantity, rather than bare Q^2, that survives arbitrary subdivision.

For a fixed stationary source, μQ = GM is the constant strength entering the field equation. The spherical construction below determines how the burden varies outside that source.

For an evaluation sphere at r, define a separate area-scaled readout:

$$Q_0(r) = 4\pi r^2 S(r). \quad (13)$$

QΩ describes the source domain; $Q_0(r)$ describes an evaluation screen. Both have units m³/s, but they answer different questions. To determine whether Q_0 stays fixed between screens, we must first solve the field.

A source-domain quantity, a screen reading and a receiving-clock rate refer to different physical operations. I keep their labels visible throughout the paper. In the compact-body calculation, M is the total source mass-energy parameter. The galaxy and cosmological applications instead use their explicitly adopted baryonic inventories.

Table 1. Source quantities and their readings

Reading	Definition and use
Source domain	$Q_\Omega^2 = 24\pi GM V_\Omega$ for the stated source volume.
Spherical exterior screen	$Q_0(r) = 4\pi r^2 S(r)$, evaluated after the field is solved.
Collective boundary	$Q_{ref} = A_{eff} S_{eff}$ at the adopted mass and reference radius.
Held receiving clock	$Q_\star = Q_0/\alpha$, with the same spatial reading expressed per held-clock second.
Horizon screen	Q_H is the limiting source reading at the specified outer cap.
Cosmological material domain	$Q_m^2 = 24\pi G M_m V_m$; the retained evolution law is $dV_m/dT = Q_m$.

For the isolated exterior, T is normalized by the unloaded limit at infinity. For the expanding homogeneous cosmological material domain, the volume law in Section 10 defines a source-volume clock through $dT_Q = dV_D / Q_D$ at fixed baryonic mass. The tested realization identifies $T = T_Q$ and relates it to the comoving physical clock by $dt = \alpha\, dT_Q$, with α determined by its stated collective-burden prescription. The reference construction is therefore explicit in each application.

2.4 The radial motion reading

Here S is gravitational burden in speed units, T is the stationary reference time, and τ records the probe's clock progression. For a held observer, define

$$A = \alpha^2 = 1 - \frac{S^2}{c^2}, d\tau_{\mathrm{h}} = \alpha\, dT. \quad (14)$$

Use areal radius r, so a spherical screen has area $A_\Sigma = 4\pi r^2$. The dimensionless A and the area A_Σ are distinct. In the equatorial plane, write

$$V_{\mathrm{rel}}^2 = \zeta^2 \dot{r}^2 + r^2 \dot{\varphi}^2, d\ell = \zeta\, |dr|. \quad (15)$$

Dots denote d/dT. The positive factor ζ converts an areal-radius displacement to the radial distance entering the local motion measurement. It will be determined by the field construction rather than fixed at the outset.

Notation distinguishes occupancy u = S²/c², the lapse A = α², and the radial coefficient B = ζ². In the work-current formulas $\mathcal{H}$ = AB is a product of the two radial functions; only the earlier Appendix A route assumes a universal function $\mathcal{H}$(A). H denotes the cosmological rate, $\mathcal{J}$ the work-current, and J source angular momentum. Hamiltonians carry a descriptive subscript; the cosmological rate H and the radial product $\mathcal{H}$ are specified in their own sections. The fixed source-domain Qs and variable exterior screen Qscr have separate roles.

3 Physical work and the spherical response

SET takes continuous sourcing of field burden as a physical principle. The spherical calculation now asks one energy to generate the probe's motion and internal clock and to account for the complete central source. The source law fixes its charge. A local work-current principle and the displayed field-energy functional then give a precise energy-selection test for the exterior. The assumptions defining that class are stated before the proof; the field is solved before an orbit is calculated.

The calculation concerns an isolated, nonrotating, positive spherical source and a test body in its stationary vacuum exterior. The reference time is normalized at infinity, and the held-clock capacity is positive throughout the region considered. The source interior, other bodies and radiation reaction are outside this calculation.

3.1 Source law and the energy of a complete source

SET begins with matter establishing burden in the field through which motion and physical events occur. The source law gives that interaction a definite magnitude. To follow an interacting source consistently, its mass parameter must include the energy of the complete source state: material energy, binding, the field contribution and any physical support. Denote that energy by E_s and its mass equivalent by $M = E_s / c^2$.

$$Q_s^2 = 24\,\pi\, G\,\frac{E_s}{c^2}\,V_{ref}$$
$$\mu_Q \equiv \frac{Q_s^2}{24\,\pi\, V_{ref}} = \frac{G\,E_s}{c^2} = GM \qquad (16)$$

For a spherical source with fixed physical areal radius R_0, the specified reference volume is $V_{ref} = 4\,\pi\, R_0^3 / 3$. It is the volume used in the source law; it is not silently replaced by an integral over an unknown radial measure. The actual source surface reads $Q_s = 4\,\pi\, R_0^2\, S(R_0)$. Squaring this surface relation and using (16) gives

$$Q_s^2 = 32\,\pi^2\,\mu_Q\, R_0^3$$
$$S_0^2 = \frac{2\,\mu_Q}{R_0},\; A_0 = 1 - \frac{2\,\mu_Q}{c^2\, R_0} \qquad (17)$$

Here Q_s labels this source domain. An exterior evaluation screen is a different object: its readout will be calculated from the solved field. The source law fixes the charge μ_Q before any orbital speed or perihelion advance is used.

For the fixed spherical source domain the notations agree as Qs = QΩ and R_0 = R. The exterior screen reading Qscr(r) is the quantity $Q_0(r)$ used in the earlier algebra retained below; it is distinct from the fixed domain label.

3.2 The same energy generates motion and the internal clock

The capacity relation assigns a clock rate to a local motion state. In a stationary description with no transport shift, let q_{ij} specify the physical spatial reading and let $v^i = d\,x^i / dT$. Then

$$V_{rel}^2 = q_{ij} v^i v^j,\ A = \alpha^2 = 1 - \frac{S^2}{c^2}$$
$$\beta^2 = A - \frac{q_{ij} v^i v^j}{c^2},\ d\tau = \beta\, dT \qquad (18)$$

I require a physical internal clock and the translational motion of the same test body to be generated by one energy. The mechanical framework retained here is regular canonical evolution, with additive test-body energy and momentum in a supplied field. The positive massive branch has $A > 0$ and $\beta > 0$. These conditions make capacity restrictive: the clock and motion derivatives of that common energy must agree with (18).

Let θ be an actual internal phase, I its conjugate action, and $h(I)$ its local internal energy. Write $E_0 = E_{base} + h(I)$ and $E_p = E_p(x, p, E_0)$ for the reference energy of the body. Canonical evolution and the physical clock give, respectively,

$$v^i = \frac{\partial E_p}{\partial p_i}$$
$$\frac{d\theta}{dT} = \frac{\partial E_p}{\partial E_0}\frac{dh}{dI} \qquad (19)$$
$$\frac{d\theta}{dT} = \beta \frac{dh}{dI}$$

For a functioning phase with $dh/dI \neq 0$, the last two readings must be the same. Thus the internal clock imposes a derivative of the very energy that also generates motion:

$$\left.\frac{\partial E_p}{\partial E_0}\right|_{x,p} = \beta$$
$$S^2 + q_{ij}\frac{\partial E_p}{\partial p_i}\frac{\partial E_p}{\partial p_j} + c^2\left(\frac{\partial E_p}{\partial E_0}\right)^2 = c^2 \qquad (20)$$

The derivative compares neighboring internal-energy states at fixed canonical momentum; it does not require the body to change its rest energy during an orbit. Additivity at test-body order supplies degree-one homogeneity. Euler's identity then writes the energy as its internal-clock and translational contributions:

$$E_p(x, \lambda p, \lambda E_0) = \lambda E_p(x, p, E_0),\ \lambda > 0$$
$$E_p = E_0 \beta + p_i v^i \qquad (21)$$

Differentiate the second expression with respect to momentum. The direct velocity terms cancel. The remaining relation and the capacity derivative are

$$\frac{\partial v^i}{\partial p_j}\left(p_i + E_0\frac{\partial \beta}{\partial v^i}\right) = 0$$
$$\frac{\partial \beta}{\partial v^i} = -\frac{q_{ij} v^j}{c^2 \beta} \qquad (22)$$

On the regular massive branch the momentum Hessian is invertible, so the bracket must vanish. This fixes momentum and reference energy together:

$$p_i = \frac{E_0 q_{ij} v^j}{c^2 \beta}$$
$$E_p = \frac{E_0 A}{\beta} = \alpha \sqrt{E_0^2 + c^2 q^{ij} p_i p_j} \qquad (23)$$
$$L_p = p_i v^i - E_p = -E_0 \beta$$

The clock term therefore becomes the reduced probe action through the common-energy construction. At fixed internal action, $I_p = -E_0 \int \beta \, dT = -E_0 \int d\tau$. The complete phase-space action also contains $I \, d\theta / dT$. The internal phase is a physical clock variable, not a claim that every particle emits an observable train of ticks at a universal frequency.

In an unloaded field, the low-speed momentum identifies the measured inertial mass as $m = E_0 / c^2$. The same expression gives the attractive weak-field energy and the familiar quadratic motion term:

$$p_i \simeq m v_i, E_0 = m c^2$$
$$E_p - E_0 = \frac{1}{2} m V_{rel}^2 - \frac{1}{2} m S^2 + O\left(\frac{m \left(S^2 + V_{rel}^2 \right)^2}{c^2} \right) \qquad (24)$$
$$L_p = -m c^2 + \frac{1}{2} m \left(V_{rel}^2 + S^2 \right) + O\left(\frac{m \left(S^2 + V_{rel}^2 \right)^2}{c^2} \right)$$

The full differential keeps the exchange visible when burden, motion or the spatial reading changes. With $m = E_0 / c^2$, differentiation of the selected energy gives

$$d E_p = v^i d \, p_i + \beta \, d E_0 - \frac{m}{2\beta} d \left(S^2 \right) - \frac{m}{2\beta} v^i v^j d \, q_{ij} \qquad (25)$$

No energy term has been set constant in this identity. Once the common canonical evolution is applied, a stationary background conserves E_p; a changing field contributes the corresponding explicit time derivative. For fixed rest energy, the spatially covariant momentum equation in the stationary description is

$$J_{Bi} = -\frac{1}{2} \partial_i S^2, \frac{D \, p_i}{dT} = -\frac{m}{\beta} J_{Bi} \qquad (26)$$

The current J_B in the momentum equation and the held-work current introduced below describe the same gradient with different clock normalization. Their factors must be retained when moving between energy, work and motion readings.

The selected action retains its measured inertial calibration. Expanding it at low speed gives the same potential and weak-field acceleration:

$$L_p = -m_p c^2 + \frac{m_p}{2} \left(V_{\mathrm{rel}}^2 + S^2 \right) + O\left(\frac{m_p \left(S^2 + V_{\mathrm{rel}}^2 \right)^2}{c^2} \right). \qquad (27)$$

At leading order in the unloaded spatial measure, the specific potential and the acceleration are

$$\Phi = -\frac{S^2}{2}, a = -\nabla \Phi = \frac{1}{2} \nabla S^2. \qquad (28)$$

The attractive response follows because the spherical burden decreases outward. The higher-order orbital terms require the full action and the radial response, which we now determine together with the exterior field.

The same action can be expressed in phase cycles. Define νm = mc²/h and dNm = νm dτ, with h Planck's constant. Then −h∫dNm = −h(mc²/h)∫dτ = −mc²∫dτ. With φm = 2πNm and $\hbar$ = h/(2π), it is also −$\hbar$∫dφm.

The connection to capacity is dNm/dT = νmβ: the same β controls this phase accumulation relative to T. Nm is a phase normalization; N in Section 2.1 counts recorded clock cycles. Observing phase requires a physical comparison sensitive to phase differences. This rewriting supplies neither an internal energy spectrum nor a quantization rule, and it assigns no observable tick train at νm to every particle.

3.3 A Q-based field system and its stationary work current

The source law supplies the charge; a field system specifies how the complete source state evolves and how that charge establishes the response sampled by a probe. The completion used here retains one spatial tensor q_{ij}, its canonical momentum π^{ij}, a common lapse N and a transport shift ξ^i. Field changes and material events use these same clocks and spatial readings. The local class has constant coefficients, is quadratic in field momenta and contains at most two spatial derivatives, with no independent additional clock or directional field and a flat unloaded state.

The following canonical equations use $c = 1$; ordinary units are restored for the source and static energy formulas below. Let $q = det\, q_{ij}$, $\pi = q_{ij}\,\pi^{ij}$, and let $R[q]$ be the scalar curvature of the operational spatial reading. The local quadratic class is

$$\mathcal{C}_f = \frac{a_K \pi^{ij}\pi_{ij} + b_K \pi^2}{\sqrt{q}} + c_R \sqrt{q}\, R[q] \qquad (29)$$
$$D_{fi} = -2\, q_{ij} D_k \pi^{jk}$$

Two descriptions of successive normal field changes must compose into the same spatial change. With $\mathcal{C}[n] = \int n\, \mathcal{C}\, d^3x$, the retained normal-composition condition and the coefficient comparison give

$$\{\mathcal{C}[n], \mathcal{C}[m]\} = D[q^{ij}(n\partial_j m - m\partial_j n)] \qquad (30)$$
$$a_K c_R = -1,\ a_K + 2b_K = 0$$

The second relation removes an independent trace-gradient contribution. Matching the remaining normalization to the retained static source energy sets $C_0 = 1/(16\pi G)$ in these units. The resulting field constraint is

$$\mathcal{C}_f = \frac{\pi^{ij}\pi_{ij} - \pi^2/2}{C_0\sqrt{q}} - C_0\sqrt{q}\, R[q] \qquad (31)$$
$$C_0 = \frac{1}{16\pi G}$$

Include material and support contributions, denoted by subscripts m, in the same generator. The complete field equation consists of the constraints and their common evolution:

$$H_{tot} = \int \left[N(\mathcal{C}_f + \mathcal{C}_m) + \xi^i (D_{fi} + D_{mi}) \right] d^3x + E_\partial$$
$$\mathcal{C}_f + \mathcal{C}_m = 0,\ D_{fi} + D_{mi} = 0 \qquad (32)$$
$$\partial_T q_{ij} = \frac{\delta H_{tot}}{\delta \pi^{ij}},\ \partial_T \pi^{ij} = -\frac{\delta H_{tot}}{\delta q_{ij}}$$

The material variables evolve with this generator as well. Source matter, stress, initial data and boundary data specify the state being solved. The vanishing bulk constraints do not make an isolated source's energy zero: its normalized boundary energy is E_s. Restoring c, that is the complete energy entering (16). This presents the source law inside the field system rather than treating Q_s as an unrelated label appended to an orbit.

The static form needed for the spherical proof is particularly direct. In isotropic spatial coordinates, $q_{ij} = \psi^4 \delta_{ij}$ and the retained field and material energies are

$$W_f = -\frac{c^4}{2\pi G} \int \nabla \psi \cdot \nabla (\alpha \psi) \, d^3 x \qquad (33)$$
$$W_m = \int \alpha \varepsilon \, dV \, , dV = \psi^6 \, d^3 x$$

Here ε is proper material and support energy density. The displayed gradients and $d^3 x$ use the isotropic Euclidean coordinates; dV is physical volume. The energy is specified before a field solution is inserted. In particular, a simplified expression that agrees with it only on the final spherical solution cannot replace it when comparing other profiles.

Stationary variations of this common energy, including physical stress σ, give the work-current relation

$$J_W = c^2 D\alpha = -\frac{D(S^2)}{2\alpha} \qquad (34)$$
$$D \cdot J_W = \frac{4\pi G}{c^2} \alpha (\varepsilon + \operatorname{tr} \sigma)$$

The derivative D uses physical spatial lengths. For an isolated stationary source, the complete material, field and support accounting identifies $E_s = \int \alpha (\varepsilon + \operatorname{tr} \sigma) \, dV$. Integrating the current equation therefore gives the source charge in Q units:

$$\int_\Sigma J_W \cdot d A = \frac{4\pi G E_s}{c^2} = 4\pi \mu_Q = \frac{Q_s^2}{6 V_{ref}} \qquad (35)$$

This is the stationary source-current field principle used below. In the complete construction it follows from the field and support variations; in the exterior comparison it will be retained while the radial functions remain unsolved. The kinetic form, source-energy functional and composition rule specify the field class in which the selection is proved. The proof does not require the scalar capacity identity by itself to select every conceivable field content.

$$E_{complete} = E_{background} + \varepsilon_s E_p + O(\varepsilon_s^2)$$
$$\Delta \left(\frac{Q_{complete}^2}{V_{ref}} \right) = \frac{24\pi G}{c^2} \varepsilon_s E_p + O(\varepsilon_s^2) \qquad (36)$$

Equation (36) connects the probe energy to complete sourcing. Scale the added body's rest energy and momentum together by ε_s, fixing independent inner-source and support changes. The first variation of the retained complete energy has coefficient E_p; background field readjustment is included through the stationary field equations and boundary terms. Thus the same energy that generates the probe's clock and motion also gives its first-order contribution to the complete source charge. The expansion is in added source strength, not in orbital speed or background burden.

3.3.1 Held work and the radial current reading

Combining equations (1) and (15) gives $\beta^2 = A - \left(\zeta^2 \dot{r}^2 + r^2 \dot{\varphi}^2\right)/c^2$. Differentiation of the action yields

$$p_r = \frac{m_p \zeta^2 \dot{r}}{\beta}, \; p_\varphi = \frac{m_p r^2 \dot{\varphi}}{\beta}. \quad (37)$$

$$E = p_r \dot{r} + p_\varphi \dot{\varphi} - L_p = \frac{m_p c^2 A}{\beta}. \quad (38)$$

The final equality uses $\zeta^2 \dot{r}^2 + r^2 \dot{\varphi}^2 = c^2\left(A - \beta^2\right)$. Locally, the motion factor is $\gamma = \alpha/\beta$, so $E = \alpha\, m_p c^2 \gamma$. A held probe therefore has $E_{\text{hold}} = m_p c^2 \alpha$.

For a quasistatic outward lift, local work $F_{\text{hold}}\, d\ell$ contributes $\alpha\, F_{\text{hold}}\, d\ell$ to the reference energy. Equating this to $d\, E_{\text{hold}}$ gives

$$\alpha\, F_{\text{hold}} \zeta\, dr = m_p c^2 \frac{d\alpha}{dr}\, dr. \quad (39)$$

$$a_{\text{hold}} = \frac{c^2}{\zeta}\frac{d\ln\alpha}{dr}, \; \mathcal{J} \equiv \alpha\, a_{\text{hold}} = \frac{c^2}{2\sqrt{\mathcal{H}}}\frac{dA}{dr}. \quad (40)$$

$\mathcal{J}$ is the outward redshifted support acceleration. Defining the radial increment $d\chi = \alpha\, d\ell = \sqrt{\mathcal{H}}\, dr$ for outward displacements also gives $\mathcal{J} = -\frac{1}{2} d\, S^2/d\chi$. The quantity has units of acceleration; it is not the volume-per-time readout Q.

Sections 4.2 and 4.3 derive the finite and differential forms of this same current from the capacity relation and the held-clock normalization. They keep the distance χ explicit until the radial response has been selected.

We retain the spherical physical-current law of the earlier SET construction [10, 1]: every complete vacuum screen carries the same source charge in this work-current measure,

$$\oint_{\Sigma_r} \mathcal{J} \cdot d\, A_\Sigma = 4\pi GM, \; r^2 \mathcal{J} = GM. \quad (41)$$

Equation (41) is the retained stationary field-response principle: the physical work-current carries the charge fixed by the source law through a complete vacuum sphere. Together with the displayed complete-energy functional, it defines the field problem used in the following selection proof. Here ζ^2=B and $\mathcal{H}$=AB are local radial readings; no universal $\mathcal{H}$(A) dependence is required in that proof.

3.4 Complete energy selects the spherical response

The next task is to determine both the burden profile and the radial motion reading. Write the stationary spherical spatial reading as $q = B(r)\, dr^2 + r^2\, d\Omega^2$, so the earlier radial factor is $\zeta^2 = B$. Let $A = \alpha^2$, $b = \mu_Q/c^2 = G\, E_s/c^4$, and work in a regular positive-clock exterior $r \geq R_0 > 2b$. The source surface fixes $A(R_0) = A_0 = 1 - 2b/R_0$, and the isolated reference clock has $\alpha(\infty) = 1$.

The source energy calibrates the isolated spatial charge as well as the clock charge. The admissible exterior falloff is $B = 1 + 2b/r + O\left(r^{-2}\right)$. The stationary spherical current has the form

$$J_W=\frac{c^2\alpha'}{\sqrt{B}},\ r^2J_W=\mu_Q$$
$$\kappa_r\equiv\sqrt{AB}>0,\ A'=\frac{2b\kappa_r}{r^2} \qquad (42)$$

A prime in this subsection initially denotes d/dr. Current conservation makes A monotone, but it has not set κ_r to one. Under the stated falloff it also gives $A=1-2b/r+O(r^{-3})$. We compare exterior profiles obeying this source boundary, current and isolated energy normalization, while keeping the complete source state fixed.

3.4.1 The source energy budget without a radial choice

Use a supported thin spherical shell to calibrate the budget independently of B. Denote its proper material plus support energy by E_m. The interior fields are constant and therefore contribute no gradient energy in (33). At fixed physical R_0, the common-clock loading condition and source-surface law are

$$g=\frac{G}{R_0c^4},\ \frac{\partial E_s}{\partial E_m}=\alpha_0=\sqrt{1-2gE_s} \qquad (43)$$
$$E_s(0)=0$$

This derivative describes the same supported source after its field has readjusted. It includes the support energy rather than treating a holding device as a free external reservoir. To integrate it, set $z=\sqrt{1-2gE_s}$. Then $dz/dE_m=-g$ and $z(0)=1$. Hence

$$\alpha_0=1-gE_m,\ E_s=E_m-\frac{gE_m^2}{2},\ 0<gE_m<1$$
$$W_{req}=E_s-\alpha_0E_m=\frac{gE_m^2}{2}=E_s\frac{1-\alpha_0}{1+\alpha_0} \qquad (44)$$
$$W_{total}=\alpha_0E_m+W_{ext}$$

The material contribution to reference energy is α_0E_m. Equation (44) fixes what the exterior field must contribute. It contains no choice of B and no orbital observable. An exterior deficit cannot be repaired by silently changing the shell or its support, because that would change the specified complete source.

3.4.2 Evaluate the full energy before selecting the field

Let $f_B=B^{-1/2}$. Introduce the isotropic radius ρ used by the functional, with $r=\rho\psi^2$. Matching the radial and angular spatial readings gives

$$f_B=1+\frac{2\rho\psi_\rho}{\psi},\ dr=\psi^2f_B\,d\rho \qquad (45)$$

Substitute these identities directly into (33) and perform the angular integral. The exterior energy becomes

$$W_{ext}=-\frac{c^4}{2G}\int_{R_0}^{\infty}\left[\frac{\alpha(f_B-1)^2}{f_B}+2r\alpha'(f_B-1)\right]dr \qquad (46)$$

The current equation says $\alpha' f_B=b/r^2$. Using it, the integrand can be rewritten without assuming $AB=1$:

$$\frac{\alpha(f_B-1)^2}{f_B}+2r\alpha'(f_B-1)=\frac{A}{\kappa_r}+\kappa_r+\frac{2b}{r}-2(r\alpha)' \qquad (47)$$

The complete expression is finite. Its separately integrated pieces need not be, so the calculation retains a finite outer cutoff until their difference is assembled. As a comparison, evaluate $A_*=1-2b/r$ and $B_*=A_*^{-1}$. Direct integration gives

$$W_*=2E_s\int_{\alpha_0}^{1}\frac{d\alpha}{(1+\alpha)^2}=E_s\frac{1-\alpha_0}{1+\alpha_0}=W_{req} \qquad (48)$$

This verifies that one profile supplies the correct source budget. It does not impose that profile on any other candidate. Uniqueness follows from the complete difference of their energies.

3.4.3 The exact positive gap

Because A is strictly increasing, use $a=A(r)$ as the integration variable. Here a is a local mathematical variable, not the cosmological scale factor. Define

$$y(a)=\ln\frac{2b}{r(a)},\ y_*(a)=\ln(1-a)$$
$$\eta(a)=y-y_*=\ln\frac{2b}{r(a)(1-a)} \qquad (49)$$

The profile difference is $\eta=0$ for the comparison field. The change of variable gives $dr/da=-r\,y_a$ and $\kappa_r=-1/(e^y y_a)$. Consequently the non-boundary part of the energy takes the simple form

$$F[y]\equiv\int\left(\frac{A}{\kappa_r}+\kappa_r\right)dr=2b\int\left[a\,y_a^2+e^{-2y}\right]da \qquad (50)$$
$$(a\,y_*')'=-e^{-2y_*}$$

Expand $y=y_*+\eta$ and integrate the quadratic cross term once. The second identity in (50) cancels the term linear in η in the interior. The remaining bulk expression is

$$I[\eta]=\int_{A_0}^{1}\left[a\left(\frac{d\eta}{da}\right)^2+\frac{e^{-2\eta}-1+2\eta}{(1-a)^2}\right]da \qquad (51)$$

The fixed source surface gives $\eta(A_0)=0$. The isolated energy normalization gives $\eta=O((1-a)^2)$ at infinity. The combined endpoint contribution therefore vanishes; Appendix K keeps the cutoff terms together and shows this limit explicitly. Since $c^4b/G=E_s$, the complete result is

$$\frac{W_{total}-E_s}{E_s}=-\int_{A_0}^{1}\left[a\left(\frac{d\eta}{da}\right)^2+\frac{e^{-2\eta}-1+2\eta}{(1-a)^2}\right]da \qquad (52)$$

Every term inside the integral is nonnegative: $a>0$, the derivative is squared, and $e^{-2\eta}\geq 1-2\eta$, with equality only at $\eta=0$. Requiring the complete field-plus-source energy to equal the independently calibrated E_s therefore forces η to vanish throughout a regular exterior. From its definition and the current equation,

$$r(1-A)=2b,\ A'=\frac{2b}{r^2}=\frac{2b\kappa_r}{r^2}$$
$$A=1-\frac{2\mu_Q}{c^2 r},\ \kappa_r=1,\ B=A^{-1} \qquad (53)$$

The radial motion response and exterior burden have now been selected by the same complete-energy accounting. Neither $AB=1$ nor an orbital precession was imposed on the candidates. The result is an exact consistency theorem within the retained static field construction, rather than a separate maximization rule or a claim about the stability of arbitrary time-dependent perturbations.

3.4.4 From the shell calibration to an exterior around a star

The shell made the global budget calculable without knowing B. To apply the same construction to a star, retain its actual interior and support contributions and use the local stationary equations of the same functional. For compact lapse variations at fixed spatial and material canonical data,

$$\delta_\alpha W_f=\frac{c^4}{2\pi G}\int \psi\,\Delta\psi\,\delta\alpha\,d^3x$$
$$\delta_\alpha W_m=\int \psi^6\,\varepsilon\,\delta\alpha\,d^3x \qquad (54)$$
$$\Delta\psi=-\frac{2\pi G}{c^4}\psi^5\varepsilon$$

The conformal spatial identity and its spherical form are

$$R[q]=-8\psi^{-5}\Delta\psi=\frac{16\pi G}{c^4}\varepsilon$$
$$R[q]=\frac{2}{r^2}\frac{d}{dr}\left[r\left(1-B^{-1}\right)\right] \qquad (55)$$

In the exterior material and support vacuum, the bracket is constant. Its isolated complete-energy normalization sets it to $2b$, giving $B=(1-2b/r)^{-1}$. The same current fixes α with the infinity clock:

$$\alpha'=\frac{b}{r^2\sqrt{1-2b/r}},\ \alpha(\infty)=1$$
$$\alpha=\sqrt{1-2b/r},\ S^2=\frac{2\mu_Q}{r} \qquad (56)$$

This extends the selected exterior to admissible isolated spherical interiors satisfying the retained source and stationary field equations. It uses a local consequence of the same energy functional; that local spatial equation was deliberately not imposed on the competing profiles in the global gap proof. The Sun can therefore be treated through its complete spherical source energy without modeling it as a thin shell.

The exterior can now be read directly in source variables. Before selection, the current equation retained the unknown radial factor; after selection it becomes

$$-\frac{r^2}{2\sqrt{AB}}\frac{d}{dr}\left[\frac{Q_{scr}(r)}{4\pi r^2}\right]^2=\frac{Q_s^2}{32\pi^2 R_0^3} \qquad (57)$$
$$AB=1,\ -\frac{r^2}{2}\frac{dS^2}{dr}=\mu_Q$$

Here $Q_{scr}(r)=4\pi r^2 S(r)$ is the evaluation-screen readout. The following exterior formulas and normalized-clock identities use this solved response; they do not replace the fixed source-domain Q_s by a constant flux through every screen.

3.5 The exterior burden and screen reading

With AB=1 selected by the complete-energy equality, the field equation becomes

$$\frac{dA}{dr}=\frac{2GM}{c^2 r^2}, A(\infty)=1. \quad (58)$$

Integrating from r to infinity gives the local profile:

$$1-A(r)=\int_r^\infty \frac{2GM}{c^2 u^2}du=\frac{2GM}{r c^2}, \quad (59)$$

$$A(r)=1-\frac{2GM}{r c^2}, S(r)^2=\frac{2GM}{r}. \quad (60)$$

The same calculation supplies the inverse-area work-current and the burden-gradient relation:

$$\mathcal{J}=\frac{GM}{r^2}=-\frac{1}{2}\frac{dS^2}{dr}, -\frac{r^2}{2}\frac{dS^2}{dr}=GM. \quad (61)$$

Before selection, the physical gradient contained the factor 1/√(AB). Both that radial response and the burden profile are now determined. The older carrier relation follows algebraically:

$$S^4=\frac{4G^2M^2}{r^2}, \mathcal{J}=\frac{S^4}{4GM}. \quad (62)$$

Define the local spherical screen readout, distinguishing it from the fixed source-domain quantity:

$$Q_0(r)=4\pi r^2 S(r)=4\pi\sqrt{2GMr^3}. \quad (63)$$

$$\frac{Q_0(r)^2}{24\pi V_{\text{ref}}(r)}=\frac{rS(r)^2}{2}=GM, V_{\text{ref}}(r)=\frac{4\pi r^3}{3}. \quad (64)$$

The source charge read this way is independent of the exterior screen. This constancy is a result of the solved field. $Q_0(r)$ itself grows as $r^{3/2}$ and is not a conserved volume discharge across successive spheres. At the material surface, $Q_0(R)=Q_\Omega$.

In terms of the screen quantity itself, the selected gradient equation can also be written as follows. This is the source-current relation expressed through Q_0 rather than a second field equation.

$$-\frac{r^2}{2}\frac{d}{dr}\left[\frac{Q_0^2}{(4\pi r^2)^2}\right]=\frac{Q_\Omega^2}{24\pi V_\Omega}. \quad (65)$$

For Earth, M = 5.9722×10²⁴ kg, R = 6.371×10⁶ m and G = 6.67430×10⁻¹¹ m³ kg⁻¹ s⁻² give QΩ ≈ 5.70567×10¹⁸ m³/s, S ≈ 11.19 km/s and u ≈ 1.39226×10⁻⁹. The weak-field gradient below gives 9.8203 m/s² at that mean spherical radius. These are normalization checks; Earth's rotation and latitude-dependent surface acceleration are outside this spherical example.

The surface of a body marks a change in the source distribution, not the end of its gravitational burden. Outside Earth, the burden continues according to the field law. Nothing special happens merely because we draw another enclosing surface ten kilometres higher. The field responds to the actual distribution of matter and its physical evolution; it does not respond to the boundary we choose for a calculation. That boundary tells us where we are reading the field.

3.6 The general static motion equation

The spherical calculation uses one radial response. The same action can be differentiated before specializing the static spatial form q_{ij}. In the following formulas m denotes the probe rest mass m_p, $v^i = dx^i/dT$, and $u_p{}^i = dx^i/d\tau$.

Set $v^i = dx^i/dT$ and write the Lagrangian using the invariant:

$$L = -mc^2\sqrt{1-u-\frac{q_{ij}v^iv^j}{c^2}}. \quad (66)$$

The required derivatives are:

$$\frac{\partial L}{\partial v^i} = m\,q_{ij}u_p^j. \quad (67)$$

$$\frac{\partial L}{\partial x^i} = \frac{mc^2}{2\beta}\partial_i u + \frac{m}{2\beta}(\partial_i q_{jk})v^jv^k. \quad (68)$$

Apply $d(\partial L/\partial v^i)/dT - \partial L/\partial x^i = 0$. Since $d/dT = \beta\, d/d\tau$ and $v^i = \beta u_p{}^i$, division by $m\beta$ gives:

$$\frac{d(q_{ij}u_p^j)}{d\tau} - \frac{1}{2}(\partial_i q_{jk})u_p^ju_p^k = \frac{c^2}{2\beta^2}\partial_i u. \quad (69)$$

Raising the index groups the spatial-metric derivatives into its connection. With $Du_p{}^i/d\tau = du_p{}^i/d\tau + \Gamma^i{}_{jk}u_p{}^ju_p{}^k$ and $D^iu = q^{ij}\partial_j u$:

$$\beta^2\frac{Du_p^i}{d\tau} = \frac{c^2}{2}D^iu = \frac{1}{2}D^iS^2. \quad (70)$$

Using equation (8) and $D^i\ln\alpha = -D^iu/(2\alpha^2)$ gives the equivalent form:

$$\frac{Du_p^i}{d\tau} = -(c^2+u_p^2)D^i\ln\alpha. \quad (71)$$

This is the static motion law used in SET. The source contributes the burden gradient, β records the probe event progression relative to T, and the probe mass cancels. Here its action is the reduced action selected by the common clock/energy construction in Section 3.2, varied for the supplied static spatial response.

$Du_p{}^i/d\tau$ is a spatial covariant acceleration. It is not identical to coordinate acceleration d^2x^i/dT^2 or to the acceleration measured by an accelerometer. A freely falling probe has zero spacetime proper acceleration. The equation describes its trajectory relative to the static spatial readout. Time-dependent, rotating and drifting fields require the corresponding additional terms.

3.7 Energy from the same action

For a stationary field, the canonical energy is $E = v^i(\partial L/\partial v^i) - L$. Substituting equations (66)–(67):

$$E = \frac{m V_{rel}^2}{\beta} + m c^2 \beta = \frac{m c^2 \alpha^2}{\beta}. \quad (72)$$

Use equation (9) to obtain:

$$E = \alpha m c^2 \sqrt{1 + \frac{u_p^2}{c^2}} = \alpha \gamma_{local} m c^2. \quad (73)$$

The energy measured by a local static observer is $E_{local} = E/\alpha = \gamma_{local} mc^2$. At fixed location it diverges as v_{local} approaches c from below for $m > 0$. In flat unloaded space the same action gives $p = \gamma mv$ and $E_{kin} = (\gamma - 1)mc^2$. Thus the limiting-speed energy cost follows from the adopted dynamics.

The conservation check can be shown directly. Metric compatibility gives $d(u_p^2)/d\tau = 2u_{pi}Du_p^i/d\tau$. Contracting equation (71) gives:

$$\frac{d(u_p^2)}{d\tau} = -2(c^2 + u_p^2)\frac{d\ln\alpha}{d\tau}. \quad (74)$$

Consequently:

$$\frac{d\ln(E/(mc^2))}{d\tau} = \frac{d\ln\alpha}{d\tau} + \frac{1}{2(c^2 + u_p^2)}\frac{d(u_p^2)}{d\tau} = 0. \quad (75)$$

The reference energy stays constant while the probe's local kinetic energy and field contribution change along its path. This checks that the adopted motion law conserves energy; the action and spatial response supplied the dynamics.

Expanding equation (73) for $S^2/c^2 \ll 1$ and $v_{local}^2/c^2 \ll 1$ gives:

$$E - mc^2 \approx \frac{1}{2} m v_{local}^2 - \frac{1}{2} m S^2. \quad (76)$$

For a spherical source this becomes $E_{kin} + U_{probe} \approx \tfrac{1}{2}mv^2 - GMm/r$. Define $\Psi = S^2/2$, the positive energy depth per unit mass. The conventional attractive potential is $\Phi = -\Psi$, so $U_{probe} = -m\Psi$. I retain S^2 for the capacity budget and use Ψ where it simplifies the energy calculation.

If the sources move, gravitational binding energy can change as the system exchanges energy among motion, internal processes, the field and radiation. Conservation applies to the complete energy balance. The source sum $\Sigma Q_a^2/V_a = 24\pi G\Sigma M_a$ has units m^3/s^2 and supplies a separate mass normalization. A collective law must account for both.

4 Clock normalization, propagation and the orbital gradient

4.1 The receiving clock and normalized burden

The same screen readout can be stated per held-clock second. Keeping its spatial normalization unchanged, define

$$Q_\star \, d\tau_h = Q_0 \, dT \, , Q_\star = \frac{Q_0}{\alpha}. \quad (77)$$

The solved exterior then gives the normalized expression used in the earlier SET algebra:

$$Q_\star(r) = 4\pi r^{3/2} \sqrt{\frac{2GM}{1 - 2GM/(rc^2)}}. \quad (78)$$

The physical source charge has not changed. A slower receiving clock assigns a larger rate to the same specified readout. If a starting rate were defined per source proper time instead of per T, its conversion would also include the source clock factor.

Introduce a speed-valued normalized burden, distinguished from the local capacity burden S:

$$s_\star \equiv \frac{Q_\star}{A_\Sigma} = \frac{S}{\alpha}. \quad (79)$$

The held capacity relation alone gives the following exact identities:

$$\frac{s_\star^2}{c^2} = \frac{1 - A}{A} \, , A = \frac{1}{1 + s_\star^2 / c^2}. \quad (80)$$

$$S^2 = \frac{s_\star^2}{1 + s_\star^2 / c^2}. \quad (81)$$

In the solved spherical profile, this becomes

$$s_\star^2 = \frac{2GM/r}{1 - R_s/r} = \frac{2GM}{r - R_s}. \quad (82)$$

This expression is not a new areal radius or a replacement for S in the invariant. Its denominator comes from the receiving clock's normalization. Section 5.1 derives its role in an equivalent orbital energy equation rather than treating it as an additional force.

For a moving receiving clock the same rate bookkeeping uses $Q_{\text{rec}} = Q_0 / \beta$. The factors α and β coincide only for a held observer. No literal transport or Doppler interception law is inferred solely by this change of clock units.

This is Q expressed through the proper observer's event rate. For example, at α = 0.8 one proper second spans 1.25 reference seconds, giving Qnormalized = $1.25Q_0$. The source QΩ, its strength μQ and the reference-time Q_0 are unchanged by this clock conversion.

Define the dimensionless readout z = Qnormalized/(AΣc), with AΣ = $4\pi r^2$. Since $Q_0/A\Sigma = S$, we have $z^2 = u/\alpha^2$. Substitute $\alpha^2 = 1 - u$ and rearrange: $z^2(1 - u) = u$, so $u(1 + z^2) = z^2$. Therefore:

$$\alpha^2 = \frac{1}{1 + z^2} \, , u = \frac{z^2}{1 + z^2}. \quad (83)$$

Substituting equations (63) and (60) gives equation (78). To recover the occupancy, use $u = z^2/(1 + z^2)$, or $S = cz/\sqrt{(1 + z^2)}$. The static exterior therefore has $0 \le u < 1$ and $0 \le S < c$. Qnormalized/AΣ includes the observer's clock normalization; it is not a local material or signal velocity.

4.2 Finite burden differences

Consider two exterior locations $r_1 < r_2$. Before setting $\mathscr{H} = 1$, their separation in the work-current measure is

$$\Delta\chi = \chi_2 - \chi_1 = \int_{r_1}^{r_2} \sqrt{\mathscr{H}(A(r))}\, dr > 0. \quad (84)$$

Subtracting the capacity invariant at the two locations gives an exact acceleration-valued difference after division by $2\Delta\chi$:

$$\frac{S_1^2 - S_2^2}{2\Delta\chi} = \frac{V_{\mathrm{rel},2}^2 - V_{\mathrm{rel},1}^2 + c^2(\beta_2^2 - \beta_1^2)}{2\Delta\chi}. \quad (85)$$

For held readings, $V_{\mathrm{rel},i} = 0$ and $\beta_i = \alpha_i$. Define the outward interval current by

$$\mathscr{J}_{12} \equiv \frac{S_1^2 - S_2^2}{2\Delta\chi} = \frac{c^2(\alpha_2^2 - \alpha_1^2)}{2\Delta\chi}. \quad (86)$$

Its connection to the physical current uses the work relation of Section 3.3. Integrating $\mathscr{J} = -\frac{1}{2} dS^2/d\chi$ shows that $\mathscr{J}_{12}$ is the interval average, not in general the local current at either endpoint:

$$\mathscr{J}_{12} = \frac{1}{\Delta\chi} \int_{\chi_1}^{\chi_2} \mathscr{J}(\chi)\, d\chi. \quad (87)$$

Now express the same burden difference through $s_{\star i} = S_i/\alpha_i$. Applying equation (81) separately at the two endpoints gives

$$S_1^2 - S_2^2 = \frac{c^4(s_{\star 1}^2 - s_{\star 2}^2)}{(c^2 + s_{\star 1}^2)(c^2 + s_{\star 2}^2)}. \quad (88)$$

Because $\alpha_i^2 = c^2/(c^2 + s_{\star i}^2)$, the exact conversion is

$$\mathscr{J}_{12} = \alpha_1^2 \alpha_2^2 \frac{s_{\star 1}^2 - s_{\star 2}^2}{2\Delta\chi}. \quad (89)$$

The product of the two endpoint factors is fixed by the algebra. No midpoint approximation or new coefficient has entered. For positive held capacity the conversion is reversible. The factor $1/2$ and the distance measure have their physical meaning from the retained work coupling; the finite identity itself follows from the capacity relation and the definitions.

4.3 The differential work current

Take the differential limit, or differentiate equation (81) directly:

$$\frac{dS^2}{d\chi} = \frac{c^4}{(c^2 + s_\star^2)^2} \frac{ds_\star^2}{d\chi} = \alpha^4 \frac{ds_\star^2}{d\chi}. \quad (90)$$

The physical work-current can therefore be written in either burden variable:

$$\mathscr{J} = -\frac{1}{2}\frac{dS^2}{d\chi} = -\frac{\alpha^4}{2}\frac{ds_\star^2}{d\chi} = -\frac{\alpha^4}{2\sqrt{\mathscr{H}}}\frac{ds_\star^2}{dr}. \quad (91)$$

This identity holds before response selection. In general $\Delta\chi$ is not r_2-r_1. Once Section 3.4 has selected AB=1, the two radial increments coincide and the solved spherical profile gives

$$\frac{d\,s_\star^2}{dr}=-\frac{2GM}{(r-R_s)^2},\alpha^4=\frac{(r-R_s)^2}{r^2}. \quad (92)$$

The apparent enhancement in the normalized gradient cancels exactly in the physical current:

$$\mathcal{J}=-\frac{1}{2}\frac{(r-R_s)^2}{r^2}\left(-\frac{2GM}{(r-R_s)^2}\right)=\frac{GM}{r^2}. \quad (93)$$

This is an exact exterior identity for $r>R_s$, not a weak-field expansion. For two finite radii the corresponding check is

$$\mathcal{J}_{12}=\frac{2GM/r_1-2GM/r_2}{2(r_2-r_1)}=\frac{GM}{r_1r_2}. \quad (94)$$

The same value follows from equation (89). It is the average of GM/r^2 over the radial interval and tends to the local current as the endpoints coincide.

These relations connect the invariant-derived burden difference to the current used in the field calculation. They express one current in two normalizations; its conservation across complete vacuum screens remains the physical field law stated in Section 3.3. The held factors cannot simply be replaced by moving-clock factors β_i: the velocity terms in equation (85) must then be retained.

The orbital use of $s_\star$ in Section 5.1 also changes the trajectory parameter and retains the radial kinetic coefficient. Its potential gradient must be interpreted together with those changes. Appendix H records the corresponding exact conversion for squared gradient expressions.

4.4 Light propagation and escape

In the selected exterior, the radial motion coefficient has a compact expression in normalized burden:

$$\zeta^2=\frac{1}{A}=1+\frac{s_\star^2}{c^2}. \quad (95)$$

Its equality to 1/A follows from the complete-energy selection in Section 3.4. The identities in Sections 4.1–4.3 express that selected response through s⋆; they do not select it a second time.

Use the retained propagation coupling in which a radial light signal has zero clock progression in the same interval. Setting $\beta=0$ in the radial capacity equation yields

$$\left|\frac{dr}{dT}\right|=\frac{c\alpha}{\zeta}=cA=\frac{c}{1+s_\star^2/c^2}. \quad (96)$$

$$\frac{d\ell}{dT}=c\alpha,\frac{d\ell}{d\tau_{\mathrm{h}}}=c. \quad (97)$$

The local clock and the signal propagation participate in the same response. For two nearby radial locations, the reference-clock round-trip time is

$$dT_{\mathrm{round}}=\frac{2|dr|}{cA}=\frac{2|dr|}{c}\left(1+\frac{s_\star^2}{c^2}\right). \quad (98)$$

The local radar distance obtained with the emitter's held clock is $c\alpha\, dT_{\text{round}}/2 = |dr|/\alpha$. A finite path requires integration through its varying capacity; the clock at a fixed emitter multiplies the complete reference-time interval by that emitter's lapse.

The distinction between S and $s_\star$ is also fixed by the escape calculation. The probe energy from Section 3.3 can be written

$$\varepsilon \equiv \frac{E}{m_p c^2} = \frac{A}{\beta} = \alpha\gamma\,, \gamma = \frac{\alpha}{\beta}. \quad (99)$$

Marginal escape to rest at infinity has $\varepsilon = 1$. For the local speed measured by a held observer, this gives

$$\gamma_{\text{esc}} = \frac{1}{\alpha}\,, v_{\text{esc}}^2 = c^2(1-\alpha^2) = S^2. \quad (100)$$

Thus S remains the local marginal escape speed in this branch. The normalized quantity is $s_\star = \gamma_{\text{esc}} v_{\text{esc}}$; it is not another local light-speed limit. Appendix D gives the optional free-fall representation after the response has been selected.

The observer's clock participates in the field it is used to measure. In SET, its ticks are physical events, and their progression depends on the available field interaction capacity. Gravitational burden therefore enters both the propagation of light and the clock used to time it.

A local static observer measures radial distance $d\ell r = |dr|/\alpha$ and records time $dt_{local} = \alpha dT$. Combining these relations with equation (96) gives:

$$\frac{d\ell_r}{dT} = c\alpha\,, \frac{dt_{local}}{dT} = \alpha. \quad (101)$$

Light covers less local distance per reference second where α is smaller. The local clock also records fewer ticks per reference second, by the same factor. Consequently, the observer measures:

$$\frac{d\ell_r}{dt_{local}} = \frac{c\alpha}{\alpha} = c. \quad (102)$$

The common factor cancels in the local measurement. The gravitational effect remains accessible through comparisons between clocks at different burdens and through signal travel times expressed in the reference time T. SET interprets this consistency between propagation and clock progression as their participation in the same limited field interaction capacity. The radial coordinate speed in equation (96) includes both the clock and spatial relations.

For fixed screens at r_1 and r_2, the radial round-trip coordinate time is:

$$\Delta T = \frac{2}{c}\left[r_2 - r_1 + R_s \ln\left(\frac{r_2 - R_s}{r_1 - R_s}\right)\right], R_s = \frac{2GM}{c^2}. \quad (103)$$

The excess over $2(r_2 - r_1)/c$ for solar screens at 1 and 2 AU is approximately 13.656 μs. This is an ideal coordinate-time comparison. A clock at the first screen records $\alpha(r_1)\Delta T$, and its comparison with a chosen flat reference must include that clock factor. The example isolates propagation delay without requiring a bent spatial ray.

S therefore determines the marginal local escape speed in this spherical branch. In the weak field, the energy condition $\tfrac{1}{2}v_{escape}^2 - \tfrac{1}{2}S^2 = 0$ gives the same result. In SET's interpretation the source establishes the burden and energy depth; the escape speed follows from them.

4.5 Dilution and the inward response

Move a probe farther from the spherical source. Doubling r reduces S by $1/\sqrt{2}$ and S^2 by 1/2. The gradient falls faster, by 1/4. To obtain the pull, we must calculate that spatial change.

For a slowly moving probe in a weak static field, $\beta \approx 1$, q_{ij} is approximately Euclidean, and equation (70) becomes:

$$a = \frac{1}{2}\nabla S^2. \quad (104)$$

Although S^2 is positive, its radial derivative is negative. With $S^2 = 2GM/r$:

$$\nabla S^2 = -\frac{2GM}{r^2}\hat{r}, a = -\frac{GM}{r^2}\hat{r}. \quad (105)$$

The burden is greater on the inner side, so its gradient points inward. Equivalently, α increases outward and $a \approx -c^2\nabla \ln\alpha$ points toward the slower held-clock rate. The energy gives the same direction: $a = -\nabla(U_{probe}/m) = \nabla\Psi$.

I picture this as a difference in event progression across the body. The variational calculation specifies its center-of-mass response without assigning an uncalculated thrust to its internal processes. Tides and internal stresses require an extended-body treatment.

A balance point helps separate burden from pull. Place two equal masses m at distance d on opposite sides of a midpoint. Their ordinary summed potential gives $S^2 = -2\Phi = 4Gm/d$ there, while symmetry makes the gradient zero. The burden remains even though the accelerations balance. This distinction is already present in ordinary weak-field gravity. General Lagrange points also involve rotating-frame terms. The additional collective response proposed below requires its own source-arrangement rule.

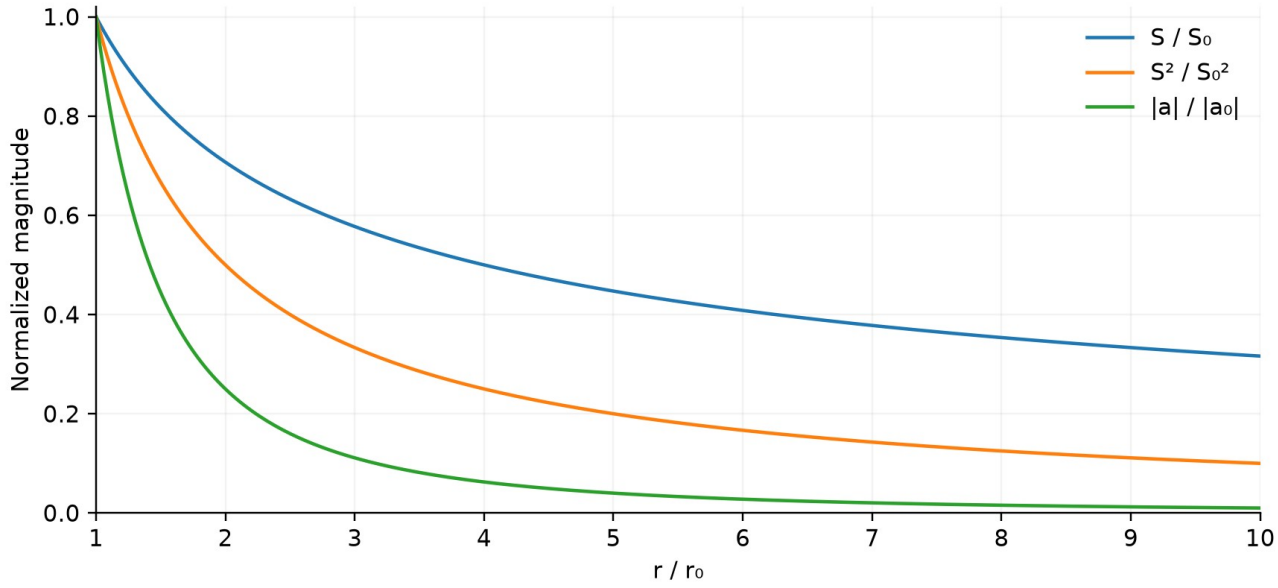


Figure 2. Exterior spherical scalings normalized at r_0. The source equation gives $S \propto r^{-1/2}$ and $S^2 \propto r^{-1}$; the motion law gives $|a| \propto r^{-2}$.

4.6 Orbital speed and the local gradient

Now ask what holds a probe in a circular orbit. The orbit responds to the local gradient of the same field. In a weak, stationary, axisymmetric field, the equatorial radial acceleration is $-\mathrm{vorb}^2/R$, giving:

$$v_{orb}^2 = -\frac{R}{2}\frac{\partial S^2}{\partial R}. \quad (106)$$

Define the local equatorial logarithmic slope k(R), with $S^2 > 0$:

$$k(R) = -\frac{\partial \ln S^2}{\partial \ln R}, v_{orb}^2 = \frac{k(R) S^2}{2}. \quad (107)$$

For the inverse-radius spherical field k = 1. The familiar conversion follows:

$$v_{orb} = \frac{S}{\sqrt{2}}, \frac{Q_{ref}^2}{V_{ref} R} = 24\pi v_{orb}^2, R = \frac{GM}{v_{orb}^2}. \quad (108)$$

This derives the speed conversion for the spherical inverse-radius field. More generally, equation (107) shows precisely when it applies: the local logarithmic slope must be k = 1. I use it as the orbital calibration for the collective boundary calculation in Section 7.3. A weak field with another slope must be evaluated through equation (106).

More generally, with S^2 tending to zero at infinity, integrating that weak-field gradient along the equatorial radius gives:

$$S^2(R) = 2\int_R^\infty \frac{v_{orb}^2(r)}{r} dr. \quad (109)$$

Here $\mathrm{vorb}^2(r)$ denotes the circular-speed function supplied by the radial force, wherever such an orbit is defined. The integral requires a convergent exterior continuation. It connects the potential depth at a boundary to the force outside it, rather than determining that depth from one velocity measurement alone. Appendix G illustrates a finite flat-speed region joined to an inverse-radius exterior.

Over a measured interval the same relation can be written without specifying the unobserved exterior:

$$S^2(R_1) - S^2(R_2) = 2\int_{R_1}^{R_2} \frac{v_{orb}^2(r)}{r} dr. \quad (110)$$

Thus circular-speed observations constrain burden differences under this motion law. They do not alone determine its absolute value. If a boundary value $S^2(R_1)$ is proposed independently, it must leave a nonnegative $S^2(R_2)$ after this subtraction, with an admissible continuation beyond R_2. This is a necessary consistency check, not a sufficient source solution.

5 Perihelion from the spherical field

The selected source charge in the orbital equation

The field selection is now upstream of the orbit. The central source enters through $\mu_Q = Q_s^2/(32\pi^2 R_0^3)$, while the probe has $E_0 = mc^2$. Stationarity and spherical symmetry of the common probe energy give the constants

$$\varepsilon = \frac{E_p}{mc^2} = \frac{A}{\beta}, \; j = \frac{p_\varphi}{m} = r^2 \frac{d\varphi}{d\tau} \qquad (111)$$

Before the radial selection is inserted, the capacity relation gives the exact radial energy equation

$$AB\left(\frac{dr}{d\tau}\right)^2 + A\left(c^2 + \frac{j^2}{r^2}\right) = c^2\varepsilon^2 \qquad (112)$$

Using (53), the product AB becomes one and $A = 1 - 2\mu_Q/(c^2 r)$. Thus the radial equation contains the additional angular term $-2\mu_Q j^2/(c^2 r^3)$. The existing Binet calculation then yields the exact spherical orbit equation in source variables:

$$\frac{d^2 w}{d\varphi^2} + w = \frac{Q_s^2}{32\pi^2 R_0^3 j^2} + \frac{3Q_s^2 w^2}{32\pi^2 R_0^3 c^2}, \; w = \frac{1}{r} \qquad (113)$$

The direct radial momentum equation supplies the same result at an apsis, where division by $dw/d\varphi$ would not be justified. The leading weak-field advance, with a_{orb} the orbital semimajor axis and e its eccentricity, is

$$\Delta\varpi = \frac{6\pi\mu_Q}{a_{orb}(1-e^2)c^2} = \frac{3Q_s^2}{16\pi R_0^3 a_{orb}(1-e^2)c^2} \qquad (114)$$

The source radius and orbital radius retain distinct roles. R_0 specifies the source-law reference domain; the orbital elements specify the measured trajectory. The combination Q_s^2/R_0^3 is the fixed central charge, so a consistent change of source reference does not change the perihelion prediction. The existing Mercury benchmark and exact-orbit checks follow from this same equation.

The selected field also fixes the radial null reading through the same capacity relation:

$$dT = \sqrt{\frac{B}{A}}\frac{|dr|}{c} = \frac{|dr|}{cA}$$
$$A = 1 - \frac{Q_s^2}{16\pi^2 R_0^3 c^2 r} \qquad (115)$$

The light section uses the common photon Hamiltonian to obtain the ray direction and deflection. Equation (115) establishes the radial delay; the deflection coefficient follows from the ray calculation rather than from matching a numerator alone.

5.1 Orbital energy in normalized burden

We now derive the normalized-burden form of the motion equations. Temporarily leave $\zeta(r)$ explicit so that the transformation can be checked without presupposing the selected response. The action has no explicit dependence on T or φ. Its constants are

$$\varepsilon = \frac{A}{\beta},\ j = \frac{r^2\dot{\varphi}}{\beta} = r^2\frac{d\varphi}{d\tau}. \quad (116)$$

The parameter j is angular momentum per unit probe mass. Introduce an auxiliary parameter along this fixed-energy orbit:

$$d\sigma = \varepsilon\, d\tau = A\, dT,\ \ell_\sigma = \frac{j}{\varepsilon} = r^2\frac{d\varphi}{d\sigma}. \quad (117)$$

σ is a computational parameter. It is neither the reference clock T, the held clock τ_{h} nor the probe clock τ. Since ε is constant, it is a constant rescaling of the probe's proper time along this orbit.

Divide the capacity equation by A^2 and use $\beta = A/\varepsilon$:

$$\frac{1}{\varepsilon^2} = \frac{1}{A} - \frac{1}{c^2}\left[\zeta^2\left(\frac{dr}{d\sigma}\right)^2 + r^2\left(\frac{d\varphi}{d\sigma}\right)^2\right]. \quad (118)$$

Now use $s_\star^2 = c^2(1/A - 1)$ and rearrange:

$$\frac{1}{2}\left[\zeta^2\left(\frac{dr}{d\sigma}\right)^2 + r^2\left(\frac{d\varphi}{d\sigma}\right)^2\right] - \frac{s_\star^2}{2} = E_\sigma, \quad (119)$$

$$E_\sigma = \frac{c^2}{2}\left(1 - \varepsilon^{-2}\right),\ \Phi_\sigma = -\frac{s_\star^2}{2}. \quad (120)$$

E_σ is an auxiliary conserved energy per mass; physical reference energy remains $E = m_p c^2 \varepsilon$. The original radial coefficient is still present. In the selected spherical field, the effective potential is

$$\Phi_\sigma(r) = -\frac{GM}{r - R_s}. \quad (121)$$

We have obtained this potential variable by transforming the retained action's conserved quantities. To establish full equivalence, the next step verifies its motion equation, not only its energy integral.

5.2 Equivalence of the motion equations

For compact notation on this page, write $q = \zeta^2$, $r_\tau = dr/d\tau$ and $r_{\tau\tau} = d^2r/d\tau^2$; a subscript r denotes a field derivative. Variation of L_p gives

$$\frac{d}{dT}\left(\frac{q\dot{r}}{\beta}\right) = -\frac{c^2 A_r}{2\beta} + \frac{q_r \dot{r}^2}{2\beta} + \frac{r\dot{\varphi}^2}{\beta}. \quad (122)$$

Use $d/dT = \beta\, d/d\tau$ and $\dot{r} = \beta r_\tau$. Divide by β, expand the derivative on the left, and move one half of the q_r term to that side:

$$q r_{\tau\tau} + \frac{q_r}{2} r_\tau^2 - r\varphi_\tau^2 = -\frac{c^2 A_r}{2\beta^2} = -\frac{c^2\varepsilon^2 A_r}{2A^2}. \quad (123)$$

Since $d\sigma = \varepsilon\, d\tau$ and ε is constant, each quadratic velocity or second derivative contributes the same factor ε^2. After division,

$$q\,r_{\sigma\sigma}+\frac{q_r}{2}r_\sigma^2-r\,\varphi_\sigma^2=-\frac{c^2A_r}{2A^2}. \quad (124)$$

Differentiate the normalized-burden identity:

$$\frac{1}{2}\frac{ds_\star^2}{dr}=\frac{c^2}{2}\frac{d}{dr}\left(\frac{1}{A}-1\right)=-\frac{c^2A_r}{2A^2}. \quad (125)$$

Equation (91) identifies the relation to the physical work-current, before or after selecting $\mathscr{H}$:

$$\frac{1}{2}\frac{ds_\star^2}{dr}=-\frac{\sqrt{\mathscr{H}}}{\alpha^4}\mathscr{J}. \quad (126)$$

The right side of the auxiliary motion equation is therefore not the physical work-current itself. The conversion to σ and the retained radial kinetic coefficient q belong to the same transformed equation. Multiplying its right side by an additional α^4 would change the orbit rather than complete the normalization.

Equations (124) and (125), together with $r^2\varphi_\sigma=\ell_\sigma$, are the equations of the auxiliary Lagrangian

$$\frac{L_\sigma}{m_p}=\frac{1}{2}\left(q\,r_\sigma^2+r^2\varphi_\sigma^2\right)+\frac{s_\star^2}{2}. \quad (127)$$

This establishes equivalence of both energy and motion for each orbit. We transformed the original Euler–Lagrange equation before identifying L_σ; a trajectory-dependent substitution of time directly into an action need not preserve its variational problem.

Normalized burden therefore has an exact role as the potential variable for these parameterized trajectories. Its radial derivative and the radial motion coefficient must be used together. Adding a second enhancement to the force because $Q_\star>Q_0$ would count the same clock normalization twice.

5.3 The exact orbit equation

Use the selected response $q=1/A$ in equation (119). Put $w=1/r$ and let a prime on w denote $d/d\varphi$. Since $r_\sigma=-\ell_\sigma w'$, the energy equation becomes

$$\frac{\ell_\sigma^2}{A}(w')^2+\ell_\sigma^2w^2=s_\star^2+c^2\left(1-\varepsilon^{-2}\right). \quad (128)$$

Multiply by A and use $A\,s_\star^2=c^2(1-A)$:

$$\ell_\sigma^2(w')^2+A\,\ell_\sigma^2w^2=c^2\left(1-A\,\varepsilon^{-2}\right). \quad (129)$$

Finally multiply by ε^2, recalling $j=\varepsilon\,\ell_\sigma$:

$$j^2(w')^2=c^2\varepsilon^2-A\left(c^2+j^2w^2\right). \quad (130)$$

This is also obtained directly from the original capacity action using $dT/d\tau=\varepsilon/A$ and $dr/d\tau=-j\,w'$. The normalized-burden and original-clock routes therefore agree before any weak-field expansion.

Substitute the solved $A=1-2GMw/c^2$ and expand the product:

$$j^2(w')^2=c^2\left(\varepsilon^2-1\right)+2GMw-j^2w^2+\frac{2GM\,j^2w^3}{c^2}. \quad (131)$$

Differentiate with respect to φ:

$$2\,j^2 w' w'' = 2\,GM\,w' - 2\,j^2 w\,w' + \frac{6\,GM\,j^2 w^2 w'}{c^2}. \quad (132)$$

Where $w' \neq 0$, divide by $2\,j^2 w'$. The smooth second-order equation extends to isolated turning points by continuity:

$$w'' + w = \frac{GM}{j^2} + \frac{3\,GM\,w^2}{c^2}. \quad (133)$$

The coefficient 3 comes from differentiating the cubic term generated by the solved response and the angular-motion term. It has not been adjusted to a measured advance. Circular motion follows directly from the radial Euler–Lagrange equation and does not require division by w'.

5.4 The leading perihelion advance

Define $\ell_g = GM / c^2$ and $p_0 = j^2 / (GM)$. The unperturbed solution is

$$w_0 = \frac{1 + e\cos\varphi}{p_0},\ w_0'' + w_0 = \frac{1}{p_0}. \quad (134)$$

Write $w = w_0 + \delta w$ and retain first order in ℓ_g / p_0. In the correction term, replace w by w_0:

$$\delta w'' + \delta w = 3\,\ell_g w_0^2. \quad (135)$$

$$3\,\ell_g w_0^2 = \frac{3\,\ell_g}{p_0^2}\left(1 + \frac{e^2}{2} + 2\,e\cos\varphi + \frac{e^2}{2}\cos 2\varphi\right). \quad (136)$$

The constant and double-frequency terms give bounded changes. The term proportional to $\cos\varphi$ is resonant with the unperturbed radial oscillation. Use

$$\left(\frac{d^2}{d\varphi^2} + 1\right)(\varphi\sin\varphi) = 2\cos\varphi, \quad (137)$$

to obtain a particular correction:

$$\delta w = \frac{3\,\ell_g}{p_0^2}\left(1 + \frac{e^2}{2} + e\varphi\sin\varphi - \frac{e^2}{6}\cos 2\varphi\right). \quad (138)$$

The homogeneous terms shift the fitted eccentricity and initial phase. To read the accumulated advance, compare the resonant term with

$$\frac{e}{p_0}\cos[(1-\delta)\varphi] \simeq \frac{e}{p_0}\cos\varphi + \frac{e\delta}{p_0}\varphi\sin\varphi. \quad (139)$$

$$\delta = \frac{3\,\ell_g}{p_0},\ \Delta\varphi = \frac{2\pi}{1-\delta} - 2\pi \simeq \frac{6\pi\,\ell_g}{p_0}. \quad (140)$$

To this order $p_0 = a(1-e^2)$, giving the advance per radial period:

$$\Delta\varphi = \frac{6\pi GM}{a(1-e^2)c^2} + O\left(\frac{G^2M^2}{a^2(1-e^2)^2c^4}\right). \quad (141)$$

The final formula assumes a weak-field bound orbit. Equation (133) itself is exact within the selected stationary spherical test-body model.

5.5 Exact orbit verification

For an exact check, specify the turning radii by $r_p = p/(1+e)$ and $r_a = p/(1-e)$, with $P = p/\ell_g$. Setting equation (130) to zero at each turning point and subtracting the two equations gives

$$\frac{j^2}{GM\ell_g} = \frac{P^2}{P-3-e^2}, \varepsilon^2 = \frac{(P-2)^2 - 4e^2}{P(P-3-e^2)}. \quad (142)$$

Here p is defined by the exact turning radii. It differs from $p_0 = j^2/(GM)$ beyond leading order. Substitute $w = (1+e\cos\xi)/p$ into equation (131):

$$(w')^2 = \frac{e^2\sin^2\xi}{p^2}\left[1 - \frac{2(3+e\cos\xi)}{P}\right]. \quad (143)$$

For $P > 6+2e$, cancellation against $(dw/d\xi)^2$ gives the total-angle integral and advance:

$$\Delta\varphi = \int_0^{2\pi}\left[\left(1 - \frac{2(3+e\cos\xi)}{P}\right)^{-1/2} - 1\right]d\xi. \quad (144)$$

Independent numerical quadrature and direct integration of equation (133) agree. The values below use $e = 0.2$ and express p in units of GM/c^2. They are checks of the equations, not additional astronomical observations.

Table 2. Exact orbit checks at eccentricity 0.2

P	Advance from quadrature rad	Advance from orbit integration rad
100	0.1974475160	0.1974475160
1000	0.0189349966	0.0189349966
10000	0.0018858061	0.0018858061

The same calculation with the alternative radial response is carried through in Appendix B. The normalized energy and motion transformation was also checked with exact rational arithmetic before the numerical integrations.

The inequality specifies the stable bound-orbit domain of this expression. The e = 0 case is the small radial-oscillation limit of a circular orbit. Equation (141) is the first-order expansion of equation (144), using the appropriate weak-field orbital elements.

5.6 The Mercury calculation

Use the solar gravitational parameter and constants from NASA JPL [12], together with JPL's approximate J2000 Mercury elements [9]:

$$G M_{\odot} = 1.3271244004127942 \times 10^{20} \ \mathrm{m^3/s^2}, \quad (145)$$

$$c = 299792458 \ \mathrm{m/s}, 1 \ \mathrm{au} = 149597870700 \ \mathrm{m}, \quad (146)$$

$$a = 0.38709927 \ \mathrm{au}, e = 0.20563593. \quad (147)$$

These give the orbital scale and expansion parameter

$$p = a(1 - e^2) = 5.546046913 \times 10^{10} \ \mathrm{m}, \quad (148)$$

$$\frac{G M_{\odot}}{p c^2} = 2.662482056 \times 10^{-8}. \quad (149)$$

Equation (141) then gives

$$\Delta\varphi_{\mathrm{orbit}} = 5.018660440 \times 10^{-7} \ \mathrm{rad} = 0.1035173023 \ \mathrm{arcsec}. \quad (150)$$

With 36525 days per Julian century [12] and the listed sidereal period 87.969 days [13],

$$\Delta\varphi_{\mathrm{century}} = 0.1035173023 \frac{36525}{87.969} \ \mathrm{arcsec} = 42.98070 \ \mathrm{arcsec}. \quad (151)$$

This is the standard approximately 43 arcseconds per century spherical relativistic contribution [14]. The source and response coefficients were fixed before this numerical substitution. The approximate elements are benchmark inputs, not a new fit to Mercury's observations; their precision does not support interpreting all displayed digits as empirical accuracy.

The calculation neglects the other planets, source rotation and multipoles, and test-body backreaction. Recovering the spherical contribution does not constitute a calculation of Mercury's complete observed precession. Appendix C uses the same inputs to evaluate the normalized readout during the finite time spent near perihelion.

The earlier calculation retained with Draft 48 uses the slightly different period 87.96946593 days. It reports 42.98048057 arcseconds per Julian century from exact quadrature. The main benchmark above uses the listed 87.969-day convention. Both give the same approximately 42.98 arcsecond spherical contribution; the period convention and the exact-versus-leading comparison are retained with the numerical checks in Appendix I.

6 Strong fields and rotating black hole exteriors

The capacity calculation can also be carried into a stationary strong field. I summarize the source-cap and rotating construction here because the same framework should be followed into demanding calculations, not restricted to weak fields. The full derivation and photon calculation are given in the companion black-hole paper [29].

6.1 The source cap and rotation

$$r_s = \frac{2GM}{c^2}, Q_H(0) = 4\pi r_s^2 c = \frac{16\pi G^2 M^2}{c^3}. \quad (152)$$

The horizon screen reading is finite. Its held-clock normalization QH/α diverges as the stationary held-clock limit is approached; those are different readings. Two masses can both have S = c at their respective caps while establishing different exterior profiles, $S(r_s + d) = c\sqrt{[r_s/(r_s + d)]}$.

$$S^2 + \mathcal{C}_{\varphi}(\Omega - \omega)^2 + c^2\beta^2 = c^2. \quad (153)$$

At fixed radial and polar position, Ω is the probe angular rate, ω is the local frame rate and $\mathcal{C}$ φ is the squared circumferential lever. Rotation therefore enters through the motion term. To connect that algebra to a source state, let $AH = 4\pi RA^2$ and $mA = RAc^2/(2G)$, and retain the source-energy relation below. It is the uncharged mass-area relation in conventional black-hole variables [23].

$$M^2c^4 = m_A^2c^4 + \frac{c^2J^2}{R_A^2}, \mu = \frac{GM}{c^2}, a_J = \frac{J}{Mc}. \quad (154)$$

$$r_H^2 + a_J^2 = R_A^2 = 2\mu r_H, r_H = \mu + \sqrt{\mu^2 - a_J^2}. \quad (155)$$

$$Q_H(M, J) = \frac{8\pi G^2 M^2}{c^3}\left(1 + \sqrt{1 - \frac{c^2J^2}{G^2M^4}}\right). \quad (156)$$

For $|cJ/(GM^2)| \le 1$, this returns the nonrotating value at J = 0 and approaches half that value in the extremal limit at fixed total mass. These are different source states with different horizon areas.

6.2 From the comoving source screen to motion

The retained exterior uses confocal oblate reference screens, their local comoving area, and the reciprocal normal propagation response. With $F = r^2 + aJ^2$, their reference volume is $V_0 = 4\pi rF/3$ and their comoving area is $Acom = 4\pi F$. The source law then gives

$$Q_{ref}^2 = 24\pi GM V_0, U^2 = \left(\frac{Q_{ref}}{A_{com}}\right)^2 = \frac{2GMr}{F}. \quad (157)$$

Here U is the comoving screen burden. The external normal-frame burden S is obtained when the completed interval is reorganized into its angular square; the two readings coincide at the cap and in the nonrotating limit. Define $\Sigma = r^2 + aJ^2\cos^2\theta$ and $\Delta = F - 2\mu r$. The source and normal-response correspondences give

$$c^2 d\tau^2 = \frac{\Delta}{\Sigma}\left(c\,dT - a_J \sin^2\theta\, d\varphi\right)^2 - \frac{\Sigma}{\Delta} dr^2 - \Sigma\, d\theta^2$$

$$- \frac{\sin^2\theta}{\Sigma}\left(F\, d\varphi - a_J\, c\, dT\right)^2 . \qquad (158)$$

The two displayed lines are one interval. Expanding it gives the Kerr exterior [24]. The source-energy identification, comoving screen correspondence and normal response specify this realization; the known rotating target informed its construction. Applying the same calibrated probe action gives the equatorial circular rates

$$\Omega_{\pm} = \frac{\pm c\sqrt{\mu / r^3}}{1 \pm a_J \sqrt{\mu / r^3}} . \qquad (159)$$

The nonrotating limit returns the spherical orbit. The full calculation also derives radial stability and photon capture from this interval. In the far exterior, ω approaches 2GJ/(c²r³); the corresponding tangential frame speed has one less inverse power of radius because its lever grows with r.

6.3 An observed strong field scale

The companion photon calculation scans 54 spins from a* = cJ/(GM²) = 0 to 0.9999 and 19 viewing inclinations from 0 to 90 degrees. Its sampled shadow diameter is 9.6571–10.3923 times GM/c². Using the stellar-orbit angular scales quoted by the EHT analysis gives 49.49–53.26 microarcseconds for VLTI and 47.51–51.13 for Keck. These are parameter ranges, with source-scale errors treated separately [25].

The reference eht-imaging shadow inference is 48.9 microarcseconds, with a 68 percent interval of 43.8–54.1. It contains both calculated ranges. This comparison uses the calibrated shadow inference, not the distinct 51.8 ± 2.3 microarcsecond emission-ring measurement. The stellar-orbit scales assume GR, and the shadow inference includes emission calibration. The result checks the strong-field scale of the specified Kerr-equivalent SET construction [25, 27].

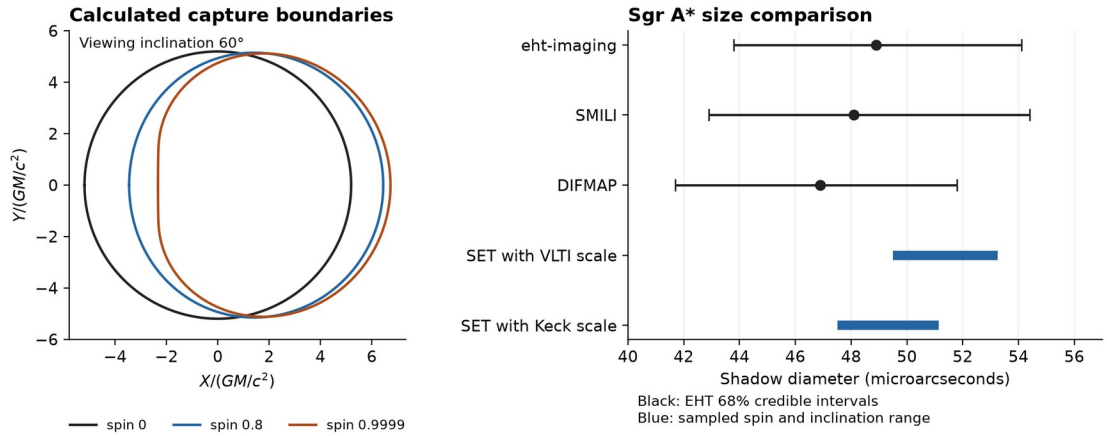


Figure 3. Left: null capture contours calculated from the completed SET exterior at a common inclination. Right: EHT shadow inferences using the GRMHD calibration [25] and our sampled parameter ranges at the central stellar-orbit scales. Blue segments are parameter ranges; black bars are 68 percent credible intervals. The source-scale errors are listed in the companion black-hole paper. No emission brightness model is calculated here.

For a concrete source reading, the quoted VLTI mass 4.297 × 10⁶ M⊙ gives QH(0) = 6.067 × 10²⁹ m³/s. At a* = 0.8 the rotating expression gives 4.853 × 10²⁹ m³/s. These follow from the source mass and chosen spin; the observational comparison is the shadow size calculated through the propagation response [25, 28].

6.4 A source and tidal reading of the spherical field

The selected spherical solution supplies a useful distinction between burden and its spatial variation. The burden S sets the local occupancy; neighboring freely falling probes respond to its variation. The Kretschmann contraction $K = R_{abcd} R^{abcd}$ packages the resulting interval derivatives into a coordinate-independent quantity. In SET it can be read as a diagnostic of the established field response. Its geometric definition remains unchanged by that physical interpretation.

For the already selected reciprocal exterior $A = 1/B$, direct differentiation of the interval gives

$$\begin{aligned} K &= (A'')^2 + 4\left(\frac{A'}{r}\right)^2 + 4\left(\frac{1-A}{r^2}\right)^2, \\ K &= \frac{1}{c^4}\left\{[(S^2)'']^2 + 4\left[\frac{(S^2)'}{r}\right]^2 + 4\left(\frac{S^2}{r^2}\right)^2\right\}. \end{aligned} \qquad (160)$$

Substitution of $S^2 = 2\mu_Q / r$ then gives three equivalent readings:

$$K = \frac{48\mu_Q^2}{c^4 r^6} = \frac{12 S^4}{c^4 r^4} = \frac{48 \mathcal{J}_W^2}{c^4 r^2}, \mathcal{J}_W = \frac{\mu_Q}{r^2}. \qquad (161)$$

Keeping the actual spherical source radius R_0 separate from the exterior reading radius r, the source law gives

$$\begin{aligned} Q_\Omega^2 &= 32\pi^2 GM R_0^3, S_0 = \frac{Q_\Omega}{4\pi R_0^2}, \\ K(r; R_0) &= \frac{3 Q_\Omega^4}{64\pi^4 c^4 R_0^6 r^6} = \frac{12 S_0^4 R_0^2}{c^4 r^6}. \end{aligned} \qquad (162)$$

The two radii now have explicit roles: one defines the source reading, the other locates the measurement. Sources with the same complete mass can have different surface radii and different Q_Ω, while the combination Q_Ω^2 / R_0^3 and their exterior K agree. In this selected exterior the value of K and the reading radius recover the same total source energy; they do not locate material at the vacuum measurement point.

$$E_{complete} = \frac{c^4 r^3}{G}\sqrt{\frac{K}{48}}. \qquad (163)$$

A static orthonormal frame, or its radial free-fall boost, gives the radial and tangential relative-acceleration coefficients

$$\Lambda_r = \frac{2\mu_Q}{r^3}, \Lambda_t = -\frac{\mu_Q}{r^3}, K = \frac{8}{c^4}(\Lambda_r^2 + 2\Lambda_t^2). \qquad (164)$$

At the nonrotating cap $r_h = 2\mu_Q / c^2$, these expressions have finite exterior limits:

$$S(r_h) = c, K(r_h) = \frac{12}{r_h^4}, \Lambda_r(r_h) = \frac{c^2}{r_h^2}. \qquad (165)$$

A ten-times-larger horizon has the same saturated burden but one ten-thousandth of this invariant and one hundredth of the radial tidal coefficient. The common capacity limit therefore does not imply the same tidal

environment. The result concerns the exterior or freely falling limit; it does not require a massive held clock at the cap. Appendix L supplies the general radial calculation and its energy reading.

6.5 The stationary rotational contribution to the work current

The rotating construction also provides a direct connection to the field equations. Use $x^0 = cT$ and write its interval with normal lapse N, spatial ruler q_{ij} and shift v^i. For a stationary slice, define $K_{ij} = (D_i v_j + D_j v_i)/(2N)$, with units of inverse length. The normal lapse is distinct from the clock of an observer held at fixed angular coordinate.

$$\Sigma = r^2 + a_J^2 \cos^2\theta \,, \Delta = r^2 - 2br + a_J^2 \,, b = GM/c^2 \,,$$
$$A_K = (r^2 + a_J^2)^2 - a_J^2 \Delta \sin^2\theta \,, N^2 = \frac{\Delta\Sigma}{A_K} \,, \qquad (166)$$
$$q_{ij} = \mathrm{diag}\left(\frac{\Sigma}{\Delta}, \Sigma, \frac{A_K \sin^2\theta}{\Sigma}\right), v^\varphi = -\frac{2br\, a_J}{A_K} .$$

Only $K_{r\varphi}$ and $K_{\theta\varphi}$ and their symmetric partners remain; their diagonal trace vanishes. Direct differentiation verifies the stationary lapse identity

$$\Delta_q N = N K_{ij} K^{ij} \,, D_q \cdot \mathscr{J}_N = c^2 N K_{ij} K^{ij} \,, \mathscr{J}_N = c^2 D_q N \,. \qquad (167)$$

Rotation consequently contributes to the normal-lapse current even in material vacuum. It is absent in the nonrotating limit. This verifies the lapse-sector connection of the retained rotating interval to the common field construction; the source-area and comoving-screen correspondences used to construct that interval remain those stated in Section 6.2.

6.6 Causal field changes and conserved radiation

Changes of the field are physical events, so the same operational clocks and distances must enter their evolution. The retained local field realization makes that requirement calculable. Around an unloaded background, it contains constrained clock and transport variables and two transverse, trace-free directional ruler modes γ_{ij}. Its quadratic terms use first derivatives, the calibrated matter coupling and one second-order tensor propagator. Complete source changes include energy, momentum and the driving or support stresses. Retarded boundary data describe an isolated outgoing disturbance. The detailed causal construction is retained in the companion milestone [35].

Within this class, requiring the complete local tidal response to be causal cancels the inverse-spatial-Laplacian poles among the constrained and propagating sectors. The scalar rate coefficient is -3 and the vector and tensor coefficients are both 8 in the normalization $\kappa_f = c^4/(2\pi G)$. The surviving tensor signal travels at c. The field architecture is the stated completion; these coefficient relations follow inside it. Appendix M displays the completion and conservation steps.

$$L_{TT} = \frac{\kappa_f}{32}\left[\frac{\dot\gamma_{ij}\dot\gamma_{ij}}{c^2} - (\partial_k \gamma_{ij})(\partial_k \gamma_{ij})\right] + \frac{1}{2}\sigma_{ij}\gamma_{ij} \,, \qquad (168)$$
$$\left(\frac{\partial_T^2}{c^2} - \Delta\right)\gamma_{ij} = \frac{8}{\kappa_f}\sigma_{ij}^{TT} .$$

Dots here denote reference-time derivatives. The two free tensor modes have positive canonical energy. Their outgoing flux is

$$F_r = \frac{c^3}{32\pi G}\langle \dot\gamma_{ij}\dot\gamma_{ij}\rangle . \quad (169)$$

For a compact, slowly moving complete source, energy and momentum conservation relate its integrated stress to the second derivative of the energy quadrupole. The retarded wave and angular flux integral then give

$$\begin{aligned} I_{ij} &= \frac{1}{c^2}\int \varepsilon\, x_i x_j d^3x\,, \ddot I_{ij} = 2\int \sigma_{ij} d^3x\,, \\ \gamma_{ij}^{rad} &= \frac{2G}{c^4 R}\Lambda_{ij,kl}\ddot I_{kl}(T - R/c), \\ P &= \frac{G}{5c^5}\langle \dddot I_{ij}^{STF}\dddot I_{ij}^{STF}\rangle . \end{aligned} \quad (170)$$

Here $\Lambda_{ij,kl}$ projects onto transverse trace-free components in the observation direction, and STF removes the spatial trace. This is a radiation-zone calculation. The changing near-field multipole and the outgoing wave have different radial dependences. For a weak circular binary with separation d, the retained weak-field orbit has $\Omega^2 = G(m_1+m_2)/d^3$. Its quadrupole power can then be expressed directly through its two source readings:

$$\begin{aligned} \mu_i &= \frac{Q_i^2}{32\pi^2 R_i^3} = G m_i\,, \\ P &= \frac{32\mu_1^2\mu_2^2(\mu_1+\mu_2)}{5Gc^5 d^5}. \end{aligned} \quad (171)$$

Changing the chosen source reference radius changes Q_i but leaves Q_i^2/R_i^3 and this power unchanged. The tensor-field calculation fixes the power normalization before that algebraic source conversion. It is a conditional derivation of the weak, slow-source radiation law, rather than an additional binary observation fitted here.

6.7 Energy carried by the field and delayed response

The same-clock requirement also applies to an excited field mode. If I is its phase action, the common energy generates its phase rate. At fixed canonical mode state, the local energy enters through the lapse: Here the displayed Hamiltonian is its normal-energy contribution, with the shift transport term separated; it is the complete mode Hamiltonian when that shift vanishes.

$$\dot\vartheta = \frac{\partial H_w}{\partial I} = N\frac{\partial E_{loc}}{\partial I}, H_w = N E_{loc}. \quad (172)$$

Zero excitation fixes the additive constant. Variation with respect to the lapse supplies the mode energy density, while ruler and transport variations supply its stress and momentum. In burden variables, $H_w = E_{loc}\sqrt{1 - S^2/c^2}$, whose weak-field coupling is $-E_{loc}S^2/(2c^2)$. Thus field energy participates in the same source accounting. This leading field-mode construction uses a slowly varying background or the short-wavelength average; the nonlinear constraint system in Appendix M provides the finite-wavelength consistency framework.

A controlled example is the leading weak-field, short-wavelength, angle-averaged backreaction of quadratic radiation energy. It is not a spherical spin-two wave mode. Let a compact emitter lose energy at a prescribed

rate $P(T)$, with no incoming pulse. In this averaged-energy regime [38], the outgoing energy density and energy inside a sphere are:

$$\dot{E}_c(T) = -P(T), u_w(r,T) = \frac{P(T - r/c)}{4\pi c r^2}, \\ E_{enc}(r,T) = E_c(T) + \frac{1}{c}\int_0^r P(T - s/c)\,ds = E_c(T - r/c). \tag{173}$$

The source energy lost locally is present in the traveling field. A distant sphere therefore reads the earlier complete energy until the pulse reaches it. Define the tidal tensor by $\delta \ddot{x}^i = -c^2 E^i_j \delta x^j$; its eigenvalues differ by $-c^2$ from the acceleration coefficients used in Section 6.4. In the weak causal solution the pulse gives

$$E_{rr} = -\frac{2G E_c(T - r/c)}{c^4 r^3}, \\ E_{\theta\theta} = \frac{G E_c(T - r/c)}{c^4 r^3} + \frac{GP(T - r/c)}{c^5 r^2}. \tag{174}$$

The second term records the passing radiation. This is a delayed response of the conserved emitter-plus-field state, not a permanent mass left at the old source location. Static burden, complete energy conservation and the transport of a changing field are distinct readings of this construction. The example supplies a controlled mathematical account of field history without identifying every static source with a steady radiation loss.

7 Effective area and collective source organization

7.1 Boundary loading and its effective area

I am asking whether the source–boundary relation used for an individual body also applies to a collective system of disjoint masses. At Earth's approximately spherical surface, the source normalization is read through the spherical area. For a galaxy, SET retains the same normalization law and proposes that the organization of its baryonic sources enters through an effective boundary area. The physical question is whether the resulting burden, together with SET's motion law, can account for the observed orbital speeds using the baryonic inventory alone.

There is a geometrical distinction between a point among sources and a boundary enclosing them. For a convex enclosing surface, such as a sphere or an oblate spheroid, every source-to-boundary direction has a nonnegative outward-normal component. In the stationary model developed in Appendix F, the contributions from the enclosed sources to the defined source flux therefore add with the same outward sign at that surface, although their strengths and directions can differ. Between sources, directional contributions can oppose while burden remains present. This is the distinction I have in mind when considering a collective system through an enclosing boundary. I now define an effective area to describe its boundary loading.

For a nonnegative scalar loading s on a specified boundary Σ, define:

$$Q_\Sigma = \int_\Sigma s\, dA\,, I_\Sigma = \int_\Sigma s^2\, dA\,. \quad (175)$$

$$A_{eff} = \frac{Q_\Sigma^2}{I_\Sigma}\,, S_{eff} = \frac{I_\Sigma}{Q_\Sigma} = \frac{Q_\Sigma}{A_{eff}}. \quad (176)$$

The first moment QΣ has units m³/s; the second moment IΣ has units m⁴/s². If s²/c² is the local occupancy under the SET identification, IΣ/(Ageom c²) is the area-averaged occupancy. Cauchy–Schwarz gives QΣ² ≤ Ageom IΣ and therefore:

$$A_{eff} \le A_{geom}\,, \frac{A_{geom}}{A_{eff}} = 1 + \frac{Var(s)}{\langle s \rangle^2}. \quad (177)$$

Take QΣ > 0. Uniform loading gives Aeff = Ageom. To see what concentration changes, use four equal unit-area patches. The loadings (1,1,1,1), (2,2,0,0) and (4,0,0,0) all carry QΣ = 4. Their second moments are 4, 8 and 16, so Aeff is 4, 2 and 1. The same total loading occupies a smaller effective area as it becomes more concentrated. Conversely, multiplying all loadings by the same factor leaves Aeff unchanged. The area measures concentration; the normalization fixes the amplitude.

Two surface integrals now appear. QΣ = ∫Σ s dA has units m³/s. Appendix F.1's gradient flux ∮Σ $\mathcal{J}$·dA = 4πGM has units m³/s². Their conservation properties differ. The gradient-flux law leaves the boundary loading normalization undetermined: Appendix F shows that rescaling Ψ and inversely rescaling C preserves $\mathcal{J}$ while changing the force. Section 7.3 supplies the QΣ normalization used for the collective calculation.

If source amplitudes add at one location, their square contains overlap terms. The boundary identity itself leaves that physical superposition rule open. Also, Seff averages the boundary loading with weight s: it equals the local field S on a uniformly loaded boundary, but generally summarizes a distribution. Applying it to probe motion requires a relation to the local S and its gradient.

An effective boundary must describe the physical response, rather than create a response merely because we chose the boundary. Rearranging the sources can change the field distribution; merely redrawing an enclosing surface cannot change the field at a fixed location. I test a specific prescription: retain the spherical

reference Qref for the adopted mass and radius, and estimate Aeff with an oblate surface constrained by imaging. This connects the proposed collective response to observed source geometry. The collective model must determine the relevant boundary or give results independent of an arbitrary boundary choice.

7.2 Spherical reference and image geometry

For a spherical evaluation screen at R, introduce its reference volume Vref = (4/3)πR³ and write Qref = $Q_0(R)$. Equation (63) gives Qref²/Vref = 24πGM. Expanding both quantities makes the normalization explicit:

$$\frac{[4\pi\sqrt{2GMR^3}]^2}{(4/3)\pi R^3} = 24\pi GM. \quad (178)$$

Divide both sides by R:

$$\frac{Q_{ref}^2}{V_{ref}R} = \frac{24\pi GM}{R} = 12\pi S^2(R). \quad (179)$$

The geometric step is 12πVref R = 16π²R⁴ = (4πR²)². Therefore:

$$S^2(R) = \frac{Q_{ref}^2}{12\pi V_{ref}R} = \left[\frac{Q_{ref}}{4\pi R^2}\right]^2. \quad (180)$$

For positive Qref and S, taking the positive square root of equation (180) gives 4πR²S = Qref. Solving for R, or using S² = 2GM/R, gives the equivalent forms:

$$R = \frac{Q_{ref}^2}{12\pi V_{ref}S^2} = \sqrt{\frac{Q_{ref}}{4\pi S}} = \frac{2GM}{S^2}. \quad (181)$$

The same spherical relation can therefore be written in terms of radius, amplitude or source normalization. In the first form, Vref still depends on R; substituting Vref = (4/3)πR³ gives the square-root form. Qref²/Vref has units m^3/s^2, and division by R gives m^2/s^2, the units of S^2 and energy per unit mass. Since Qref was calculated using R, rearranging it does not independently determine a body's size or internal density. For galaxies I retain this spherical reference volume while changing the boundary area.

I denote this particular solution by Ssph(R) = √(2GM/R). S itself is the burden determined by the source arrangement and response. In Section 7.3 I retain the reference Q while allowing the effective area, and therefore the boundary amplitude, to change.

For an oblate spheroid with equatorial radius R, polar radius ηR, and eccentricity eη = √(1 − η²), the geometric surface is:

$$A_{geom} = f\pi R^2, f = 2\left[1 + \frac{\eta^2 \operatorname{atanh}(e_\eta)}{e_\eta}\right]. \quad (182)$$

The limits are f → 2 for a thin oblate surface and f = 4 for a sphere. A projected minor-to-major ratio pimage at inclination i gives, under the axisymmetric oblate assumption:

$$\eta^2 = \frac{p_{image}^2 - \cos^2 i}{\sin^2 i}. \quad (183)$$

This reconstructs an intrinsic oblate shape from a projected image and an adopted inclination. At i = 0 the image cannot determine η; near that orientation the inference becomes sensitive to measurement error. Bars, warps, triaxiality and varying inclination can also affect it. I flag a negative central η^2 as unresolved deprojection rather than clipping it to a thin disk. The range $2 \le f \le 4$ applies to these geometric oblate surfaces; the participation area in equation (176) can also reflect nonuniform loading.

7.3 The weak field boundary calculation

For the collective boundary calculation, take the adopted baryonic mass Mbar and comparison radius R. I retain Vref = (4/3)πR³ as the source reference volume and identify the boundary loading with Qref = Aeff Seff. The source normalization and effective area then determine the boundary amplitude together. The reference quantity and image-based reading area are:

$$Q_{ref} = 4\pi\sqrt{2 G M_{\mathrm{bar}} R^3},\ A_{image} = f_{image}\pi R^2. \quad (184)$$

With QΣ = Qref, combine $Q_{ref}^2/V_{ref} = 24\pi G M_{bar}$ with Qref = Aeff Seff:

$$\frac{A_{eff}^2 S_{eff}^2}{V_{ref}} = 24\pi G M_{\mathrm{bar}}. \quad (185)$$

Solving directly for the positive boundary amplitude gives:

Seff = √(24πGMbarVref)/Aeff.

For Aeff = fπR², the effective boundary amplitude is therefore:

$$S_{eff}^2 = \frac{32 G M_{\mathrm{bar}}}{f^2 R}. \quad (186)$$

The spherical case f = 4 returns Ssph = √(2GMbar/R). The thin-area limit f = 2 gives Seff = √(8GMbar/R). Both carry the same reference Q:

$$\begin{aligned} Q_{ref} &= \sqrt{32\pi^2 G M_{\mathrm{bar}} R^3} = 4\pi\sqrt{2 G M_{\mathrm{bar}} R^3} \\ &= f\pi\sqrt{\frac{32}{f^2} G M_{\mathrm{bar}} R^3} = (f\pi R^2)\sqrt{\frac{32 G M_{\mathrm{bar}}}{f^2 R}} = A_{eff} S_{eff}. \end{aligned} \quad (187)$$

$$4\pi R^2\sqrt{\frac{2 G M_{\mathrm{bar}}}{R}} = 2\pi R^2\sqrt{\frac{8 G M_{\mathrm{bar}}}{R}} = Q_{ref}. \quad (188)$$

These are equivalent expressions for the same reference Q at fixed baryonic mass, reference radius and reference volume. In this collective prescription, the effective area and boundary burden change reciprocally through Qref = Aeff Seff. The spherical expression √(2GM/R) is the f = 4 case; it is not held fixed when the effective area changes. Where Seff represents the local S, the energy relation in Section 4.4 gives the corresponding marginal escape speed. This is the physical interpretation proposed by SET: source organization changes the boundary burden, and that burden determines the escape-energy scale. Moving the observation radius through one field is a different operation, for which $Q_0(r)$ follows the field solution. For the observational calculation I approximate Aeff by Aimage and call the resulting amplitude Sboundary:

$$S_{boundary} = \frac{Q_{ref}}{A_{image}},\ v_{SET} = \frac{S_{boundary}}{\sqrt{2}}. \quad (189)$$

Combining the expressions makes the prediction explicit:

$$v_{SET} = \frac{4}{f_{image}} \sqrt{\frac{G M_{\text{bar}}}{R}}. \quad (190)$$

To connect the boundary reading to orbital motion, distinguish the radius Rb of the reference boundary from the equatorial orbital radius R. Write Sb = Sboundary(Rb). For a conditional fixed-mass, fixed-shape inverse-radius exterior, the continuation is

$$S^2(R) = S_b^2 \frac{R_b}{R}, R \geq R_b. \quad (191)$$

If d is distance outward from that boundary, R = Rb + d and

$$S^2(R_b + d) = S_b^2 \frac{R_b}{R_b + d}. \quad (192)$$

$$\frac{dS^2}{dd} = -\frac{S_b^2 R_b}{(R_b + d)^2}. \quad (193)$$

Measuring distance from the boundary does not remove the source's radius from the denominator. The motion law uses the orbital radius R, giving

$$v_{orb}^2(R) = \frac{S_b^2 R_b}{2R}. \quad (194)$$

At R = Rb this is vorb = Sb/√2. Substituting Sb² = 32GMbar/(f²Rb) gives vorb²(R) = 16GMbar/(f²R): the reference boundary radius cancels. Moving a reference screen through this same fixed-shape exterior therefore leaves the field at a fixed observation point unchanged. This continuation is conditional on the stated exterior profile; selecting an image radius does not impose it on the physical field.

The reference volume used to define Qref is still (4/3)πRb³. The actual oblate enclosed volume is η times that value. Replacing the reference volume by the actual oblate volume in the source formula would change Q by √η and define a different prescription. Equations (184)–(185) state which normalization is being tested.

For an arbitrary specified source distribution, I can check the approximation against the solved local field. Define vB(R) = (4/f)√(GMbar/R), SB(R) = √2vB(R), and b_s(R) = Slocal(R,0)/SB(R). With k defined in equation (107), the motion law gives

$$\frac{v_{field}}{v_B} = b_s \sqrt{k}. \quad (195)$$

The boundary estimate's fractional difference from that field prediction is therefore $|1/(b_s\sqrt{k}) - 1|$, wherever the radial force is inward. These are diagnostics of the field solution, not adjustable response factors. The stated fixed-mass, fixed-shape inverse-radius continuation has $b_s = k = 1$. For a more general source, the same calculation tests whether the boundary estimate is accurate enough for its stated use. A prospective application should specify its approximation tolerance before inspecting observed-speed agreement, then assess it over a stated radial neighborhood and allowed source reconstructions. Identifying SB with the local escape burden additionally requires b_s to be close to one; amplitude and slope deviations can compensate in the speed.

The galaxy protocol tests the identification of the image area with Aeff and the use of Sboundary in this orbital conversion. The second-moment identities quantify a known loading; they do not establish that identification from an image. Equation (190) states the particular prescription used for the comparisons.

7.4 Inverse readings of the same relation

To express the same boundary normalization in orbital variables, use f = fimage and Qref = fπR²Sboundary. Dividing the squared relation by Vref R gives:

$$\frac{Q_{ref}^2}{V_{ref} R} = \frac{3\pi f^2}{4} S_{boundary}^2 . \quad (196)$$

With the orbital calibration Sboundary² = 2vSET², the same expression becomes:

$$\frac{Q_{ref}^2}{V_{ref} R} = \frac{3\pi f^2}{2} v_{SET}^2 = \frac{24\pi G M_{\mathrm{bar}}}{R} . \quad (197)$$

Canceling common factors gives vSET² = 16GMbar/(f²R), consistent with equation (190). Its inverse forms are:

$$R = \frac{16 G M_{\mathrm{bar}}}{f^2 v_{SET}^2} , M_{required} = \frac{f^2 v_{obs}^2 R}{16 G} . \quad (198)$$

For f = 4, the inverse relation is R = GMbar/vSET². At the same mass and radius, f = 2 doubles the predicted speed relative to the spherical reference. Here R is the orbital comparison radius. Its algebraic inversion does not independently locate the source's physical edge: the reference boundary radius cancels from the fixed-shape exterior relation. A measured speed and orbital radius constrain Mbar/f² under this prescription. An independently constrained mass permits inference of f, and an independently constrained f permits inference of the required mass. These are conditional inferences to be checked against the source observations. A Newtonian disk calculation uses the actual spatial mass distribution, as shown in Section 8.4.

7.5 Observational inputs and comparison radii

At vSET = 200 km/s, the prescription gives Sboundary = 282.84 km/s and $S_{boundary}^2/c^2 \approx 8.90\times10^{-7}$. These are weak-field values; relativistic corrections of that scale cannot account for several-percent speed discrepancies. The orbital slope condition of Section 4.6 still applies.

Across a range of radii, constant Mbar and f would give $vSET \propto R^{-1/2}$. A flat circular-speed profile instead requires $S^2 = S_0^2 - 2vflat^2 \ln(R/R_0)$ over that range, where $S_0^2 = S^2(R_0)$. Appendix G joins such a finite region to an inverse-radius exterior. The galaxy table tests the prescription at specified radii; deriving the field across the measured curve is a further calculation.

For the nineteen SPARC catalogue cases in the selected set, use [5]:

$$M_{\mathrm{bar}} = 0.5 L_{disk} + 0.7 L_{bulge} + 1.33 M_{HI} . \quad (199)$$

with luminosities in solar 3.6-micron units, stellar mass-to-light ratios in their corresponding solar units, and the result in solar masses. The factor 1.33 accounts for the adopted helium convention. This inventory omits molecular gas and is not a complete enclosed census at every interior radius. The supplementary NGC 4565 calculation uses the same mass recipe with the separate published photometric decomposition and HI flux [7, 8].

The comparison radius is the outermost actual rotation-curve point bracketed by qualifying image isophotes. The rule does not search radii for the smallest velocity discrepancy. Recalculate Qref at that radius. The image pipeline retains positive intensity, finite formal ellipse errors no larger than 0.1 in axis ratio, and

surface-brightness signal-to-noise at least 3. It interpolates between bracketing valid isophotes and uses the larger bracketing formal shape error. Fine and coarse samplings are recorded separately [6].

This rule locates a usable comparison point. Whether that point admits the collective exterior approximation is a separate question about the source and its field. A galaxy need not have a sharply terminated density profile for the approximation to be useful; its accuracy is what matters.

An inferred area coefficient is useful as a diagnostic:

$$f_{inferred} = \frac{4\sqrt{G\,M_{\mathrm{bar}}/R}}{v_{obs}}. \qquad (200)$$

This inferred coefficient can be compared with the independently reconstructed image coefficient. A value outside $2 \le f \le 4$ cannot describe the geometric oblate surface of equation (182) at that radius. The more general participation area in equation (176) can be smaller under nonuniform loading, but invoking that possibility requires a corresponding field-loading calculation; it is not an image measurement of a more flattened spheroid.

Its agreement with the image coefficient is exactly related to speed agreement:

$$\frac{f_{inferred}}{f_{image}} = \frac{v_{SET}}{v_{obs}}. \qquad (201)$$

8 Galaxy calculations and observed velocities

8.1 The twenty examples

The question in these examples is whether the area estimated from a galaxy’s image corresponds to the boundary amplitude required to account for its observed orbital speed. I present twenty worked applications chosen to illustrate close agreement under the stated prescription. In each calculation, the adopted baryonic mass, comparison radius and image-derived area determine the SET speed; no separate halo mass or concentration is adjusted. The interest is the connection between an observed property of the source arrangement and the gravitational response proposed by SET. The research supplement retains the broader calculation record and controls.

Galaxy	R kpc	Mbar 10^{10} M⊙	f image	f inferred	v observed km/s	v SET km/s	Difference
NGC4051	12.19 E	5.122	3.422	3.515	153.0 ± 9.8	157.1	+2.71%
NGC5371	23.12 I	18.507	3.049	3.092	240.0 ± 3.0	243.4	+1.42%
UGC07690	2.36 I	0.095	2.664	2.827	58.8 ± 4.3	62.4	+6.13%
NGC3949	7.07 E	2.352	2.589	2.831	169.0 ± 25.5	184.8	+9.34%
NGC4013	13.47 I	4.657	2.557	2.637	185.0 ± 6.0	190.7	+3.09%
UGC06818	5.24 I	0.223	2.551	2.403	71.2 ± 5.2	67.1	-5.80%
NGC2683	11.54 I	4.315	2.371	2.536	200.0 ± 5.0	213.9	+6.97%
NGC3198	16.07 I	3.360	2.321	2.479	153.0 ± 2.0	163.4	+6.81%
NGC3992	27.52 I	13.554	2.255	2.205	264.0 ± 5.3	258.2	-2.21%
UGC08837	4.20 E	0.068	2.217	2.193	48.0 ± 3.0	47.5	-1.10%
NGC5585	6.16 I	0.371	2.201	2.313	88.0 ± 3.0	92.5	+5.13%
NGC4010	10.47 E	1.236	2.183	2.337	122.0 ± 3.9	130.6	+7.05%
NGC4565	19.75 E	6.540	2.182	2.104	226.9 ± 22.0	218.8	-3.58%
NGC4183	11.35 I	1.008	2.181	2.248	110.0 ± 5.6	113.3	+3.04%
NGC5907	27.68 I	11.568	2.178	2.342	229.0 ± 2.0	246.2	+7.51%
NGC0100	8.90 I	0.426	2.164	2.040	89.0 ± 7.8	83.9	-5.76%
NGC4100	13.94 I	3.382	2.163	2.245	182.0 ± 4.5	188.9	+3.81%
NGC5033	24.01 I	7.533	2.126	2.093	222.0 ± 2.0	218.5	-1.57%
UGC07089	9.16 E	0.341	2.117	2.023	79.1 ± 7.3	75.6	-4.46%
NGC1090	23.89 I	4.770	2.112	2.247	165.0 ± 2.2	175.5	+6.35%

Table 3. Twenty single-radius comparisons. In the radius column E denotes the endpoint of the adopted rotation dataset and I an interior comparison point. These labels describe the extent of the adopted velocity dataset, not the physical source boundary. Neither an endpoint nor an earlier measured orbit, by itself, establishes the local field's exterior character. Mbar is the adopted global stellar-plus-HI/helium inventory, in 10^{10} M⊙. The observed speeds predominantly trace gas rotation. Velocity errors are the quoted comparison-bin errors, with the separately documented figure-based estimate for NGC 4565. The central mass, radius and image coefficient are not retuned within their uncertainty ranges to obtain these entries. The input ranges, inclinations, projected axis ratios and radius status are recorded in Appendix E.

The objects are distinct disk galaxies represented by oblate surfaces. The calculation uses the outermost comparison radius jointly supported by the adopted kinematic and image data. Fourteen radii lie inside the last rotation point, while six are endpoints of their adopted datasets. UGC 7089 and UGC 8837 have no established catalogue flat speed. Table 3 therefore reports single-radius comparisons, with the radius status stated explicitly. The adopted global inventory should not be read as a verified enclosed baryonic mass at every interior radius.

The central absolute discrepancies range from 1.10% to 9.34%, with a median of 4.79%. The area pairs have Pearson r = 0.957 and the speed pairs r = 0.994. These describe the examples in Table 3. Equation (201) also shows that the area and speed residuals contain the same comparison information.

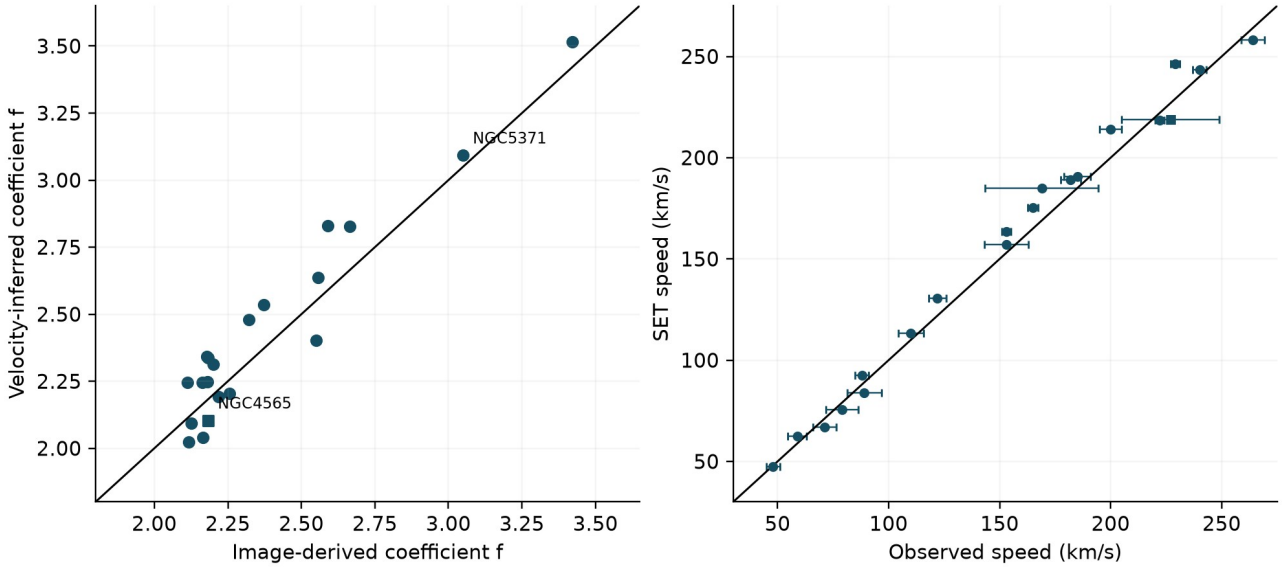


Figure 4. The twenty comparisons, with equality lines. Left: image-derived and velocity-inferred area coefficients. Right: predicted and observed speeds; horizontal bars show the comparison-bin velocity errors only. Squares identify NGC 4565, whose source provenance differs from the catalogue cases. The sample description and shared-input dependence are stated in the text.

8.2 NGC 5371

For NGC 5371, D = 39.7 Mpc gives a comparison radius R = 23.12 kpc. The adopted global baryonic inventory is approximately 1.851×10^{11} M⊙. The image axis ratio is about 0.7822; with i = 53 degrees it gives η ≈ 0.6256 and fimage ≈ 3.049. Using the retained full-precision inputs:

$$Q_{ref} = 1.67823 \times 10^{48}\ \ m^3/s\,, A_{image} = 4.87552 \times 10^{42}\ \ m^2. \quad (202)$$

$$S_{boundary} = 344.215\ \ km/s\,, v_{SET} = 243.397\ \ km/s. \quad (203)$$

The observed speed is 240.0 ± 3.0 km/s. The central difference is +1.42%, with finferred ≈ 3.092. The larger oblate area lowers the thin-area prediction toward this bin. The comparison radius is half the last measured rotation radius of 46.24 kpc; its 240 km/s speed differs from the catalogue flat speed of 209.5 km/s. This is therefore an interior comparison using the stated boundary prescription. The inferred thickness depends on the oblate geometry and adopted inclination.

8.3 NGC 4565

For nearly edge-on NGC 4565, I use D = 9.7 Mpc and the last actual plotted HI point, at 420 arcsec, in the adopted Yim et al. curve [7]. This gives R ≈ 19.75 kpc. The S4G decomposition supplies the stellar

luminosity, and the adopted HI flux is 274 Jy km/s [7, 8]. Equation (199) gives Mbar ≈ 6.540×10[10] M⊙. The image ratio of about 0.2050 and inclination of 86.6 degrees give η ≈ 0.1965 and fimage ≈ 2.182. Keeping full precision in the calculation:

$$Q_{ref} = 7.87756 \times 10^{47} \ \ m^3/s, A_{image} = 2.54625 \times 10^{42} \ \ m^2. \quad (204)$$

$$S_{boundary} = 309.379 \ \ km/s, v_{SET} = 218.764 \ \ km/s. \quad (205)$$

The figure-derived observed speed is approximately 226.9 ± 22.0 km/s, giving −3.58%. The extraction leaves a small ambiguity about whether the plotted value already includes inclination correction; omitting that correction gives −3.41%. This remains an estimate from the published figure. The calculation record retains alternative distances as sensitivity checks, with mass and radius rescaled together.

8.4 The matched Newtonian baryonic comparison

To examine whether the area prescription improves the speed estimate, I compare it with the published Newtonian baryonic mass models at the same radii. These models calculate the radial force of the resolved gas, stellar disk and bulge [5]. They already include disk geometry. Their result is not the spherical reference √(GMbar/R).

The component forces add, giving the signed-quadrature expression:

$$v_N^2 = V_{gas}|V_{gas}| + 0.5\,V_{disk}|V_{disk}| + 0.7\,V_{bulge}|V_{bulge}|. \quad (206)$$

Here Vdisk and Vbulge are the published components at unit stellar mass-to-light ratio; the coefficients apply the same 0.5 and 0.7 convention as equation (199). Vgas already includes the adopted helium correction. The sign convention preserves any outward radial component. No dark-matter halo is added, and no component normalization is fitted to these velocities. The nineteen catalogue rows were checked against the published SPARC mass-model table at each exact adopted radius. NGC 4565 remains a supplementary SET point because a comparable Newtonian spatial model with the same adopted inventory has not been established here.

For example, NGC 4013 at R = 13.47 kpc has Vgas = 31.57, Vdisk = 154.56 and Vbulge = 70.18 km/s. The calculation gives vN = √[31.57² + 0.5(154.56²) + 0.7(70.18²)] = 128.02 km/s. At those same inputs the SET boundary prescription gives 190.73 km/s, compared with 185.0 ± 6.0 km/s observed. The uncertainty quoted here belongs to the measured speed, not to the complete prediction.

Figure 5 puts both calculations on common dimensionless axes. For each galaxy define vref = √(GMbar/R). The horizontal coordinate is the predicted speed divided by vref; the vertical coordinate is vobs/vref. Thus a SET point has horizontal coordinate 4/fimage, a Newtonian point has vN/vref, and equality with the observation lies on the diagonal. The dotted line at one identifies the spherical reference only. The SET estimate is calculated from Qref and Aimage; it is not obtained by multiplying the full Newtonian disk prediction by 4/fimage.

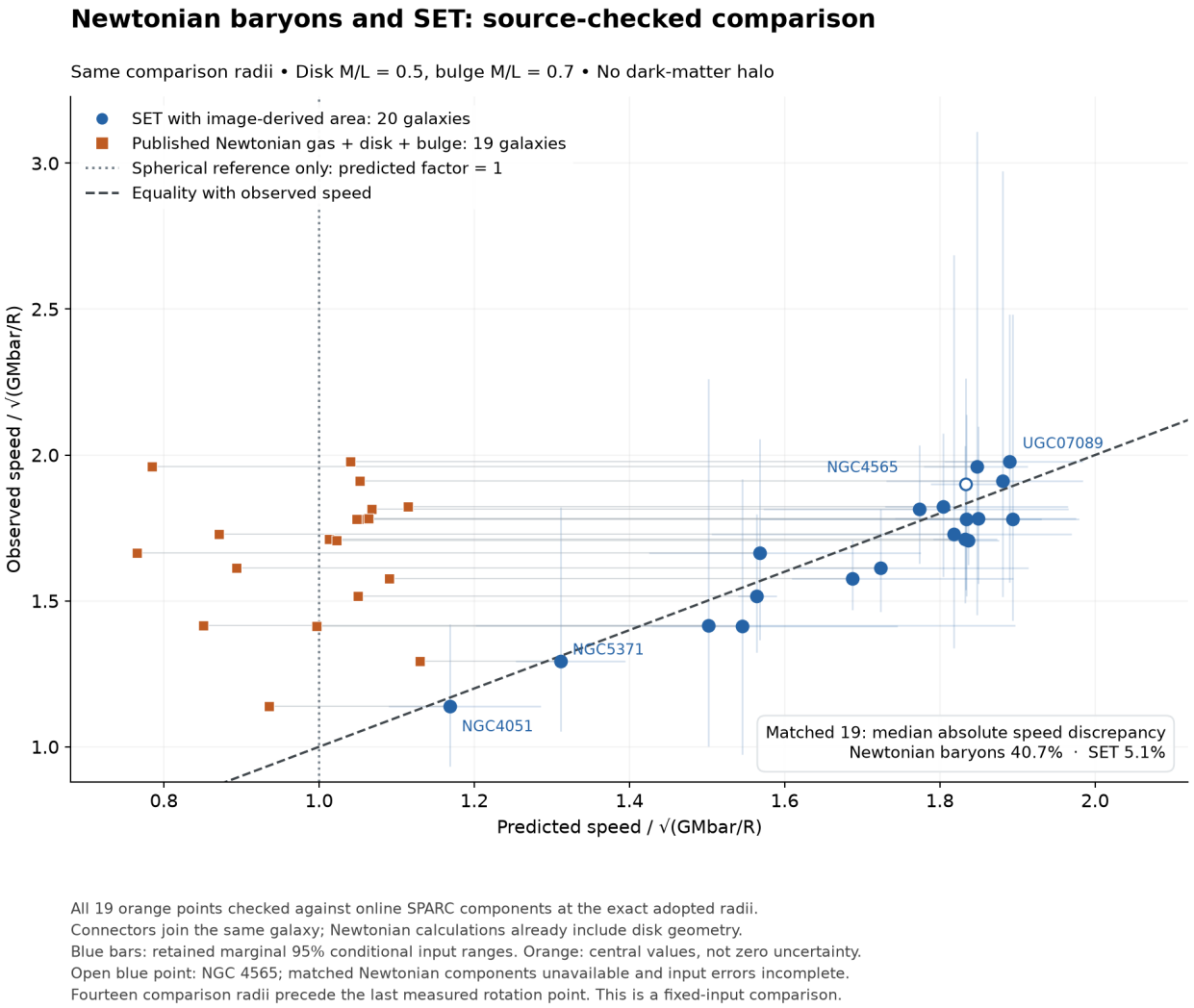


Figure 5. Published Newtonian baryonic components and the SET image-area prescription at the adopted comparison radii. Connectors join the same galaxy. Blue bars show marginal 2.5th–97.5th percentile ranges from joint input draws, with correlated axes; they are not complete prediction confidence intervals. Orange markers show central predictions, not zero uncertainty. The open blue point is NGC 4565. The matched-nineteen medians use 100|vpred/vobs − 1|. Fourteen comparison radii are interior to the last measured rotation point. The source table already accounts for the Newtonian disk geometry.

Across the nineteen matched cases, the median absolute central speed discrepancy is 40.67% for the adopted Newtonian baryons and 5.13% for SET. The SET central discrepancy is smaller in each of these nineteen comparisons. The corresponding SET median for all twenty cases is 4.79%, as reported above. These are descriptive results for the cases in Table 3, with the same central inputs retained. The comparison concerns baryons at the stated stellar normalization, rather than a fitted dark-matter model or every possible baryonic inventory.

Here the image-area calculation improves agreement relative to spatially resolved Newtonian baryons, whose disk geometry is already included. The normalized plot displays the speed comparison on common axes. Appendix E gives the conditional input ranges, and the calculation record contains all nineteen component comparisons.

8.5 Source organization and measurement sensitivity

The proposal can now be stated physically. Mass establishes the source normalization. I then ask whether source organization changes the collective response in a way that can be expressed through an effective boundary area. At the same Mbar and R, a smaller effective area gives a larger boundary amplitude. The image-based implementation makes that idea calculable.

For each displayed case, the image coefficient comes from photometry and the adopted inclination. The central mass, radius and area are retained when calculating the speed. There is no separately adjusted halo mass or concentration in this calculation.

The diagnostic finferred asks which area the observed speed would require under this prescription. Comparing it with fimage tests the proposed geometric connection. Because both depend on some shared inputs, including inclination, their errors must be propagated together. The selection and correlation interpretation are stated with Table 3.

Figure 6 keeps the same masses, radii and observed speeds and changes only the area coefficient: a common f = 2 is replaced by each fimage. This shows how much the image correction changes these twenty predictions. The broader comparison with common normalizations is retained in the research supplement.

The differential dependence is:

$$d\ln v_{SET} = \frac{1}{2} d\ln M_{\mathrm{bar}} - \frac{1}{2} d\ln R - d\ln f_{image}. \qquad (207)$$

Distance changes several inputs together. For a flux-derived inventory Mbar $\propto D^2$, while a radius obtained from angular size has R $\propto$ D. Thus vSET $\propto \sqrt{D}$ at fixed geometry. Inclination changes both the reconstructed area and the deprojected observed speed. Appendix E varies these inputs jointly and separates the stellar mass-to-light exercise. Its allowed-input envelopes show sensitivity; they are not confidence intervals or a simultaneous fit with shared systematic parameters.

The mass inventory and comparison boundary should describe the same system. Table 3 uses global inventories at the adopted comparison radii. Source reconstructions can determine how much material lies beyond each location and how that material contributes to the local burden and force. A matched mass aperture alone does not replace the field calculation for a general flattened system.

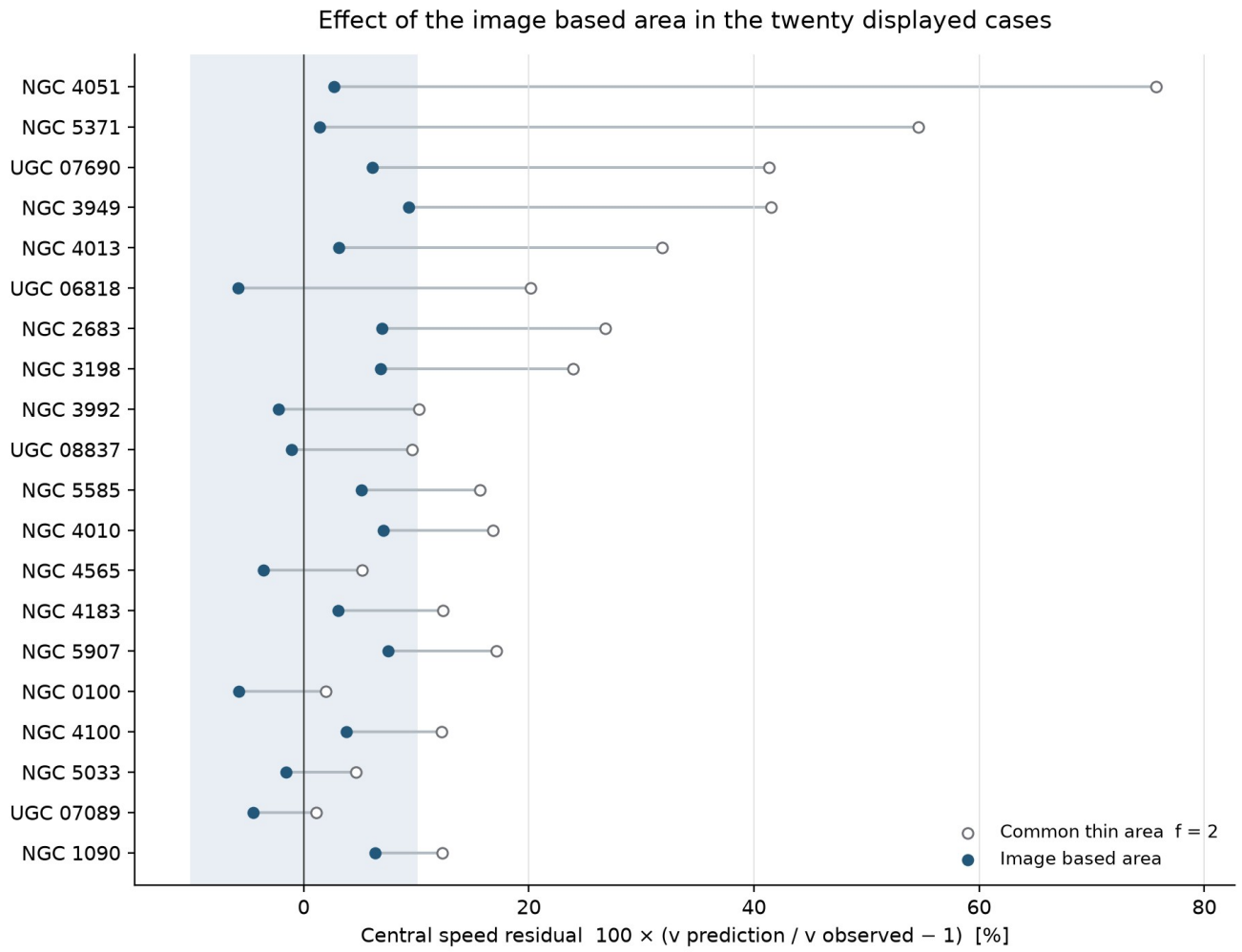


Figure 6. Signed speed residuals using f = 2 and the image-based f for the same twenty examples. Each line connects the two calculations for one galaxy. The shaded ±10% band is a descriptive reference, not a confidence interval.

8.6 Exterior continuation at farther measured radii

Table 3's comparison radii are observational evaluation locations. The local-field diagnostic in Section 7.3 tests the reduction at each location; measurements farther out separately test its proposed continuation.

As a separate consistency check, I retained the Table 3 masses, image coefficients and comparison radii and continued the fixed-shape exterior into the available outer velocity data. Fourteen galaxies have such measurements. At their last measured points, all fourteen central predictions are lower than the adopted observed speeds, with a median shortfall of 18.2%; two differ by less than 10%. The other six have no farther point in the adopted datasets. This checks that particular exterior continuation, rather than constituting twenty resolved-source field solutions. The tabulated velocity errors do not include a new complete propagation of source and geometric uncertainties. The full calculation record accompanies the single-radius comparisons.

8.7 Additional galaxy calculations

A first extension of the same source and image-area calculation supplies 42 additional worked examples, bringing the initial collection to 62 distinct galaxies. The original twenty calculations, their adopted inputs and their resolved baryonic comparisons remain unchanged. The additions comprise thirty PROBES objects, ten GHASP objects, NGC 3972 from SPARC, and UGC 06894 from the Ursa Major survey [5, 30–33]. Each

calculation uses baryonic mass, a stated comparison radius and an area reconstructed from the image and adopted inclination. No dark-matter halo or additional speed normalization is fitted.

The supplemental stellar inventory uses the S4G 3.6-micron total magnitude and published decomposition, with the same fixed disk and bulge mass-to-light ratios of 0.5 and 0.7. Atomic hydrogen enters through the published integrated HI flux or HI-flux magnitude, with the factor 1.33 for helium [6, 8, 34]. At distance D in Mpc, the adopted conversions are

$$\begin{aligned} \frac{L_{3.6}}{L_{\odot}} &= 10^{0.4\left[6.02 - m_{3.6} + 5\log_{10}(D/Mpc) + 25\right]} \\ \frac{M_{HI}}{M_{\odot}} &= 2.356\times 10^{5}\left(D/Mpc\right)^{2}\frac{F_{HI}}{Jy\,km\,s^{-1}}. \end{aligned} \qquad (208)$$

$$\begin{aligned} \frac{F_{HI}}{Jy\,km\,s^{-1}} &= 10^{0.4\left(17.40 - m_{21}\right)} \\ M_{\mathrm{bar}} &= \left(0.5 + 0.2\,b\right)L_{3.6} + 1.33\,M_{HI} \\ b &\equiv L_{bulge}/L_{3.6}. \end{aligned} \qquad (209)$$

The last expression uses solar luminosity and mass units. It is the same component inventory as Section 7.5, written using the total luminosity and bulge fraction. The actual adopted distances are retained: PROBES supplies its catalogue distances, GHASP its published distance table, SPARC its tabulated distance, and the Ursa Major example the survey value of 18.6 Mpc. The calculation archive records the individual measurements and aliases.

For the optical branch measurements, the comparison is made at the outermost actual radius covered by both sides and by qualifying infrared isophotes. The opposite side is interpolated only within its measured coverage. GHASP supplies deprojected branch speeds. The PROBES raw speeds retain their projection and redshift factors, so their equal-side mean is converted by $v_{obs} = \left(|V_{+}| + |V_{-}|\right)/\left[2\sin i\left(1+z\right)\right]$ [30–32]. The fixed image-quality and oblate-deprojection rules of Section 7.5 then give f_{image}. A failed deprojection does not initiate a search for a radius with a smaller speed discrepancy.

$$\begin{aligned} Q_{ref} &= 4\,\pi\sqrt{2\,G\,M_{\mathrm{bar}}\,R^{3}} \\ A_{image} &= f_{image}\,\pi\,R^{2} \\ S_{boundary} &= \frac{Q_{ref}}{A_{image}} \\ v_{SET} &= \frac{S_{boundary}}{\sqrt{2}} = \frac{4}{f_{image}}\sqrt{\frac{G\,M_{\mathrm{bar}}}{R}}. \end{aligned} \qquad (210)$$

This chain is the tested boundary reading of Section 7: the source law supplies Q_{ref}, the image supplies the area, and the stated exterior orbital conversion supplies the speed. The global inventory and comparison radius retain those meanings. A comparison radius locates the observation; the local field and its slope determine when the same boundary reading is also the local orbital solution. Section 9.3 makes the relation between boundary amplitude and response family explicit.

8.7.1 NGC 1425: from source measurements to speed

NGC 1425 provides a direct numerical example with small approaching–receding asymmetry. The adopted distance is $D = 17.6959$ Mpc and the shared angular comparison radius is 120.761 arcsec, giving

$R = 10.360345$ kpc. The total magnitude is $m_{3.6} = 10.806$, the decomposition gives $b = 0.049$, and the HI-flux magnitude is $m_{21} = 13.448$. These source inputs give

$$\begin{aligned} L_{3.6} &= 3.8136893 \times 10^{10}\, L_{\odot} \\ F_{HI} &= 38.0890 \;\; Jy\, km\, s^{-1} \\ M_{HI} &= 2.8100924 \times 10^{9}\, M_{\odot}. \end{aligned} \qquad (211)$$

$$\begin{aligned} M_{\star} &= 1.9442188 \times 10^{10}\, M_{\odot} \\ 1.33\, M_{HI} &= 3.7374229 \times 10^{9}\, M_{\odot} \\ M_{\mathrm{bar}} &= 2.3179611 \times 10^{10}\, M_{\odot}. \end{aligned} \qquad (212)$$

At that image radius the projected axis ratio is $p = 0.4571341$. With the adopted $i = 68.2^{\circ}$, the oblate reconstruction gives the intrinsic short-to-long ratio η and surface coefficient

$$\begin{aligned} \eta^2 &= \frac{p^2 - \cos^2 i}{\sin^2 i} = 0.08242510 \\ \eta &= 0.28709771 \\ f_{image} &= 2\left[1 + \frac{\eta^2 \operatorname{atanh}\sqrt{1-\eta^2}}{\sqrt{1-\eta^2}}\right] = 2.33038932. \end{aligned} \qquad (213)$$

Keeping those measured central inputs fixed, the source and area are

$$Q_{ref} = 1.7816439 \times 10^{47} \;\; m^3 s^{-1}, A_{image} = 7.4821755 \times 10^{41} \;\; m^2. \qquad (214)$$

$$S_{boundary} = 238.11843 \;\; km\, s^{-1}, v_{SET} = 168.37516 \;\; km\, s^{-1}. \qquad (215)$$

The projected two-side mean is 160.16691 km/s. Using the published recession velocity 1512 km/s, $z = 0.00504349$, the same radius gives $v_{obs} = 171.63770$ km/s. The branch-based descriptive velocity range is ± 7.49472 km/s, formed from the reported branch errors and their half-difference. The SET calculation is therefore 1.90% below this measured central speed. The prediction was obtained from the source mass and image area before this comparison; no halo amplitude or velocity-derived area was inserted.

$$100\left(\frac{168.37516}{171.63770} - 1\right) = -1.90083\,\%. \qquad (216)$$

8.7.2 The expanded comparison

Appendix E lists the first 42 additions and the 21 further examples with their radii, baryonic masses, image coefficients, calculated speeds and measured velocity readings. The median absolute central discrepancy is 5.99% for the first 42 additions, 5.90% for the further 21 and 5.76% for all 83 displayed examples. All displayed central discrepancies are within 10%; the unchanged original-twenty median is 4.79%. These statistics describe the displayed worked examples. The full survey and calculation ledger preserves the input coverage, exclusions and every calculated outcome.

Figure 7 extends the normalized comparison to 83 examples. Its horizontal coordinate is $x = v_{calculated} / v_{ref}$ and its vertical coordinate is $y = v_{obs} / v_{ref}$, where $v_{ref} = \sqrt{G M_{\mathrm{bar}} / R}$. Hence $x_{SET} = 4 / f_{image}$. The actual resolved Newtonian gas, disk and bulge calculations remain available for the original nineteen matched objects, NGC 3972 and the six recovered SPARC objects, giving 26 such comparisons. The vertical line at $x = 1$ marks the spherical reference. It is not used as a substitute for a resolved Newtonian disk calculation.

The green region displays the common 10% speed-agreement criterion. The purple stepped region displays each galaxy's own recorded velocity range, divided by its fixed reference speed and placed around equality at that galaxy's horizontal position. Each cell belongs to one galaxy; the steps join the cells for display and are not an interpolated uncertainty law. Forty-eight of the 83 nominal predictions fall inside their own recorded velocity ranges. The five earlier side-asymmetry flags and five new Apertif amplitude or inclination flags remain visible in the figure and tables.

8.7.3 Twenty-one further source-based comparisons

A further source audit adds 21 distinct examples: six SPARC objects whose missing S4G image products were recovered under catalogue aliases, nine Apertif objects, five WALLABY objects and one PROBES II object [5, 6, 39, 40, 42]. The original 62 calculations are retained unchanged. The same stellar mass-to-light ratios, helium factor, image-quality rules, adopted inclinations and outermost common measured-radius rule are applied before calculating each speed discrepancy. The new examples therefore extend the same source-law calculation without a new velocity normalization.

Apertif and WALLABY supply released deprojected circular-speed bins; the catalogue conventions for HI flux, distance and velocity error are retained [39, 40, 41]. UGC01551 uses the PROBES II archived curve with the original observing paper's distance and inclination [42, 43]. Already deprojected speeds receive no second inclination correction. Appendix E.3 gives the input and quality details.

NGC3835 gives a compact example of the unchanged numerical chain. Its adopted Apertif distance is $D = 39.3$ Mpc, inclination $i = 69.4^{\circ}$, comparison radius $R = 12.86089493$ kpc and projected image ratio $p = 0.35549931$. The stellar and HI-plus-helium inventories are $2.640422931 \times 10^{10}\,M_{\odot}$ and $3.620816909 \times 10^{9}\,M_{\odot}$, giving $M_{\mathrm{bar}} = 3.002504622 \times 10^{10}\,M_{\odot}$. The oblate reconstruction gives $f_{image} = 2.021320269$.

$$Q_{ref} = 2.804497932 \times 10^{47}\ \ m^{3}\,s^{-1}$$

$$A_{image} = 1.000065566 \times 10^{42}\ \ m^{2}.$$

$$S_{boundary} = 280.4314064\ \ km\,s^{-1}$$

$$v_{SET} = 198.2949491\ \ km\,s^{-1}.$$

The observed speed at the same radius is 203.6 ± 3.3 km/s, giving a signed central discrepancy of $-2.6056\,\%$. The reported bin error retains its observational meaning. This is an interior comparison in the adopted curve, obtained with the fixed source inventory and image-area prescription; the speed agreement does not by itself identify a physical galaxy edge.

8.8 Propagating uncertainty in the first 62 SET calculations

The reported-input propagation is retained for the first 62 examples: the original twenty and the first 42 additions. It asks how the measured uncertainties in the source inputs affect the SET calculation itself. Its central values remain exactly those in the tables. Distance, photometry, gas inventory, projected shape and inclination are varied through the same equations that produced each central result. No new mass-to-light scale is introduced to widen agreement. The further 21 examples enter Figure 7 with their central calculations and recorded observational ranges; Figure 8 retains the completed source-input propagation for the first 62.

$$M_{\star}, M_{gas} \propto D^{2}$$
$$R \propto D$$
$$v_{SET} \propto D^{1/2} \qquad (217)$$
$$Q_{ref} \propto D^{5/2}$$
$$A_{image} \propto D^{2}.$$

Here f is held fixed. One distance draw therefore changes both component masses and the physical radius together. Where the same inclination underlies the image deprojection and measured speed, that draw also changes the observed reading by $\sin i_0 / \sin i$. This preserves the shared-input dependence. UGC 06894 is handled separately: its image uses the global inclination $83^{\circ} \pm 3^{\circ}$, whereas its outer HI speed uses the local ring value 89 degrees. The global image draw is not applied to that distinct ring reading.

For the 42 additions, 250000 attempts per galaxy use the reported marginal errors as Gaussian sensitivity inputs. Positive source quantities and physical oblate deprojections are retained, with their acceptance fractions recorded. Inclinations are folded to equivalent plane orientations, as in the original-twenty calculation. The original twenty use their archived 65536 source-input draws, with independent observed-speed noise removed for this figure. The resulting segments are conditional input percentiles: thick segments contain the central 68% and thin segments the central 95% of retained draws. The separate observational ranges are displayed without resampling their heterogeneous errors or dispersions.

The source-input segments overlap the individual recorded observational ranges for 58 of 62 examples at the central 68% level and 60 of 62 at the central 95% level. At the latter level the two separated cases are NGC 5907 and NGC 4651. These counts describe overlap of the displayed ranges, not a common-confidence significance test. For NGC 1425, the propagated central 68% speed interval is 163.72–173.50 km/s and the central 95% interval 159.49–178.88 km/s, while the nominal prediction remains 168.38 km/s.

Figure 8 shows what the uncertainty calculation adds: agreement can be assessed using measured input variation rather than central values alone. The intervals are partial because some measurements have no reported errors. Among the 42 additions, distance errors are supplied for 31, stellar photometry errors for 42, HI errors for 41, image-shape errors for 42 and global inclination errors for 12. Missing errors, the adopted mass-to-light ratios and unreported decomposition or calibration terms remain fixed and identified. Conditioning on a valid oblate geometry can produce asymmetric intervals; NGC 2543, for example, retains 53.45% of its geometry draws. Appendix E and the machine-readable inventory specify these conditions for each object.

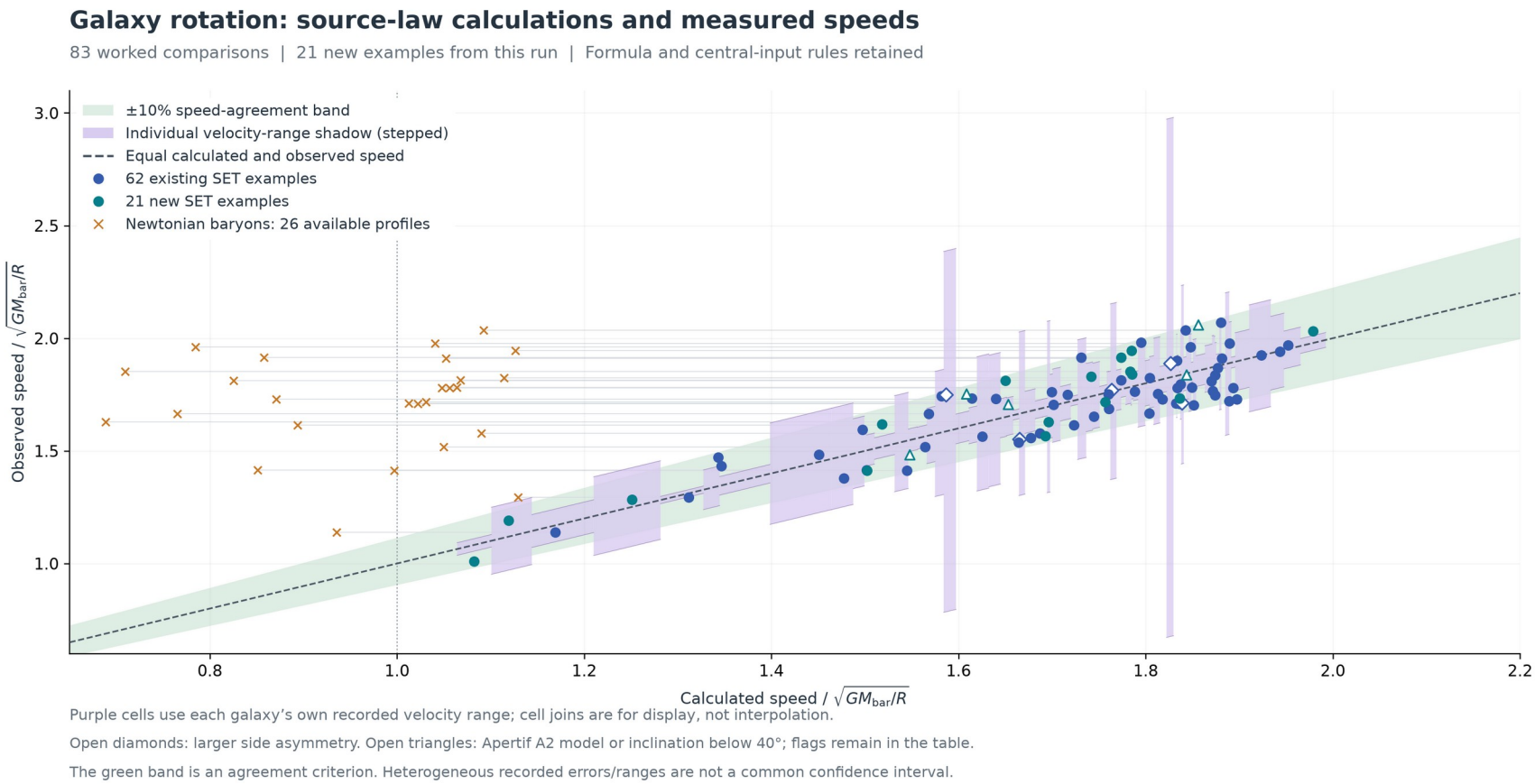


Figure 7. Galaxy rotation under the fixed SET source/image-area prescription: 83 worked examples and 26 available resolved Newtonian baryonic calculations. Blue and teal distinguish the earlier 62 and the further 21 examples. Connectors identify the same galaxy. The green band is $|v_{SET}/v_{obs}-1|\leq 0.10$, or $x/1.1\leq y\leq x/0.9$. Purple cells place each recorded velocity range around equality at fixed reference normalization; their joins are graphical only. Open diamonds retain the five larger side-asymmetry flags. Open triangles mark four Apertif A2 cases and IC0758, whose adopted inclination is below 40 degrees. The ranges retain their individual source definitions. The new eligible screen contains 71 calculations; 21 meet the displayed agreement criterion.

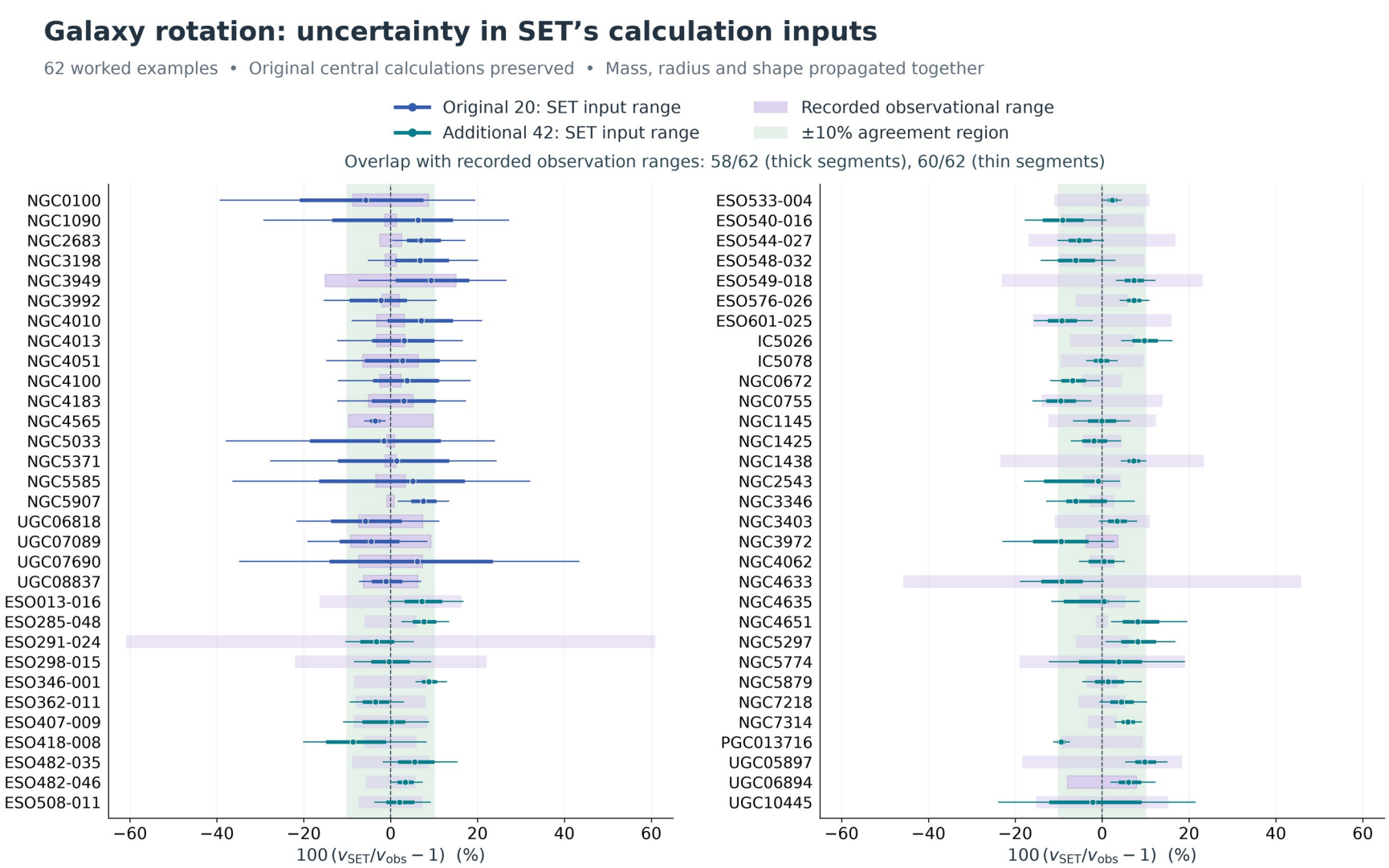


Figure 8. Source-input propagation for the first 62 galaxy calculations. The 21 further examples are not included in this propagation. Dots retain each nominal residual 100(vSET/vobs−1); thick and thin segments show the central 68% and 95% of conditional source-input draws. Purple cells show each object’s recorded observational range; outlined cells use the listed rotation-bin error, while pale cells use the descriptive branch-error/dispersion and side-difference combination. Distance changes mass and radius together, and a shared inclination changes both image geometry and speed deprojection. The green background marks ±10%. Reported-input segments overlap the observational cells for 58/62 and 60/62 examples, respectively. These are partial input-sensitivity intervals, with fixed adopted M/L and unreported errors, conditional on a physical oblate deprojection.

9 Collective fields and energy accounting

The boundary reading and the stationary field describe connected but distinct parts of the calculation. I now put the source contribution into the invariant explicitly, retaining the specified response tensor and boundary condition. The same formulation supplies a check on the work associated with changes in source organization.

Appendix F states a stationary local-field formulation with a prescribed spatial response. Its equation is $-\nabla\cdot(C\nabla\Psi) = 4\pi G\rho\text{bar}$, with $\Psi = S^2/2$. The dimensionless symmetric positive tensor C describes the spatial response associated with that source organization. It is distinct from the remaining capacity $1 - u$ and the displacement metric q_{ij}.

Setting C to the identity recovers the ordinary weak-field source equation in burden variables.

The energy calculation identifies the work term associated with changes in the response.

9.1 The sources inside the invariant

We can now put the sources themselves into the invariant. For this stationary calculation I use Appendix F's spatially constant, symmetric positive-definite response tensor C and the boundary condition $S^2 \to 0$ at infinity. C specifies the response to source geometry; q_{ij} describes spatial displacement and α^2 the remaining capacity.

Let source a have mass M_a, reference radius R_a and volume $V_a = (4/3)\pi R_a^3$. Its nonnegative spatial profile $w_a(x')$ is normalized by $\int w_a(x')d^3x' = 1$, so $\rho\text{bar}(x') = \Sigma_a M_a w_a(x')$. The reference radius belongs to the source normalization; the observation position is x. Write $d = x - x'$. Substituting the stationary source solution into the capacity budget gives:

$$\frac{1}{12\pi\sqrt{\det C}}\sum_a\Big\{\frac{[4\pi\sqrt{2GM_aR_a^3}]^2}{(4/3)\pi R_a^3}\times\int\frac{w_a(x')}{\sqrt{d^TC^{-1}d}}d^3x'\Big\}+V_{rel}^2+c^2\beta^2=c^2. \quad (218)$$

The first term is $S^2(x)$, written with its sources visible. The source distribution and specified response establish this field; the probe shares the remaining capacity between translation and event progression. A ship need not increase its intrinsic source normalization for a larger $Vrel^2$ contribution to leave less capacity for local events. The distinction is between source output and the probe's participation in the field. Here the ship is treated as a test probe in a prescribed field; its motion contribution is already represented by $Vrel^2$. All terms have units m^2/s^2.

The normalization simplifies directly:

$$\frac{[4\pi\sqrt{2GM_aR_a^3}]^2}{(4/3)\pi R_a^3}=24\pi GM_a. \quad (219)$$

Using $\rho\text{bar} = \Sigma_a M_a w_a$ therefore gives:

$$S^2(x)=\frac{2G}{\sqrt{\det C}}\int\frac{\rho_{\text{bar}}(x')}{\sqrt{d^TC^{-1}d}}d^3x'. \quad (220)$$

The reference radius cancels from the normalized coefficient, while the source positions and density profiles remain in the integral. Subdividing the same density distribution into more numerical pieces leaves this field unchanged. For C = I and an isolated spherical exterior, equation (220) gives $S^2(r) = 2GM/r$. Equation (218) then reduces to:

$$V_{rel}^2 + c^2\beta^2 = c^2 - \frac{2GM}{r}. \quad (221)$$

This recovers the spherical amplitude as one solution for S. For the collective boundary proposal, I retain $Q\Sigma = Qref$ and $Aeff = f\pi R^2$. Where the boundary amplitude equals the local field, $S = Seff = (4/f)\sqrt{(2GM/R)}$. The fixed-Q identity in equation (187) follows at the same M and reference R. Here $f \to 2$ is the formal thin-area limit.

Writing the source term inside the invariant retains the specified response and boundary conditions. In this static timelike description $S^2 + Vrel^2 < c^2$. Sections 3 and 4 give the spherical action, clock and propagation relations.

9.2 Response and mechanical energy

For $\Psi = S^2/2$, the stationary source equation is $-\nabla\cdot(C\nabla\Psi) = 4\pi G\rho bar$. Appendix F gives its integral solution and the boundary orientation argument. For finite smooth sources, the binding energy is $W = -\frac{1}{2}\int\rho\Psi\, d^3x$. When the prescribed response changes, the energy balance is

$$\frac{d(E_{kin}+W)}{dT} = \frac{1}{8\pi G}\int \nabla\Psi\cdot\frac{\partial C}{\partial T}\nabla\Psi\, d^3x. \quad (222)$$

A fixed response conserves this mechanical energy. A changing response supplies or removes work through the displayed term. If the response depends on source coordinates, its variation also enters their force. Appendix F retains the full derivation, including that source-dependent contribution. This accounting makes the physical meaning of a change in collective response explicit.

9.3 Source-normalized amplitude and the collective work term

The image-area calculation can be connected to the stationary field equation through an explicit amplitude result. Work within the collective weak-field construction of Appendix F, with $\Psi = S^2/2$, a symmetric positive response tensor C, a fixed physical source density and a fixed physical boundary Σ. The response tensor describes the relation between the source current and the burden gradient; it is distinct from the displacement metric. Suppose a specified family member satisfies

$$-\nabla\cdot(C_0\nabla\Psi_0) = 4\pi G\rho_{\text{bar}}, \Psi_0 \to 0 \text{ at infinity}. \quad (223)$$

Multiply the potential by one positive constant λ and divide the tensor by the same constant. The source current and the complete density on the right-hand side remain unchanged:

$$\Psi_\lambda = \lambda\Psi_0, C_\lambda = \lambda^{-1}C_0, J_\lambda = -C_\lambda\nabla\Psi_\lambda = -C_0\nabla\Psi_0 = J_0. \quad (224)$$

To read the burden on the boundary, use the same surface moments that define effective area. All areas here are evaluated on the same physical Σ:

$$\begin{aligned} Q_\Sigma &= \int_\Sigma \sqrt{2\Psi}\, dA \\ I_\Sigma &= \int_\Sigma 2\Psi\, dA \\ A_{eff} &= \frac{Q_\Sigma^2}{I_\Sigma}. \end{aligned} \quad (225)$$

$$\begin{aligned} Q_{\Sigma,\lambda} &= \sqrt{\lambda}\, Q_{\Sigma,0} \\ I_{\Sigma,\lambda} &= \lambda\, I_{\Sigma,0} \\ A_{eff,\lambda} &= A_{eff,0}. \end{aligned} \qquad (226)$$

Imposing the retained SET boundary correspondence $Q_\Sigma = Q_{ref}$ fixes the amplitude uniquely for $Q_{\Sigma,0} > 0$:

$$\lambda = \left(\frac{Q_{ref}}{Q_{\Sigma,0}} \right)^2 , Q_{ref}^2 = 24\,\pi\, G\, M_{\mathrm{bar}}\, V_{ref}. \qquad (227)$$

This is an amplitude selection before comparison with orbital data. It uses the source law and the specified response family, while leaving the family's effective area unchanged. In the observational implementation, Section 7 supplies the further identification $A_{eff} \simeq A_{image}$. For a uniformly loaded boundary this gives $S_{boundary} = Q_{ref} / A_{image}$. The family's spatial solution then supplies the gradient that a material orbit measures.

$$\begin{aligned} v_c^2(R) &= -R\frac{\partial \Psi}{\partial R} = -\frac{R}{2}\frac{\partial S^2}{\partial R} = \frac{k(R)}{2} S^2(R) \\ k(R) &= -\frac{d\ln S^2}{d\ln R}. \end{aligned} \qquad (228)$$

The fixed-shape inverse-radius exterior has $k = 1$ and returns the boundary conversion $v_c = S/\sqrt{2}$. Its spatial condition can be stated directly. For similar ellipsoidal screens $s^2 = x^T D\, x$ and $\Psi = K/s$, with constant positive C, the source-free equation is equivalent to

$$3\,DCD = \mathrm{tr}(CD)\, D, \text{ hence } C = c_0\, D^{-1}. \qquad (229)$$

Thus the homothetic exterior fixes the tensor shape, and the boundary correspondence fixes its remaining amplitude. The roles are separate and calculable: the source density and response family determine the spatial field; the source law calibrates its boundary amplitude; the probe law reads the resulting local gradient. Equal boundary amplitudes in different families do not, by themselves, imply equal local gradients. The response-family audit provides an explicit same-source check of that distinction.

The same construction determines the mechanical cost of a response change. With finite sources and the isolated boundary condition, integration by parts gives

$$W = -\frac{1}{2}\int \rho_{\mathrm{bar}}\, \Psi\, d^3x = -\frac{1}{8\pi G}\int \nabla\Psi \cdot C\, \nabla\Psi\, d^3x,\; W_\lambda = \lambda\, W_0. \qquad (230)$$

A change of amplitude therefore has a definite binding-energy contribution. For the quasistatic evolution used in Section 9.2, define the response power by the explicit tensor variation:

$$P_{response} = \frac{1}{8\pi G}\int \nabla\Psi \cdot \dot{C}\, \nabla\Psi\, d^3x, \frac{d(E_{kin}+W)}{dT} = P_{response}. \qquad (231)$$

$$C = C_0/\lambda, \Psi = \lambda\,\Psi_0 \Rightarrow P_{response} = \lambda\, P_{response,0} + W_0\,\dot{\lambda}. \qquad (232)$$

At fixed source and fixed C_0, this reduces to $P_{response} = W_0\,\dot{\lambda}$. Where C depends on source coordinates X_a, its variation supplies the corresponding generalized-force contribution as well:

$$F_{a,response} = -\frac{1}{8\pi G}\int \nabla\Psi \cdot \frac{\partial C}{\partial X_a}\, \nabla\Psi\, d^3x. \qquad (233)$$

These equations tie an organization-dependent response to a source normalization and a work balance. They belong to the stated C-weighted collective completion. A physical rule for the response must use the density and relevant source motion or stress state, so that dividing the same source into more numerical pieces does not change its field. The invariant then carries the resulting S^2 into clocks and material motion. In this way the galaxy calculation has an explicit mathematical route from its source law to amplitude, local response and mechanical accounting.

10 Cosmological source evolution and physical clocks

The cosmological question follows the same source–capacity reasoning in a different setting. Instead of reading a stationary exterior at successive radii, I follow an evolving material domain and the collective region included in its burden. The observable will be the brightness received at a measured redshift. The volume evolution and front prescription are stated before that brightness is calculated.

10.1 Source volume and receiving clock

Space Emanation Theory proposes that matter establishes a burden in the field through which physical interactions occur. Gravitational burden, motion and the progression of local events participate in one finite capacity. A clock measures that progression through its physical ticks. We examine a homogeneous cosmological realization of these principles and calculate how its source and clock relations affect the brightness received from distant supernovae.

The source law and the local capacity relation are

$$Q^2 = 24\,\pi\, GMV\,,\; S^2 + V_{\mathrm{rel}}^2 + c^2\beta^2 = c^2. \qquad (234)$$

Here M is the baryonic mass assigned to the source domain and V is its volume. The law specifies the positive source quantity Q, with units $\mathrm{m^3\,s^{-1}}$, using reference time T. The gravitational burden S is expressed in speed units; S^2 is its contribution to the capacity budget. V_{rel} describes local motion relative to the field. The dimensionless factor β describes the progression of a physical clock relative to T, and c sets the capacity scale. Volume V and velocity V_{rel} are distinct quantities.

For a spherical domain, $V = 4\,\pi\, R^3/3$ and $M = \rho\, V$. The source law gives

$$Q = 4\,\pi\,\sqrt{2GM\,R^3}\,,\; H_b \equiv \frac{Q}{3V} = \sqrt{\frac{8\,\pi\, G\,\rho}{3}}. \qquad (235)$$

The second equality is direct algebra. Take a material domain containing a fixed collection of baryonic matter, with mass M_m and physical volume V_m. The subscript m distinguishes it from the advancing collective region introduced in Section 10.2. Define its source-volume clock by

$$d\,T_Q \equiv \frac{d\,V_m}{Q_m}\,,\; Q_m = \sqrt{24\,\pi\, G\, M_m V_m}. \qquad (235\,a)$$

On the expanding branch this differential can be integrated directly:

$$T_Q = \int_0^{V_m} \frac{dV}{\sqrt{24\,\pi\, G\, M_m V}} = \sqrt{\frac{V_m}{6\,\pi\, G\, M_m}} = \frac{1}{\sqrt{6\,\pi\, G\,\rho}}. \qquad (235\,b)$$

The displayed expression chooses the formal zero-volume origin; it does not assert that the physical front begins there. A constant time shift leaves the differential unchanged. Rescaling the homogeneous mass and volume together leaves T_Q unchanged, so the clock is independent of the size chosen for a representative

material domain. Write $V_m \propto a^3$, where a is the common scale factor. Conservation of the material mass gives

$$\rho = \frac{1}{6\pi G T_Q^2}, H_b = \frac{2}{3T_Q}, \frac{a}{a_0} = \left(\frac{T_Q}{T_{Q0}}\right)^{2/3}. \quad (235c)$$

For this cosmological realization we use T_Q as the reference T in the source law, with $a_0 = 1$ at reception. Thus $\rho = \rho_0 a^{-3}$ and

$$H_T \equiv \frac{1}{a}\frac{da}{dT} = \frac{Q_m}{3V_m} = H_b(a) = H_{b0} a^{-3/2}. \quad (236)$$

The factor three comes from differentiating a volume proportional to a^3. At any given epoch, $Q/(3V)$ has the same value for domains with the same uniform density. A fixed material mass does not require a constant Q_m: the source law also contains the evolving volume. Present-day values carry the subscript zero.

The same process read by a physical clock

The variable t records the reading of the comoving physical clock. For any expanding homogeneous material-volume history, differentiation of the source-volume clock gives

$$\alpha_Q \equiv \frac{dt}{dT_Q} = \frac{H_b}{H}, H \equiv \frac{1}{a}\frac{da}{dt}. \quad (236a)$$

This is a reading of the supplied volume history. Identifying α_Q with the capacity factor α is the physical correspondence retained in this calculation. SET interprets clock ticks as events realized within the available field capacity: a smaller factor means fewer physical ticks per reference second.

A comoving clock follows the local homogeneous matter and has $V_{\text{rel}} = 0$ in this construction. The capacity relation therefore gives $\beta = \alpha$ and

$$u \equiv \frac{S^2}{c^2}, \alpha = \sqrt{1-u}, dt = \alpha\, dT. \quad (237)$$

For $0 \le u < 1$, the clock factor is positive. The same source-driven volume increment can be expressed using either time reading:

$$dV_m = Q_m dT = Q_{\text{normalized},m}\, dt. \quad (238)$$

Dividing by the physical clock interval gives the clock-normalized source rate,

$$Q_{\text{normalized},m} = Q_m \frac{dT}{dt} = \frac{Q_m}{\alpha}. \quad (239)$$

The source law is expressed per reference time. A clock within the burdened field registers a smaller interval over the same process, so expressing that process per second of this clock gives a larger numerical rate. The amount assigned to a given physical interval is unchanged. For example, if $\alpha = 1/2$ throughout an interval, one reference second corresponds to half a clock second and the clock-normalized rate is $2Q_m$. The factor follows from the clock conversion; it is not an extra source contribution.

We use this reading to calculate the evolution measured by the comoving clock:

$$H \equiv \frac{1}{a}\frac{da}{dt} = \frac{1}{3V_m}\frac{dV_m}{dt} = \frac{Q_{\text{normalized},m}}{3V_m} = \frac{H_b}{\alpha}. \quad (240)$$

For any selected domain in this homogeneous prescription, we write $Q_{\text{normalized}} = Q/\alpha$; the domain label can be omitted when it is clear. As the collective burden changes, α changes, and the source rate is read differently by physical clocks. Section 10.2 calculates this history. Section 11.1 uses it in the distance and brightness calculation. The inverse clock factor is already included in $H(z)$, so it is not applied again to the final brightness.

10.2 The collective burden history

To calculate α through time, we specify the collective source region that supplies the burden. This test uses uniform density inside a moving spherical front of radius R_h, with $M_h = 4\pi\rho R_h^3/3$. In the retained flat homogeneous spatial reading, $d\ell^2 = a^2 d\chi^2$ along a radial path. An outward front moving locally at c therefore obeys

$$\frac{d\chi_h}{dt} = \frac{c}{a}, R_h = a\chi_h. \quad (240a)$$

Differentiating the physical radius, with $H = \dot{a}/a$, gives

$$\frac{dR_h}{dt} = HR_h + c. \quad (241)$$

The first term carries the front with the evolving scale; the second advances it locally at c. We identify this null-front region with the domain included in the central burden readout. Its enclosed mass changes as existing matter enters the front, whereas the material domain used to define the source-volume clock has fixed mass.

For the burden readout we retain SET's isotropic static source kernel: a mass element dm at distance r contributes $dS^2 = 2G\,dm/r$. At the centre of a uniform sphere, a shell of thickness dr contains $dm = 4\pi\rho r^2 dr$. Adding these shell contributions gives

$$S_c^2 = 2G\int_0^{R_h}\frac{\rho}{r}4\pi r^2 dr = 4\pi G\rho R_h^2 = \frac{3GM_h}{R_h}. \quad (242)$$

This is the central burden of the collective sphere. The spherical exterior expression $S^2 = 2GM/r$ describes a different evaluation location; it is not substituted for this interior sum. Since $H_b^2 = 8\pi G\rho/3$, the central result becomes

$$u = \xi\frac{H_b^2 R_h^2}{c^2}, \xi = \frac{3}{2}. \quad (243)$$

The factor $3/2$ is the central uniform-source factor, fixed by this readout rather than adjusted numerically. We apply the static uniform-source kernel to the evolving domain as the burden prescription for this calculation. Together with the front evolution and source-volume law, it specifies the physical realization being tested.

The physical balance is visible before changing time variables. Material conservation gives $\dot{\rho} = -3H\rho$. Differentiating $u = 4\pi G\rho R_h^2/c^2$ and using (241) gives

$$\frac{du}{dt} = -Hu + \frac{8\pi G\rho R_h}{c}. \quad (243a)$$

The first term is the declining contribution of shells already included as their density dilutes and their radii grow. The second is the contribution of existing matter newly included by the advancing front. The evolving front must therefore be followed together with the changing density.

Substitution of the source-clock density from (235c) puts the same burden directly in radius and clock variables:

$$u = \frac{2R_h^2}{3c^2T_Q^2}. \quad (243b)$$

Now multiply (241) by $dt/dT_Q = \alpha$, use $\alpha H = H_b = 2/(3T_Q)$, and substitute $\alpha = \sqrt{1-u}$. The explicit source-radius equation is

$$\frac{dR_h}{dT_Q} = \frac{2R_h}{3T_Q} + c\sqrt{1 - \frac{2R_h^2}{3c^2T_Q^2}}. \quad (243c)$$

Define the dimensionless radius $\eta_h = R_h/(cT_Q)$. Differentiating this ratio yields

$$\frac{d\eta_h}{d\ln T_Q} = -\frac{\eta_h}{3} + \sqrt{1 - \frac{2\eta_h^2}{3}}. \quad (243d)$$

Since $u = 2\eta_h^2/3$ and $d\ln a = (2/3)\,d\ln T_Q$, the radius equation gives the same burden history used in the numerical calculation. Equivalently, direct differentiation of the density and radius gives

$$\frac{du}{d\ln a} = u\left[-3 + 2\left(1 + \frac{c}{HR_h}\right)\right] = -u + 2\sqrt{\xi u(1-u)}. \quad (244)$$

The substitution $y = \sqrt{u/(1-u)}$ makes the numerical evolution especially simple:

$$\frac{dy}{d\ln a} = \frac{1+y^2}{2}\left(2\sqrt{\xi} - y\right), H = H_{b0}a^{-3/2}\sqrt{1+y^2}. \quad (245)$$

This is the response equation used for the brightness calculation. The source law supplies the density-dependent rate; capacity supplies the clock conversion; the stated uniform-domain and front prescriptions supply the burden history. Once those prescriptions are retained, the differential equation follows without an adjustable dimming exponent.

The balance also has a definite limit within this prescription. Setting the right-hand side of (243d) to zero gives

$$\eta_{h*} = \frac{3}{\sqrt{7}}, u_* = \frac{6}{7}, \alpha_* = \frac{1}{\sqrt{7}}. \quad (245a)$$

The right-hand side is positive below this radius ratio and negative above it. On the retained lower branch the burden therefore rises toward $6/7$ while the material density continues to fall. This is the accumulation–dilution balance of the stated source and front construction, rather than a universal capacity bound. Appendix J relates the regular front launch to the finite redshift domain of this branch.

10.3 Connecting the material domain and collective source

Two domains enter this history. Write Q_D, V_D, M_b for the material-domain quantities denoted Q_m, V_m, M above. Its fixed collection of baryonic matter has $V_D \propto a^3$ and $\rho_b = M_b / V_D$. The advancing collective region has radius R_h, volume $V_h = 4\pi R_h^3/3$ and mass $M_h = \rho_b V_h$. Its boundary includes additional matter as it advances. These domain labels distinguish both source quantities from a moving probe's clock reading.

The source-volume clock also makes the changing enclosed mass explicit. Apply the density expression $T_Q = \sqrt{V_h/(6\pi G M_h)}$ to the growing homogeneous region and differentiate both V_h and M_h. With $Q_h^2 = 24\pi G M_h V_h$, the result is

$$d T_Q = \frac{d V_h - (V_h/M_h)\, d M_h}{Q_h}. \quad (246)$$

The mass term removes the volume increase associated with matter newly swept into the front. What remains is the material-volume increment at the common density, so it gives the same T_Q as the fixed material domain. At a zero-radius front, this clock is read from the finite-density material domain. The central shell result (242), together with the material-domain source law, also puts the burden amplitude directly in source variables:

$$S_c^2 = \frac{Q_D^2 R_h^2}{6 V_D^2}, \frac{S_c^2}{c^2} = \frac{3}{2}\frac{H_b^2 R_h^2}{c^2}. \quad (247)$$

The front and material domain consequently obey different mass bookkeeping. With $H_{\cos} = d\ln a/dt$, material conservation gives $d\rho_b/dt = -3 H_{\cos}\rho_b$. Differentiating $M_h = 4\pi\rho_b R_h^3/3$ and using the stated front law yields

$$\frac{d M_h}{dt} = 4\pi\rho_b R_h^2 c, \frac{d V_h}{dt} = 3 H_{\cos} V_h + 4\pi R_h^2 c. \quad (248)$$

The added mass is the matter swept into the collective region. Define its source reading by $Q_h^2 = 24\pi G M_h V_h$. Uniform density gives $Q_h/(3V_h) = Q_D/(3V_D) = H_b$. Converting the front-volume equation to reference time therefore gives

$$\frac{d V_h}{dT} = Q_h + 4\pi\alpha R_h^2 c. \quad (249)$$

The second term is the moving-boundary contribution. Thus the material-domain volume law and the collective front can be used together without identifying their volumes or assuming that the front encloses a fixed mass. The same distinction supplies the derivative in Section 10.2: $u \propto \rho_b R_h^2$ contributes $-3 + 2\, d\ln R_h / d\ln a$, while $d\ln R_h / d\ln a = 1 + c/(H_{\cos} R_h)$. These substitutions lead to the displayed u and y history equations.

These relations connect the material clock, source radius and collective burden within the retained luminosity prescription. Appendix N connects the material-domain clock to the common field constraint and states the further correspondence that the actual source evolution must preserve. The light calculation below uses the unchanged history and source inputs.

11 Supernova brightness and observations

11.1 From the source history to received light

11.1.1 Source exposure and the photon response

To turn the source history into received brightness, use the photon coupling of the common probe system. In the zero-shift coordinates used here, let p_i be the canonical spatial momentum and q_{ij} the physical spatial reading. The reference-time photon Hamiltonian and its local energy are

$$H_\gamma = \alpha c P\,,\; P = \sqrt{q^{ij}\,p_i\,p_j}\,,\; E_\gamma = \frac{H_\gamma}{\alpha}. \qquad (250)$$

Defining the unit propagation direction $e^i = q^{ij}\,p_j / P$, Hamilton evolution and the null capacity relation give

$$\frac{d\,x^i}{dT} = \alpha c\,e^i\,,\; q_{ij}\,e^i\,e^j = 1\,,\; \frac{d\ell}{dT} = c\alpha\,. \qquad (251)$$

Define the accumulated source exposure along a ray by $dX = Q_D\,dT$. Its units are volume. Eliminating reference time with (251) gives the precise residence-time weighting:

$$dX = \frac{Q_D}{\alpha}\frac{d\ell}{c} = Q_{normalized\,,\,D}\,\frac{d\ell}{c}. \qquad (252)$$

A physical path increment $d\ell$ therefore carries the normalized source reading Q_D/α. Here the clock conversion belongs to the local field and material observers used to express the propagation law; no proper clock is assigned to the photon. The probe coupling determines the observable response to this exposure.

That response can be displayed before imposing homogeneity. Along a Hamiltonian ray, $d\,H_\gamma / dT = \partial_T H_\gamma$. Differentiating the inverse spatial tensor and then $E_\gamma = H_\gamma/\alpha$ yields

$$\frac{d\ln E_\gamma}{dT} = -\,\alpha c\,e^i\,\partial_i \ln\alpha - \frac{1}{2}\,e^i\,e^j\,\partial_T\,q_{ij}. \qquad (253)$$

$$\frac{d\ln E_\gamma}{dX} = -\,\frac{1}{Q_D}\left(\alpha c\,e^i\,\partial_i \ln\alpha + \frac{1}{2}\,e^i\,e^j\,\partial_T\,q_{ij}\right). \qquad (254)$$

The first term reads a spatial clock gradient and the second reads the changing spatial response. This gives the photon-energy change per unit source exposure. Its distribution through the field is carried into the calculation by α and q_{ij}.

11.1.2 The homogeneous source-to-energy relation

In the realization used for the supernova calculation, $q_{ij} = a^2\,\delta_{ij}$ and α is spatially uniform. Spatial translation symmetry conserves p_i. Consequently

$$E_\gamma = \frac{c\,|p|}{a}\,,\; \frac{d\ln E_\gamma}{dT} = -\,\frac{1}{a}\frac{da}{dT} = -\,\frac{Q_D}{3\,V_D}. \qquad (255)$$

The retained volume law identifies $dX = Q_D\,dT = d\,V_D$. With V_e denoting the material volume at emission, integration is then elementary:

$$\frac{d\ln E_\gamma}{dX}=-\frac{1}{3(V_e+X)},\ln\frac{E_{\gamma,o}}{E_{\gamma,e}}=-\frac{1}{3}\ln\frac{V_e+X}{V_e}. \quad (256)$$

$$\frac{E_{\gamma,o}}{E_{\gamma,e}}=\left(\frac{V_e}{V_e+X}\right)^{1/3},Z=1+z=\frac{a_o}{a_e}=\left(1+\frac{X}{V_e}\right)^{1/3}. \quad (257)$$

This is a direct source-exposure form of the photon-energy result. The clock factor cancels from this endpoint energy ratio, while its history remains in the ray's travel and distance integral. Rescaling a representative material domain at fixed density rescales V_D, Q_D and X together, leaving X/V_e and the observable ratio unchanged.

11.1.3 Arrival intervals and the area of the light beam

Received brightness also depends on how rapidly photons arrive and how the beam spreads. For a radial ray between fixed comoving endpoints, the null equation gives

$$d\chi=\frac{c\alpha\,dT}{a}=\frac{c\,dt}{a}=\frac{c\,dV_D}{a(V_D)[Q_D(V_D)/\alpha(V_D)]}. \quad (258)$$

Apply the same path integral to two neighboring emission times. Their comoving endpoint separation is unchanged, so differentiation of $\chi=c\int_{t_e}^{t_o}dt/a(t)$ gives

$$0=\frac{c\,dt_o}{a_o}-\frac{c\,dt_e}{a_e},\frac{dt_o}{dt_e}=\frac{a_o}{a_e}=Z. \quad (259)$$

The arrival relation also has an exact exposure form. Let $T_o=f(T_e)$ and let $X(T_e)$ be the exposure accumulated by the corresponding ray. Because $Q_D(T)$ is spatially uniform in this realization,

$$X(T_e)=\int_{T_e}^{f(T_e)}Q_D(T)\,dT,\frac{dX}{dT_e}=Q_{D,o}f'(T_e)-Q_{D,e}. \quad (260)$$

$$Z=\frac{\alpha_o}{\alpha_e}f'(T_e)=\frac{\alpha_o}{\alpha_e}\frac{dX/dT_e+Q_{D,e}}{Q_{D,o}}. \quad (261)$$

The derivative compares neighboring light histories. It connects the change in accumulated exposure to the spacing of arriving wavecrests. Equations (257) and (259) give the same Z for the homogeneous history.

For flat spatial slices, a narrow bundle emitted into solid angle $d\Omega_e$ occupies reception area $dA_o=a_o^2\chi^2\,d\Omega_e$. Denote the corresponding area distance by $D_G^2=dA_o/d\Omega_e$. Each photon contributes an energy factor Z^{-1} and the arrival rate contributes a second Z^{-1}. For isotropic proper luminosity L_e and transparent propagation,

$$D_G=a_o\chi,F_o=\frac{L_e}{4\pi D_G^2Z^2},D_L=Z\,D_G. \quad (262)$$

Combining the beam result with (258) puts the luminosity distance directly in source and clock variables:

$$D_L=Z\,a_o\,c\int_{V_e}^{V_o}\frac{dV_D}{a(V_D)[Q_D(V_D)/\alpha(V_D)]},F_o=\frac{L_e}{4\pi D_L^2}. \quad (263)$$

The normalized source response enters the complete distance kernel. Photon energy, arrival spacing and beam area are already included when this kernel is converted to flux. The calculation therefore uses this expression directly, without an additional exposure-dependent attenuation factor.

11.1.4 From the light calculation to the catalogue contrast

Set $a_o = 1$ and use $1 + z = a^{-1}$. The same distance can now be written in the redshift form used for the catalogue calculations:

$$D_L(z) = (1+z)\, c \int_0^z \frac{dz'}{H_{\cos}(z')}. \qquad (264)$$

To retain the released catalogue convention, the integral uses the Hubble-diagram redshift z_{HD} and its prefactor uses $1 + z_{HEL}$. Define the dimensionless rate $E(z) = H_{\cos}(z) / H_{\cos,0}$ and the dimensionless distance $d_L = H_{\cos,0} D_L / c$. The history equation in Section 10.2 then gives

$$E(z) = (1+z)^{3/2} \sqrt{\frac{1 + y(z)^2}{1 + y_0^2}}, d_L = (1 + z_{HEL}) \int_0^{z_{HD}} \frac{dz'}{E(z')}. \qquad (265)$$

The reference calculation keeps a constant clock factor in its normalized history. Its integral is analytic:

$$E_0(z) = (1+z)^{3/2}, d_{L,0} = 2(1 + z_{HEL})\left[1 - (1 + z_{HD})^{-1/2}\right]. \qquad (266)$$

For each target group, let L_i be the frozen weighted target-minus-nearby-reference contrast used in the calculation. Its weights sum to zero. If m denotes the released standardized supernova magnitude, the observed contrast and the SET expectation are

$$B_{obs,i} = 10^{-0.4 L_i [m - 5 \log_{10} d_{L,0}]}, \qquad (267)$$
$$B_{SET,i} = 10^{-0.4 L_i [5 \log_{10} (d_L / d_{L,0})]}, B_0 = 1.$$

The zero-sum contrast cancels a common magnitude or absolute-distance zero point. The result is a weighted geometric-mean brightness contrast between standardized supernovae at different redshifts. It is evaluated at every contributing catalogue redshift before the original group weights are applied. This gives the precise observable represented by the tables and figures that follow.

For the worked group, the unchanged reference, SET and observed contrasts are 1, 0.7066664790 and 0.7006560778, respectively. The source-exposure integration above recovers all fifteen archived group predictions, with maximum numerical differences 6.70×10^{-12} for Pantheon+ and 3.31×10^{-12} for DES-Dovekie. The catalogue inputs, source parameters, initial conditions, reference weights and covariance are unchanged. The bridge is therefore an explicit derivation of the same brightness calculation.

A separate integration of the source-radius equation and the null-distance relation recovers the same eight Pantheon+ and seven DES-Dovekie group predictions, with maximum absolute differences 7.06335×10^{-12} and 3.71803×10^{-12}, respectively. This checks a second mathematical route to those archived predictions; it introduces no new observations or adjusted inputs.

For the homogeneous spatially flat propagation coupling just derived, photon-number conservation, the endpoint energy ratio and the arrival-interval ratio give the following luminosity distance and bolometric flux. Normalize a_o=1:

$$D_L(z)=(1+z)c\int_0^z\frac{dz'}{H(z')},F_o=\frac{L_e}{4\pi D_L^2}. \quad (268)$$

In catalogue calculations, the integral uses z_{HD} and the prefactor uses $1+z_{\mathrm{HEL}}$, following the published redshift convention. The released brightnesses already include light-curve standardization and observational corrections [15, 16, 17, 18]. The test compares these standardized brightnesses across different supernovae, rather than the fading of one explosion over several days.

Define $E(z)=H(z)/H_0$ and a dimensionless luminosity distance $d_L=H_0D_L/c$. The tested history gives

$$E(z)=(1+z)^{3/2}\sqrt{\frac{1+y(z)^2}{1+y_0^2}},d_L=(1+z_{\mathrm{HEL}})\int_0^{z_{\mathrm{HD}}}\frac{dz'}{E(z')}. \quad (269)$$

The initial source-volume calculation has a constant clock factor, so its normalized history is $E_0(z)=(1+z)^{3/2}$. Its distance is

$$d_{L,0}=2(1+z_{\mathrm{HEL}})\left[1-(1+z_{\mathrm{HD}})^{-1/2}\right]. \quad (270)$$

This is the matter-only, spatially flat baseline with a constant clock factor. Its normalized distance shape is also the classical dust cosmology shape. The comparison asks whether the evolving SET source–capacity prescription accounts for the observed brightness departure from this baseline while the source inputs remain fixed. Agreement is assessed against catalogue-standardized brightness and redshift; the physical interpretation is considered separately. The plotted baseline is not a fitted ΛCDM model.

For each target group, let $L_i[x]$ mean its weighted mean of x minus the weighted mean in the fixed nearby reference group, $0.025<z_{\mathrm{HD}}<0.10$. Its weights sum to zero. Therefore an unknown constant luminosity or distance zero point cancels. If m denotes the released standardized magnitude or distance modulus, the plotted quantities are

$$B_{\mathrm{obs},i}=10^{-0.4L_i[m-5\log_{10}d_{L,0}]}, \quad (271)$$

$$B_{\mathrm{SET},i}=10^{-0.4L_i[5\log_{10}(d_L/d_{L,0})]}. \quad (272)$$

The initial prediction is $B_0=1$. A value below one means dimmer than the low-redshift-anchored initial curve. The predicted contrast uses the catalogue redshifts and the fixed source calculation; observed brightness does not enter its numerator. Every object is evaluated at its own redshift before averaging. These are geometric-mean relative fluxes, because averaging is performed in magnitudes.

11.2 Why the brightness departure motivated dark energy

Type Ia supernovae allow us to compare received brightness with the brightness predicted at a measured redshift. Their light curves are standardized to account for known luminosity differences, and nearby supernovae establish a common reference. The question is then whether the predicted distance–redshift relation accounts for the brightness of the more distant sample.

The original supernova studies found distant events to be systematically fainter than expected in the matter-only cosmologies they examined. With standardized luminosity L, the flux relation $F=L/(4\pi D_L^2)$ assigns a larger luminosity distance to a fainter source. Fits to the full distance–redshift relation favored histories with recent accelerated expansion, which a positive cosmological constant can produce in GR cosmology. Acceleration was a quantitative inference from the brightness data under those models, with observational systematics assessed [21, 22]. SET investigates whether its evolving collective burden, clock relation and

source-driven history can account for the brightness departure without inserting an additional dark-energy density.

The modern comparison below uses a specified matter-only, spatially flat baseline. It illustrates the brightness departure with the present catalogues; it is not a reconstruction of the original discovery analysis. For the Pantheon+ group near $z = 0.731$, the baseline predicts relative brightness $B_0 = 1$, while the observed value is $B_{\text{obs}} = 0.700656$. The observation is therefore 29.93% dimmer than this baseline after the nearby-reference normalization.

The SET history predicts $B_{\text{SET}} = 0.706666$, or 29.33% dimmer than the same baseline. The corresponding effective relative distance factors are

$$B_{\text{obs}}^{-1/2} = 1.1947\,, B_{\text{SET}}^{-1/2} = 1.1896\,. \quad (273)$$

These are group contrasts relative to the nearby reference, not absolute distances to a single supernova. The baseline predicts too much brightness, and the SET calculation reduces it by almost the amount required. Section 11.3 shows the source inputs and numerical steps. Its 42.72% and 0.86% discrepancies use observed brightness as the denominator; the dimming percentages above instead use the baseline.

The brightness departure that motivated dark-energy cosmology is also the observational target of this SET calculation. Here, the evolving collective burden and its clock conversion change the predicted distance–redshift relation without inserting an additional dark-energy density. The same prescription moves the predictions toward observation in all fifteen redshift groups. Across the full grouped comparison, the median absolute discrepancy falls from 38.53% to 2.09% in Pantheon+ and from 36.39% to 2.96% in DES.

11.3 One complete calculation

Measured inputs and the source calculation

Use the Pantheon+ group $0.65 \le z < 0.80$: 63 supernovae with weighted mean redshift 0.73085168, compared with the fixed 311-object nearby reference. This is the closest central group match in the run; the full sample follows in Figure 9 and Table 6. The present-day source inputs are

$$M_0 = 1.5005279083 \times 10^{53}\ \text{kg}\,, R_0 = 4.3992396698 \times 10^{26}\ \text{m}\,. \quad (274)$$

The radius is 46.5 billion light-years. These frozen benchmark inputs use a cosmology-conditioned radius and imply

$$V_0 = 3.5663295987 \times 10^{80}\ \text{m}^3\,, Q_0 = 5.1893886387 \times 10^{62}\ \text{m}^3\text{s}^{-1}\,, \quad (275)$$

$$H_{b0} = 14.96662955\ \text{km}\,\text{s}^{-1}\text{Mpc}^{-1}\,, u_0 = \frac{3GM_0}{R_0 c^2} = 0.7598922704\,. \quad (276)$$

Thus $\alpha_0 = 0.4900078873$, $y_0 = 1.7789876897$, and $H_0 = 30.54365029\ \text{km}\,\text{s}^{-1}\text{Mpc}^{-1}$. The initial condition is fixed before calculating the brightness contrast.

Expected matter only baseline

Evaluate the initial distance expression at every target and reference redshift. After the common reference normalization, this baseline has $\Delta m_0 = 0$ and expected relative brightness $B_0 = 1$. The value one is a defined comparison scale, not an absolute flux measurement.

Expected SET calculation

Integrate the burden history from the fixed initial condition, calculate the distances, and average the magnitude differences using the same weights. This gives

$$\Delta m_{\mathrm{SET}} = L[5\log_{10}(d_L / d_{L,0})] = 0.37696377 \ \text{mag}, \quad (277)$$

$$B_{\mathrm{SET}} = 10^{-0.4(0.37696377)} = 0.70666648. \quad (278)$$

Observed brightness and comparison

The catalogue gives $\Delta m_{\mathrm{obs}} = 0.38623777 \pm 0.03403476$ mag, equivalent to $B_{\mathrm{obs}} = 0.70065608$ with marginal 1σ interval 0.67903317–0.72296754. Define the signed discrepancy as $100(B_{\mathrm{pred}} / B_{\mathrm{obs}} - 1)$.

Table 4. Initial prediction SET prediction and observed brightness

Calculation	Relative brightness	Discrepancy
Initial matter-only baseline	1.00000000	+42.72%
SET source-clock calculation	0.70666648	+0.86%
Observed group	0.70065608	—

The remaining difference is 0.00927399 mag, or 0.27 marginal standard deviations. The SET calculation reduces a 42.72% central discrepancy to 0.86%, placing the prediction comfortably inside the observed 1σ interval.

11.4 The same calculation across both catalogues

Figure 9 shows the unchanged calculation for every fixed group. No group is removed for disagreeing. The orange line connects the group predictions; it is not a curve fitted through the observations.

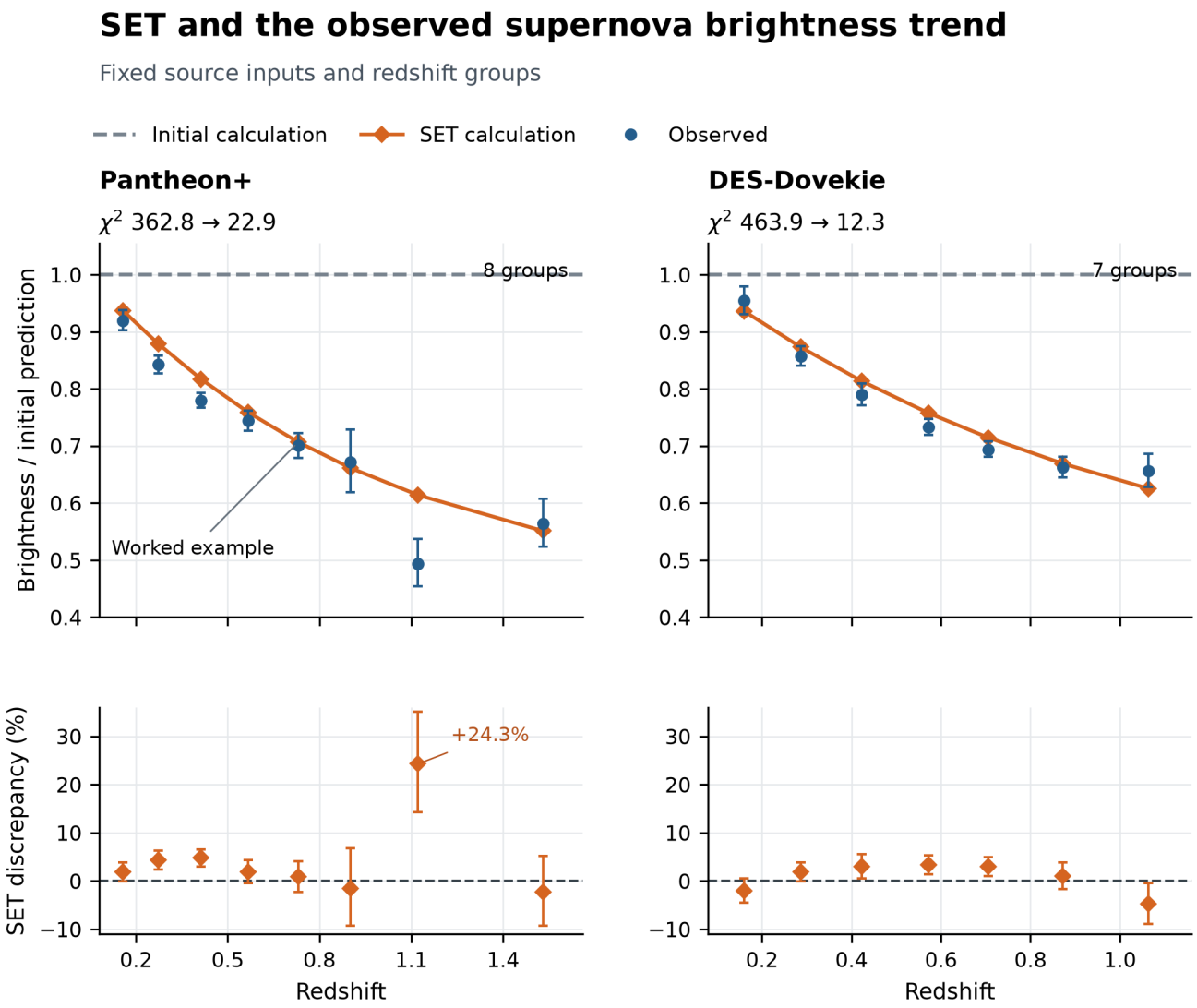


Figure 9. Relative supernova brightness and remaining SET discrepancy. Blue bars are marginal 1σ observational intervals. The initial source-volume prediction is one after reference normalization. The lower panels retain the sign of the remaining discrepancy. Formal comparisons use the full correlated magnitude covariance, including the common nearby reference. Source inputs are held fixed.

Table 5. Grouped comparison statistics with the fixed source prescription

Comparison	Pantheon+	DES-Dovekie
Initial χ^2	362.75	463.91
SET χ^2	22.87	12.34
Median absolute error before	38.53%	36.39%
Median absolute error after	2.09%	2.96%

All fifteen central discrepancies decrease in magnitude. Eight group predictions lie within their marginal 1σ intervals: five of eight in Pantheon+ and three of seven in DES. These counts describe grouped means, not the fraction of individual supernovae explained. The two catalogue statistics must not be added as independent evidence.

11.5 Complete grouped results

Table 6 retains all plotted values. N is the number of distinct catalogue object identifiers in the target group; $\bar{z}$ is its weighted mean redshift. The observed interval is marginal 1σ. The initial prediction is one in every row. Before and after are signed percentage discrepancies relative to the observed central brightness.

Data	$\bar{z}$	N	Observed and 1σ interval	SET	Before	After
P	0.1575	207	0.919799 [0.902394, 0.937540]	0.936955	+8.72%	+1.87%
P	0.2729	380	0.842757 [0.827165, 0.858642]	0.879009	+18.66%	+4.30%
P	0.4131	163	0.779774 [0.766711, 0.793059]	0.817031	+28.24%	+4.78%
P	0.5675	117	0.744370 [0.726833, 0.762331]	0.758733	+34.34%	+1.93%
P	0.7309	63	0.700656 [0.679033, 0.722968]	0.706666	+42.72%	+0.86%
P	0.9018	5	0.671954 [0.619381, 0.728991]	0.661347	+48.82%	-1.58%
P	1.1217	9	0.494130 [0.454592, 0.537105]	0.614079	+102.38%	+24.27%
P	1.5302	16	0.563875 [0.523682, 0.607152]	0.551170	+77.34%	-2.25%
D	0.1608	44	0.954695 [0.930655, 0.979356]	0.935815	+4.75%	-1.98%
D	0.2863	309	0.857211 [0.840490, 0.874264]	0.873224	+16.66%	+1.87%
D	0.4225	404	0.789908 [0.771064, 0.809211]	0.813755	+26.60%	+3.02%
D	0.5726	467	0.733203 [0.719298, 0.747377]	0.757433	+36.39%	+3.30%
D	0.7059	265	0.694030 [0.680790, 0.707528]	0.714577	+44.09%	+2.96%
D	0.8721	113	0.662420 [0.644568, 0.680767]	0.669145	+50.96%	+1.02%
D	1.0639	17	0.656663 [0.628014, 0.686618]	0.625400	+52.29%	-4.76%

Table 6. Full results of the retained central uniform-burden calculation. P denotes Pantheon+ and D denotes DES-Dovekie. Values are rounded for presentation; calculations use the full stored precision.

The target intervals are [0.10, 0.20), [0.20, 0.35), [0.35, 0.50), [0.50, 0.65), [0.65, 0.80), [0.80, 1.00), [1.00, 1.30), and [1.30, 2.40). DES has no object in the final interval. The nearby references contain 311 Pantheon+ objects and 201 DES objects. Pantheon+ rows below $z = 0.025$ are not used in this fixed-reference plot.

Within each group and reference, weights use the published marginal inverse variance, with duplicate rows sharing object weight. The full statistical and systematic covariance is then propagated. This weighting was adopted in a documented amendment after an equal-object calculation exposed extremely uncertain DES entries; the redshift boundaries were unchanged. No weight or inclusion rule depends on a SET residual.

The sparse Pantheon+ group near $z = 1.122$ remains a visible exception to close central agreement: the discrepancy falls from +102.38% to +24.27%. Its nine objects remain in the table and figure.

11.6 Covariance and verification

The plotted measurements come from released standardized supernova products, not raw photon counts. Pantheon+ uses the published SN-only cut $z_{\mathrm{HD}} > 0.01$ before the fixed reference and group definitions are applied. DES uses the 1,820-row Dovekie release. The plotted target and reference groups contain 1,271 distinct Pantheon+ identifiers and all 1,820 DES identifiers, with overlap between catalogues.

For the formal statistic, stack the weighted contrasts into a matrix L. If C is the released magnitude covariance and r the residual vector, the calculation is

$$C_{\mathrm{contrast}} = L\,C\,L^{\mathrm{T}},\ \chi^2 = r^{\mathrm{T}} C_{\mathrm{contrast}}^{-1} r. \quad (279)$$

The DES release supplies an inverse covariance, which is solved against to form the contrast covariance. It is not treated as the covariance itself. Error bars in Figure 9 are the marginal diagonal uncertainties, whereas this statistic includes correlations. The uncertainty bands are conditional on the frozen SET inputs; they are not model-parameter uncertainty bands.

A separate unbinned check retained the full released covariance and analytically profiled one constant luminosity intercept. Its initial and SET statistics were 2101.15 and 1426.45 for Pantheon+, and 2558.52 and 1647.64 for DES. This is a different check from the fixed-reference figure: it fits one nuisance zero point, but no SET parameter. It supports a broad improvement without establishing that every object improves.

The companion verification script recomputes all fifteen predictions directly from the fixed history and the catalogue redshifts, then checks them against the archived table. The forward predictions agree to better than 10^{-12} in relative brightness. The source history has also been cross-checked against an analytic integral of its evolution equation. The source records, plot and input hashes are retained with this section.

The research compared several physical prescriptions, including central and boundary burden readouts, before retaining this expression. Source parameters were held fixed, but this was exploratory development rather than a blind prediction. The reported χ^2 values describe these comparisons and are not corrected for the search across candidate histories.

The comparison uses relative brightness, for which the common absolute distance scale cancels. It therefore tests the redshift dependence of the prediction without independently validating its absolute scale. Additional distance–redshift and domain checks for this prescription are documented in Appendix J.

The original research also checked exact catalogue overlap. Removing the 339 shared identifiers from both catalogues, while retaining the same response and redshift intervals, leaves a reduction in the correlated brightness statistic from 265.77 to 18.54 in Pantheon+ and from 187.09 to 8.82 in DES. The latter reference then contains only ten objects, and common calibration remains. This supports persistence of the improvement, but is not a blind independent replication.

11.7 The unchanged calculation on additional supernovae

The preceding distance expression can be tested on supernovae that were absent from the earlier inputs. Union2.1 supplies released standardized distances and a full statistical-plus-systematic covariance for 580 objects [45, 46]. An identifier, alias and sky-position audit finds 302 objects distinct from the earlier Pantheon+ and DES-Dovekie inputs. Excluding one object previously used in the spectral-aging calculation and three predeclared redshift-quality conflicts gives 298 eligible objects: 23 nearby references, 260 targets

and 15 below the fixed reference window. Union2.1 is an older compilation; the additional information here consists of distinct objects newly evaluated with the retained SET calculation.

The source mass, radius, $\xi = 3/2$, initial condition, evolution equation and distance kernel remain those of Sections 10 and 11.3. The original fifteen predictions were reproduced before the additional calculation. The nearby interval $0.025 < z < 0.10$, target intervals and marginal inverse-variance weighting rule also remain fixed. No SET parameter or target intercept is fitted. Union2.1 supplies one CMB-frame redshift, used as released in both the integral and prefactor; it does not supply the separate z_{HD} and z_{HEL} columns used above.

The observed quantity is the released standardized distance modulus μ, transformed with the same target-minus-reference contrast L. Thus B_{obs} is a geometric relative-brightness contrast inferred from the standardized magnitudes. The SET value is calculated from the unchanged distance history at each contributing redshift, and only then averaged with the fixed weights. Published light-curve standardization, calibration and selection are inherited. These distinct objects therefore extend the test while retaining some common calibration and standardization information.

11.7.1 The same redshift interval with distinct objects

Take the interval $0.65 \le z < 0.80$ used in the earlier worked example. It now contains 21 distinct target objects with weighted mean redshift 0.72225118, compared with the 23 distinct nearby references. Evaluating the unchanged distance ratio at each object gives

$$\Delta m_{SET} = L\left[5\log_{10}\left(d_L / d_{L,0}\right)\right] = 0.35956113 \ \ mag,$$

$$B_{SET} = 10^{-0.4\left(0.35956113\right)} = 0.71808449.$$

The released distances give $\Delta m_{obs} = 0.41580043 \pm 0.07715345$ mag, or $B_{obs} = 0.68183585$ with marginal 1σ interval $[0.63506540, 0.73205078]$. The baseline remains $B_0 = 1$. Using the same observed-brightness denominator as Section 11.3, the central discrepancy falls from $+46.66\,\%$ to $+5.32\,\%$. The remaining magnitude residual is 0.05623930 mag, or 0.729 marginal standard deviations. This interval is used to continue the earlier example; all seven occupied intervals are retained below.

11.7.2 Grouped results and the individual-target check

Figure 10 and Table 7 retain every populated interval. All seven grouped central discrepancies decrease. The median absolute relative-brightness discrepancy falls from $37.27\,\%$ to $4.85\,\%$. Six group predictions are within the marginal 1σ intervals; the $0.35 \le z < 0.50$ group is 1.73 marginal standard deviations from its central value. The predeclared $1.30 \le z < 2.40$ interval has no eligible object; the largest retained redshift is 1.265.

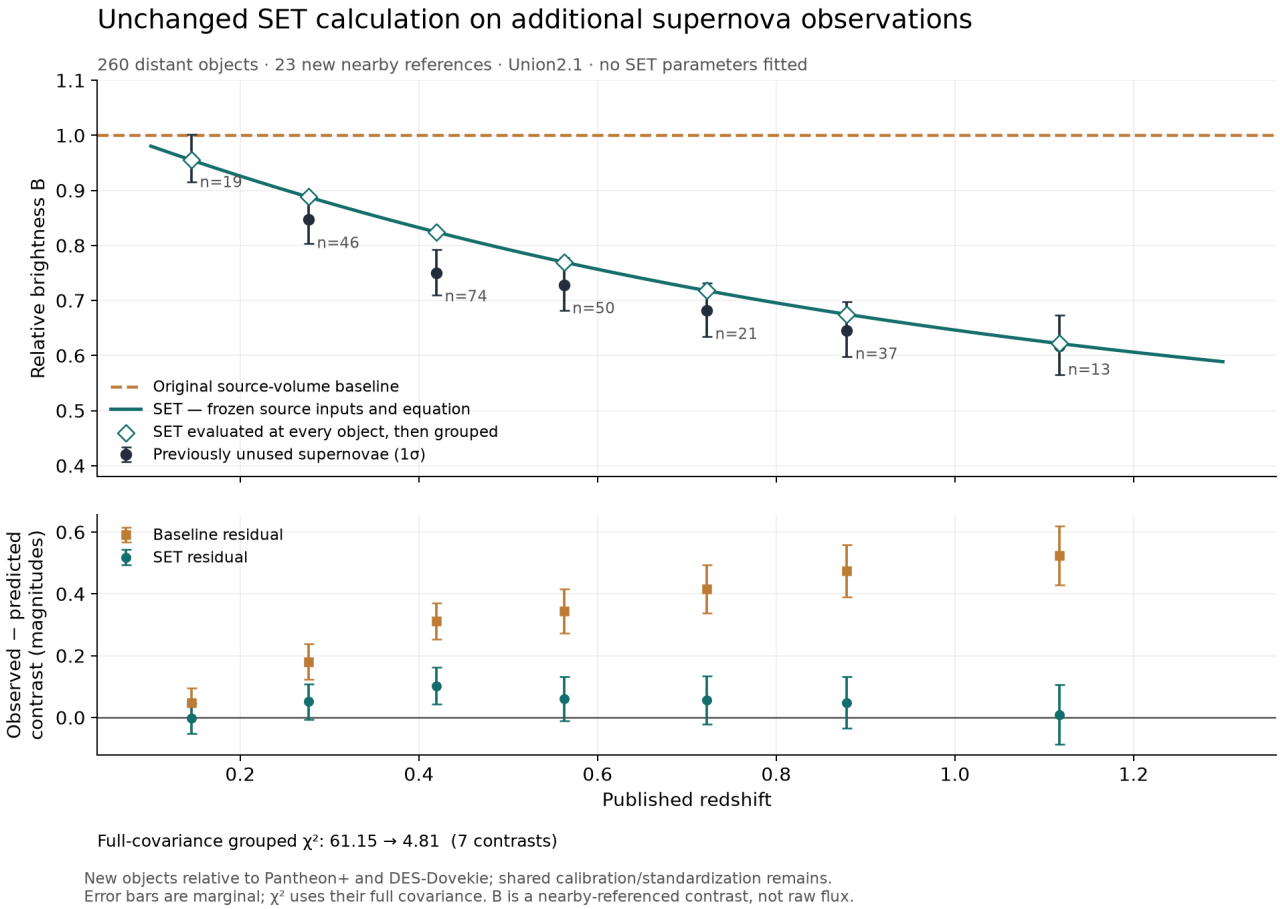


Figure 10. The unchanged SET distance calculation for 260 additional target supernovae and 23 distinct nearby references. Points and bars are standardized relative-brightness contrasts and marginal $1\,\sigma$ intervals. Diamonds use the prediction evaluated at every contributing object before grouping; the smooth line displays the same fixed distance history. The lower panel retains signed magnitude residuals. Formal statistics propagate the full covariance, including the common reference. Source parameters are held fixed.

Table 7. Additional Union2.1 objects under the unchanged SET history. The baseline is one in every row. Before and after are signed discrepancies relative to observed central brightness. Intervals include their lower boundary and exclude their upper boundary.

Redshift interval	N	Weighted z	Observed B [1σ]	SET B	Before	After
0.10–0.20	19	0.145360	0.957542 [0.915686, 1.001310]	0.955224	+4.43%	-0.24%
0.20–0.35	46	0.276420	0.847165 [0.803650, 0.893036]	0.888266	+18.04%	+4.85%
0.35–0.50	74	0.419552	0.750173 [0.710357, 0.792221]	0.824536	+33.30%	+9.91%
0.50–0.65	50	0.562441	0.728466 [0.682007, 0.778090]	0.769969	+37.27%	+5.70%
0.65–0.80	21	0.722251	0.681836 [0.635065, 0.732051]	0.718084	+46.66%	+5.32%
0.80–1.00	37	0.879155	0.645638 [0.597433, 0.697731]	0.675240	+54.89%	+4.58%
1.00–1.30	13	1.117121	0.616894 [0.564814, 0.673777]	0.622254	+62.10%	+0.87%

The full-covariance grouped statistic decreases from $\chi^2 = 61.1511$ to 4.8119 for seven contrasts. The conditional Gaussian goodness-of-fit probability for the fixed SET prediction is $p = 0.6829$. The individual-target calculation uses all 260 target-minus-reference contrasts, with their shared-reference covariance. Its

statistic decreases from 322.0996 to 255.6098, with $p = 0.5653$. Thus the retained distance shape is compatible with the released standardized distances and covariance at both aggregation levels.

For the individual targets, 202 of 260 absolute magnitude residuals decrease and 178 are inside their marginal 1σ intervals. The SET prediction is brighter than the observed central value for 158 targets and dimmer for 102. Figure 11 displays every target, including those whose residual grows. These counts describe correlated contrasts; the grouped and individual statistics summarize the same observations.

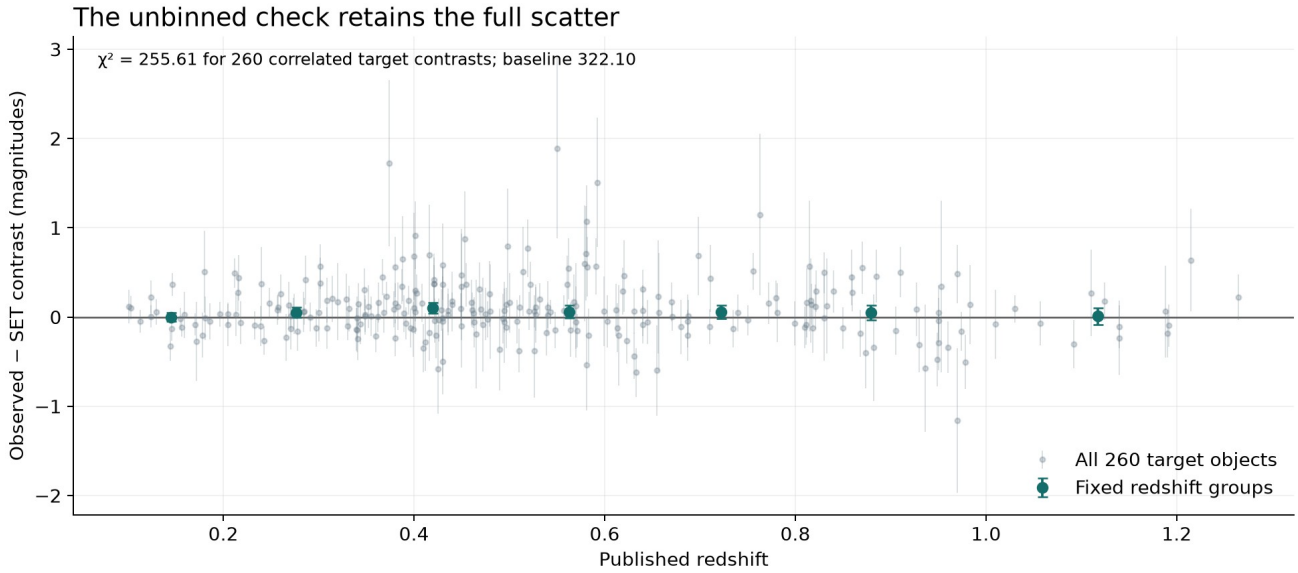


Figure 11. All 260 additional target-minus-nearby-reference magnitude residuals. Faint points retain the individual scatter and marginal errors; darker points show the fixed groups. Positive residual means the SET prediction is too bright. The covariance statistic includes the reference and released systematics.

Retaining all three redshift-conflict objects at their unchanged released values gives 263 targets, with grouped SET $\chi^2 = 5.2641$ and individual $\chi^2 = 259.2076$; the corresponding conditional probabilities are 0.6278 and 0.5545. The improvement therefore persists under the released-redshift sensitivity check.

The numerical verification reconstructs the contrasts directly from the original release and integrates the retained history by an independent separated-variable method. The per-object magnitude corrections agree within 1.26×10^{-10} mag. The calculation records retain the membership audit, all individual residuals, released covariance and original input hashes. Methodological details are given in the accompanying calculation record.

12 Physical interpretation and results

I began with clocks because every clock reading is itself a physical event. The capacity relation gives a common language for that event progression, motion and gravitational burden. The source law then connects the burden calculation to matter. What matters is following that connection all the way to a quantity that can be compared with observation.

12.1 The spherical construction

The revised spherical calculation combines the source law, common clock and probe energy, physical work-current balance and the displayed complete static energy. Within that class, the exact energy gap selects $S^2=2GM/r$ and $B=1/A$. The local vacuum variation shows how the same exterior applies outside an admissible spherical source. The resulting conserved orbital energy gives the exact orbit and perihelion equation. Finite and differential normalized-current identities preserve that motion under a change of reading.

The logical dependencies are explicit:

Table 8. Principles and derived relations in the spherical calculation

Quantity or relation	Status in this construction
Capacity relation and source law	Retained SET principles with explicit source domain
Probe energy and action	Selected within the common canonical clock/motion class
Physical work-current and complete static energy	Displayed field principle and energy functional
Radial conversion and exterior burden	Selected by the exact energy equality; local vacuum extension given
Finite and differential current normalization	Algebraic identities in the specified physical reading
Normalized orbital energy	Exact change of orbital parameter
Perihelion equation and leading advance	Derived from the selected field and probe dynamics

The solved clock and radial relations are mathematically equivalent to the Schwarzschild exterior. Section 3 reaches them through the stated SET common-energy construction and its exact selection test; Appendix A retains the earlier universal-family route. Appendix D obtains the equivalent free-fall representation after the exterior is solved.

Probe energy and field energy have complementary roles here. The common-energy requirement selects the probe action within its local canonical class. The complete static functional then supplies the field-energy term, and agreement between complete and exterior source energy selects the radial response. The full boundary-term proof is in Appendix K; the causal extension is described in Section 6 and Appendix M.

The resulting spherical derivation has a stated domain: positive-clock, stationary spherical exteriors with the displayed local energy and current construction, an asymptotically unloaded reference, and the given source matching. Within this domain the exterior response and perihelion follow without fitting an orbital observation. The capacity identity gives the common reading; the retained field principle and energy functional specify its dynamical realization.

12.2 Collective sources and received light

For collective systems, source arrangement enters through the proposed effective boundary area. The original twenty examples and 63 additions connect adopted baryonic mass and image geometry to observed speeds, giving 83 displayed examples with a median absolute central discrepancy of 5.76%. The latest 21 examples come from 71 eligible new calculations, with the complete outcomes retained. Among the original nineteen matched Newtonian comparisons the SET central calculations are closer; the expanded plot now includes 26 available resolved baryonic comparisons. The 83-example figure shows recorded observational ranges, while the coupled source-input propagation remains explicitly scoped to the first 62 examples.

Appendix F supplies a stationary source equation for a prescribed response and the work associated with changes in that response. The source-boundary algebra, the galaxy comparisons and this field formulation are the results presented here. I invite independent reproduction using the stated inputs and calculation record.

SET's physical proposal is that the baryonic sources participate in the same field capacity through which clocks, motion and light are realized. In this calculation, the source law fixes a rate per reference time. The collective burden determines how that reference rate is read by physical clocks. The moving front changes the burden history, and the history changes the integrated distance associated with a measured redshift. Photon energy, arrival rate and geometric spreading then determine the brightness.

The specific realization tested here includes a changing physical scale a. The brightness change comes from the coupled history of source-driven motion, clock progression and light propagation. No additional dark-energy density is inserted into these equations. A static delay along the same path would not by itself give this result: the operative quantity here is the evolving distance–redshift relation.

This connects to SET's broader motivation. The galaxy calculation investigates whether baryonic organization can alter the gravitational response through collective burden and effective area. Here the question concerns the history of a collective response and the light received through it. The source–capacity connection motivates both investigations, while the specific mathematical assumptions and observations differ.

The empirical value is the breadth of the improvement. A single fixed expression reduces the discrepancy in every group across two published compilations. The data were included by redshift and published quality information, rather than agreement with SET. The result concerns a trend supported by many observations, rather than one fortunate match. The median absolute central discrepancy falls from 38.53% to 2.09% in Pantheon+ and from 36.39% to 2.96% in DES.

The result presented here is that one fixed SET prescription moves the predicted relative brightness toward observation in all fifteen redshift groups. Its physical significance is the demonstrated connection between the proposed source–capacity mechanism and the observed brightness trend. Whether this mechanism changes the inferred cosmic expansion history must be assessed through its predictions for observable quantities. The same expression is now tested on 260 additional target supernovae with 23 distinct nearby references. All seven populated groups improve, and the full-covariance statistic for the individual contrasts is 255.61 for 260 contrasts, with no fitted SET parameter or target intercept.

The connection I am investigating is that the organization of baryonic sources and the history of their collective burden can affect what we infer from motion and light. In the galaxy application, the effective area enters a proposed boundary response. In the cosmological application, the source law, clock conversion and evolving collective region enter a distance–redshift history. The observables differ, and their specific response prescriptions are displayed, but the physical question is shared.

The calculations give that question numerical form: a selected spherical exterior and its perihelion motion, a rotating strong-field construction, an image-area comparison with galaxy speeds, and a source-clock

calculation of supernova brightness. I invite readers to reproduce these steps, examine the physical correspondences and investigate whether the source–capacity account can be developed across these scales.

The source-exposure formulation makes the physical calculation continuous. Baryonic matter supplies Q_D; the collective burden determines the clock factor in the stated history; and the common photon coupling converts that history into energy shifts, arrival intervals and beam spreading. The combination yields the received brightness used in the comparison. The evolving volume and the normalized source reading participate together, with no fitted dark-energy density or separately chosen dimming factor in this calculation.

This is the same research motivation as the galaxy application expressed through a different observable. Orbital motion samples a spatial collective response, whereas received supernova brightness samples the source history through clocks and light. The derivations above show exactly which source and response quantities enter each stage, so the proposed common mechanism can be examined through its equations and numerical consequences.

12.3 Data and calculation records

The expanded galaxy archive additionally preserves all 83 displayed central rows, the complete new 131-identity attempt ledger, all 71 eligible calculated outcomes, 13 separate calculated diagnostics and independent source/profile audits. The first 62 source-input ensembles remain unchanged; no additional propagated ensembles are assigned to the new 21 examples.

The additional-supernova record includes the strict 298-object input ledger, released covariance, all 260 individual target residuals, the 301-object redshift sensitivity, the frozen protocol and the independent analytic-history audit.

The research supplement accompanying Draft 48, SET-research-supplement.md, preserves the catalogue context and comparison controls. The source and calculation bundle contains the twenty central rows, sensitivity ranges, observational provenance, archived input files, figure data and the scripts used to build and check this manuscript. It also includes the nineteen matched Newtonian component calculations and the conditional input-draw record used in Figure 5. The calculation record also includes the boundary-applicability audit, distinguishing the prescribed exterior continuations from the separately reconstructed local field of NGC 3198. Appendices I and E give the numerical-check summary and the inputs required to trace the displayed examples. These materials are part of the research record; a public archival identifier can be assigned with submission.

The numerical checks accompanying this derivation include exact algebraic identities, direct orbit integrations, independent angular quadratures and the finite clock integrals in Appendix C. They validate the displayed calculations under their stated premises. They do not constitute additional astronomical measurements.

The derivation, galaxy and cosmological records remain available with their original manuscripts. The combined numbering used here is accompanied by an equation and section correspondence record. The numerical tables and source figures are retained from the stated calculations; combining them in one paper does not make their datasets independent. The primary catalogues and observational sources are listed in the references.

Appendix A Earlier source-family and probe-calibration controls

The revised main derivation selects the exterior through complete-energy accounting. The following earlier route is retained as an independent conditional control: it assumes a universal response function across a specified family of source surfaces. Its algebra remains useful, while the main proof no longer needs that family assumption.

A.1 Earlier source-surface and universal-response route

Let the source have mass parameter M and physical surface radius R. The source law uses the spherical reference volume and the surface burden:

$$V_{\mathrm{ref}}=\frac{4\pi R^3}{3},\ Q_\Omega^2=24\pi GM\,V_{\mathrm{ref}}=32\pi^2 GM\,R^3. \quad (280)$$

$$Q_\Omega=4\pi R^2 S(R). \quad (281)$$

Q_Ω is assigned to this source domain. M denotes the total source mass-energy parameter used consistently in the boundary and exterior equations. The reference volume is the volume specified by the source law; it is not replaced by an integral over an as-yet unknown physical radial measure.

For the positive source branch, equation (280) is $Q_\Omega=\sqrt{32\pi^2 GM\,R^3}=4\pi\sqrt{2GM\,R^3}$. Its squared form and the ratio $Q_\Omega^2/(24\pi V_{\mathrm{ref}})=GM$ are algebraic expressions of the same law. The surface relation places that source law in the capacity budget through $S(R)=Q_\Omega/(4\pi R^2)$. A change to receiving-clock units, treated in Section 4.1, is a separate conversion of the readout.

Squaring equation (281) and comparing with equation (280) gives

$$16\pi^2 R^4 S(R)^2=32\pi^2 GM\,R^3,\ S(R)^2=\frac{2GM}{R}. \quad (282)$$

$$A_R=1-\frac{R_s}{R},\ R_s=\frac{2GM}{c^2},\ R>R_s. \quad (283)$$

This fixes the burden at an actual source surface. We next determine how the same source establishes the exterior response. Parameterize the radial factor by

$$\zeta^2=\frac{\mathscr{H}(A)}{A},\ \mathscr{H}(A)>0,\ \mathscr{H}(1)=1. \quad (284)$$

We take $\mathscr{H}$ to be positive, continuous and universal across the specified stationary spherical source family, with the differentiability required for the motion equations. Its only independent argument in this class is the local capacity A. The class excludes separate dependence on source size, density gradients, history or an additional field. This restriction is part of the proposed spherical response law; spherical symmetry alone does not impose it.

The unloaded limit has $A\to 1$ and $\zeta\to 1$ at infinity. The allowed source family is assumed to span a continuous interval of surface capacities extending to this limit. Section A.3 uses that common family to determine $\mathscr{H}$; it does not move the boundary of one fixed source through vacuum.

A.2 Earlier stationary-event calibration

The capacity relation assigns a clock-progression factor β to each proposed probe path. For a stable clock, calibrated event accumulation along that path is proportional to $\int \beta\, dT$. We use this accumulation to define the action of a structureless test body with constant rest mass and no additional direct field coupling.

The motion principle is stationarity under path variations with fixed endpoint events. With a positive constant κ carrying energy units, write

$$I_p = -\kappa \int \beta\, dT\,, \delta I_p = 0. \quad (285)$$

Each interval contributes the same action per unit of calibrated event progression. This is the minimal coupling adopted here. The invariant determines β; stationarity determines which path the free probe follows. It does not require every physical path to maximize an event count.

Calibrate the coefficient with the probe's measured inertial mass. In the unloaded limit $S = 0$ and $\zeta = 1$, write $V_{\mathrm{rel}} = v$ and expand at low speed:

$$L_p = -\kappa \sqrt{1 - \frac{v^2}{c^2}} = -\kappa + \frac{\kappa v^2}{2c^2} + O\left(\frac{\kappa v^4}{c^4}\right). \quad (286)$$

The kinetic term must be $m_p v^2 / 2$. Therefore

$$\frac{\kappa}{c^2} = m_p\,, \kappa = m_p c^2. \quad (287)$$

The calibrated probe action is consequently

$$I_p = -m_p c^2 \int \beta\, dT\,, L_p = -m_p c^2 \beta. \quad (288)$$

This earlier calibration fixes the coefficient without using an orbital precession. Its weak-field potential and acceleration are displayed in Section 3.2 of the main derivation.

A.3 The earlier family consistency proof

Combine the work-current expression with the complete-screen law:

$$\frac{r^2 c^2}{2\sqrt{\mathscr{H}(A)}} \frac{dA}{dr} = GM. \quad (289)$$

Separate variables and integrate from the actual surface R to the unloaded exterior limit:

$$\frac{dA}{\sqrt{\mathscr{H}(A)}} = \frac{R_s}{r^2} dr\,, \quad (290)$$

$$\int_{A_R}^{1} \frac{du}{\sqrt{\mathscr{H}(u)}} = R_s \int_R^{\infty} \frac{dr}{r^2} = \frac{R_s}{R}. \quad (291)$$

The independently retained source-surface relation gives $R_s / R = 1 - A_R$. Thus agreement of the source and field requires

$$\int_{A_R}^{1} \frac{du}{\sqrt{\mathscr{H}(u)}} = 1 - A_R. \quad (292)$$

Require this identity for the continuous family of source surfaces, using the same function $\mathscr{H}$. Differentiating with respect to the family's boundary capacity gives

$$-\frac{1}{\sqrt{\mathscr{H}(A_R)}}=-1,\mathscr{H}(A_R)=1. \quad (293)$$

The fundamental theorem of calculus applies because $\mathscr{H}$ is positive and continuous. If the admissible family covers $A_{min}\le A_R<1$, it determines the response throughout that interval:

$$\mathscr{H}(A)=1,\zeta^2=\frac{1}{A}. \quad (294)$$

Each member of the family has its own actual source radius, reference volume and source readout. The differentiation compares those physical configurations. It does not equate a fixed Q_Ω to every exterior sphere's area times its local burden.

This earlier argument applies to the exterior capacities covered by its specified source family. Extending it below Amin requires a corresponding extension of that family; it does not establish the stability of arbitrarily compact sources. Section A.4 shows why agreement at only one source surface does not determine the entire universal function.

A.4 Controls on the universal family

Write $x=1-A_R$ and define the integrated response

$$F(x)=\int_0^x\frac{dt}{\sqrt{\mathscr{H}(1-t)}}. \quad (295)$$

The source-field correspondence requires $F(x)=x$. At only one chosen boundary $x=x_0$, this is one integral constraint. As an explicit control, take

$$F(x)=x+\eta x^2(x-x_0),x_0=\frac{1}{2},\eta=\frac{1}{5}. \quad (296)$$

$$F'(x)=1+\eta(3x^2-2x_0x),\mathscr{H}(1-x)=\frac{1}{[F'(x)]^2}. \quad (297)$$

This derivative is positive on $0\le x\le 1$. The response has $\mathscr{H}(1)=1$ and satisfies $F(x_0)=x_0$, yet

$$\mathscr{H}(1-x_0)=\frac{1}{(1+\eta x_0^2)^2}=0.90702948\ne 1. \quad (298)$$

This is a mathematical control, not a proposed complete alternative theory. It shows why one boundary agreement does not select a function. Requiring $F(x)=x$ on a continuous interval instead gives $F'(x)=1$ and hence $\mathscr{H}=1$ on that interval.

A second control follows an alternative through the actual field equation. Let $\mathscr{H}(A)=1-b(1-A)^2$, with $b=1/2$, and put $y=R_s/r$ and $x=1-A(r)$. The physical-current equation gives

$$\frac{dx}{dy}=\sqrt{1-bx^2},x(0)=0,x(y)=\frac{\sin(\sqrt{b}\,y)}{\sqrt{b}}. \quad (299)$$

At a source surface, the source law requires $x(y_R)=y_R$. The displayed alternative instead has $x<y$ for the relevant positive compactness values. It cannot retain both the same current law and the same continuous source-surface correspondence. If one keeps $x=y$ instead, its current is $r^2\mathcal{J}=GM/\sqrt{\mathscr{H}}$, which violates the specified current balance. The selection therefore uses the combined field and source premises, before any orbit is compared.

Appendix B Alternative radial response and exact orbits

To isolate what normalization does, hold the same exterior A=1−Rs/r but choose the flat-radial control ζ²=1. It shares the held Q⋆ reading with the selected solution while violating the complete-energy selection of Section 3.4. Appendix A.3 also tests it against the earlier source-family premises.

The exact auxiliary equation from Section 5.1 becomes

$$r_{\sigma\sigma}-r\,\varphi_\sigma^2=-\frac{GM}{(r-R_s)^2},\quad r^2\varphi_\sigma=\ell_\sigma. \qquad (300)$$

$$w''+w=\frac{GM}{\ell_\sigma^2(1-R_s w)^2}. \qquad (301)$$

This is exactly the spatial orbit of the earlier shifted-potential calculation, provided its angular momentum is identified as $\ell_\sigma=j/\varepsilon$ and its time is interpreted as σ. The match holds beyond the leading perihelion coefficient. Calling that auxiliary force a force in reference time T would be a different dynamical prescription.

For a direct comparison, choose the same two areal turning radii in both branches. Their conserved j and ε follow from the same turning-point equations because the radial velocity vanishes there. The total-angle integrand differs through the radial response:

$$\frac{d\varphi}{d\xi}=\frac{\sqrt{\mathscr{H}(A(\xi))}}{\sqrt{1-2(3+e\cos\xi)/P}},\quad \mathscr{H}=A\zeta^2. \qquad (302)$$

The selected branch has $\mathscr{H}=1$; the flat-radial control has $\mathscr{H}=A$. At $e=0.2$, independent orbit integration and quadrature give:

Table 9. Alternative radial response and exact orbit checks

P	Radial response	Advance from quadrature rad	Advance from orbit integration rad
100	$\zeta^2=1$	0.1322932660	0.1322932660
100	$\zeta^2=1/A$	0.1974475160	0.1974475160
1000	$\zeta^2=1$	0.0126295320	0.0126295320
1000	$\zeta^2=1/A$	0.0189349966	0.0189349966

The agreement of two formulations within the flat-radial branch is exact, while the two radial branches give different accumulated azimuth. The difference is not removed by a consistent change of clock parameter. It is why the field-response selection must precede the orbital interpretation of $Q_\star$.

B.1 The weak field contribution of radial response

The leading difference can also be seen directly in the action. For this comparison only, write $\zeta^2 = A^{-\lambda}$, with $\lambda = 0$ for the control and $\lambda = 1$ for the selected response. Let $U_N = GM / r$ and $v^2 = v_r^2 + v_t^2$, with these coordinate velocities taken per T. Expansion gives

$$\frac{L_p}{m_p} = -c^2 + \frac{v^2}{2} + U_N + \frac{L_1}{c^2} + O\left(c^{-4}\right), \quad (303)$$

$$L_1 = \frac{v^4}{8} + \frac{U_N v^2}{2} + \lambda U_N v_r^2 + \frac{U_N^2}{2}. \quad (304)$$

To see the advance without inserting its coefficient, expand the exact general orbit equation at fixed $p_0 = j^2 / (GM)$. With $\ell_g = GM / c^2$, the result is

$$w'' + w = \frac{1}{p_0} + \ell_g \left[(1 + 2\lambda) w^2 + \frac{2(1-\lambda) w}{p_0} + (1-\lambda)(w')^2 \right] + O\left(\frac{\ell_g^2}{p_0^3} \right). \quad (305)$$

Substitute $w_0 = (1 + e\cos\varphi) / p_0$ into the first-order terms. The resonant coefficient is $2(2+\lambda) e \ell_g / p_0^2$. The same resonance calculation used in Section 5.4 therefore gives

$$\Delta\varphi = \frac{2\pi(2+\lambda) GM}{p c^2} + O\left(\frac{G^2 M^2}{p^2 c^4} \right). \quad (306)$$

Table 10. The perihelion contribution of the radial response

Specified radial response	Leading coefficient	Mercury arcsec per century
$\zeta^2 = 1$	$4\pi GM / (p c^2)$	28.6538
$\zeta^2 = 1 / A$	$6\pi GM / (p c^2)$	42.9807

The radial term in equation (304) vanishes instantaneously at periapsis because $v_r = 0$, yet contributes to the finite orbit as the planet approaches and leaves it. A calculation confined to the peak readout cannot capture that contribution. Splitting the total advance into time and spatial pieces depends on the chosen representation; the accumulated apsidal angle is the comparison quantity.

Responses differing only at higher weak-field order can preserve the leading 6π coefficient while changing exact orbits. Agreement with the leading Mercury number alone is therefore weaker than the complete-energy selection in Section 3.4; Appendix A.3 gives the earlier family-based control.

Appendix C Receiving clock accumulation around perihelion

The normalized readout is greatest at perihelion, but its accumulated effect also depends on the time spent there. For a held clock and a moving clock, respectively, the leading rate changes are

$$\frac{Q_\star}{Q_0}-1=\frac{GM}{r\,c^2}+O\left(c^{-4}\right), \quad (307)$$

$$\frac{Q_{\mathrm{rec}}}{Q_0}-1=\frac{1}{c^2}\left(\frac{GM}{r}+\frac{v^2}{2}\right)+O\left(c^{-4}\right). \quad (308)$$

Use the leading Kepler orbit, which is sufficient for these first-order clock effects. If E denotes eccentric anomaly and $n=\sqrt{GM/a^3}$, then

$$r=a\left(1-e\cos E\right), dT=\frac{1-e\cos E}{n}\,dE. \quad (309)$$

The radius dependence cancels from the leading gravitational clock-deficit integral:

$$\Delta T_{\mathrm{h}}\equiv\int\left(1-\alpha\right)dT\simeq\frac{GM}{a\,c^2\,n}\Delta E. \quad (310)$$

Thus the deficit is uniform per eccentric-anomaly increment at this order, although the instantaneous rate change is larger near perihelion. For a full orbit of period $P_T=2\pi/n$, the gravitational and moving-clock deficits are

$$\Delta T_{\mathrm{h}}\simeq P_T\frac{GM}{a\,c^2}, \Delta T_{\mathrm{moving}}\simeq\frac{3}{2}P_T\frac{GM}{a\,c^2}. \quad (311)$$

The second expression follows from $v^2=GM\left(2/r-1/a\right)$. With the Mercury inputs of Section 5.6, the self-consistent Kepler period is 87.96947 days. The resulting full-orbit deficits are 0.193806 seconds and 0.290710 seconds. These are calculated clock comparisons in the isolated solar model, not observations of a clock on Mercury.

A sum of held-clock increments along successive orbital locations is not the reading of one clock held at a fixed radius. The moving-clock quantity does follow the model orbit. Mercury's own gravitational field, other bodies and corrections beyond the stated approximation are not included in these numerical comparisons.

C.1 The finite interval around perihelion

Define a finite passage by true anomaly between -30° and $+30^\circ$. Its conversion to eccentric anomaly is

$$\tan\frac{E}{2}=\sqrt{\frac{1-e}{1+e}}\tan\frac{\varphi}{2}. \quad (312)$$

The interval lasts 9.6021 days. The gravitational clock deficit is 26.4235 milliseconds; including orbital motion gives 42.2698 milliseconds. It contains 10.9153 percent of the orbital time and 13.6340 percent of the leading gravitational clock deficit.

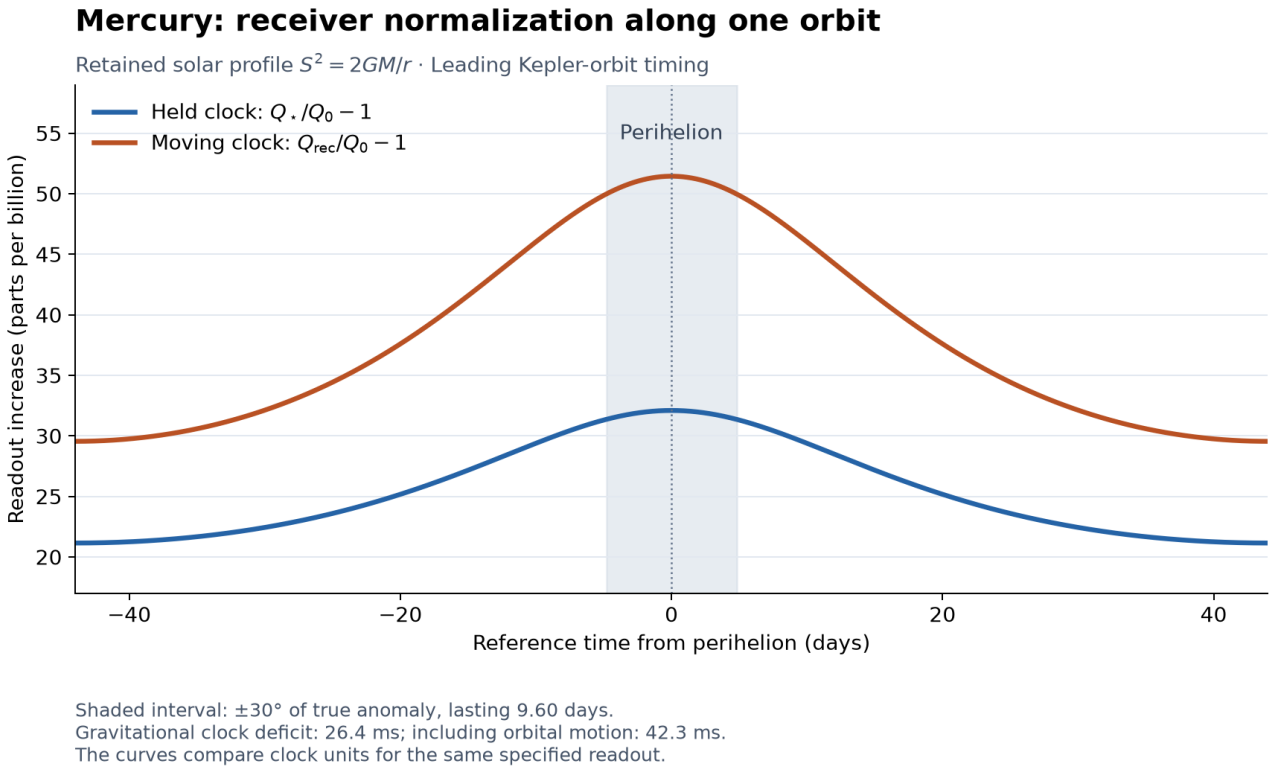


Figure 12. Receiving-clock normalization through one leading Mercury orbit in the retained solar field. The shaded region is the finite passage defined above. The held-clock increase is 32.10 parts per billion at perihelion and 21.15 at aphelion; the moving-clock increase is 51.45 and 29.55 parts per billion, respectively.

The time normalization conserves the same specified readout point by point:

$$Q_\star \, d\tau_h = Q_0 \, dT \, , \int Q_\star \, d\tau_h = \int Q_0 \, dT \, . \qquad (313)$$

This remains true when $Q_0(r)$ varies along the orbit. The larger rate per receiving-clock second does not provide an additional source quantity to add to the force. Its dynamical role is already carried by the normalized potential and time conversion derived in Section 5.1.

Appendix D The free fall representation

The main calculation used the capacity norm, physical work and source consistency. A coordinate representation provides an additional check. Before selecting $\mathscr{H}$, multiplication of the invariant by dT^2 gives

$$c^2 d\tau^2 = A c^2 dT^2 - \frac{\mathscr{H}}{A} dr^2 - r^2 d\Omega^2. \quad (314)$$

Write $h = \sqrt{\mathscr{H}}$. For inward marginal-energy radial motion, $\varepsilon = 1$ and $j = 0$ give

$$\frac{dT}{d\tau} = \frac{1}{A}, \frac{dr}{d\tau} = -\frac{S}{h}. \quad (315)$$

Define a time coordinate by

$$dt_f = dT + \frac{Sh}{A c^2} dr. \quad (316)$$

The coefficient depends only on radius, so this is locally an integrable change of stationary synchronization. Along the stated free-fall trajectories, $dt_f / d\tau = 1/A - S^2/(Ac^2) = 1$. Substitution into equation (314) gives

$$c^2 d\tau^2 = c^2 dt_f^2 - (h\,dr + S\,dt_f)^2 - r^2 d\Omega^2. \quad (317)$$

This identity holds for every positive $\mathscr{H}$ before selection. It identifies $d\chi = h\,dr = \alpha\,d\ell$ as the radial increment on these free-fall slices. The work-current defined in Section 3.3 is accordingly

$$\mathscr{J} = -\frac{1}{2}\frac{dS^2}{d\chi} = \frac{c^2}{2h}\frac{dA}{dr}. \quad (318)$$

After the source-family result sets $\mathscr{H} = 1$, equation (317) is the usual ingoing Painlevé–Gullstrand form of the selected exterior [14]. Its appearance is a consequence and coordinate crosscheck. No inference that S is a literal material outflow is required by this transformation.

The spatial volume element on this unit-energy free-fall slicing is $4\pi r^2 h\,dr$. It equals the spherical reference-shell volume after selection. That equality has not been used as an additional assumption to obtain $\mathscr{H} = 1$, and it is not asserted for all observer slicings.

D.1 The equivalent radial drift expression

The same selected exterior can be written with a signed radial drift Br. The following transformation records the equivalent representation used in Draft 48. Here it is applied after the spherical field has been determined.

$$c^2 d\tau^2 = c^2 dt_c^2 - (dr - B_r\,dt_c)^2 - r^2 d\Omega^2, B_r^2 = S^2. \quad (319)$$

Here tc is the time coordinate of the propagation representation, $d\Omega^2 = d\theta^2 + \sin^2\theta\, d\varphi^2$, and Br is a signed radial drift variable. The radial displacement is dr − Br dtc, while the angular displacement is unchanged. This is the Painlevé–Gullstrand representation [3], used here for the already selected exterior. Its square contains the gravitational contribution, so adding a separate $S^2 dtc^2$ term would count it twice.

For dr = dΩ = 0, equation (319) gives $(d\tau/dtc)^2 = 1 - Br^2/c^2$. Matching the held-clock budget fixes $Br^2 = S^2$. Now remove the mixed radial–time term by defining:

$$d t_c = dT - \frac{B_r}{\alpha^2 c^2} dr . \quad (320)$$

Because the coefficient depends only on r, this transformation is integrable in the static exterior. Expansion and substitution give:

$$c^2 d\tau^2 = \alpha^2 c^2 dT^2 - \left[1 + \frac{B_r^2}{\alpha^2 c^2}\right] dr^2 - r^2 d\Omega^2 . \quad (321)$$

The radial coefficient reduces algebraically:

$$1 + \frac{B_r^2}{\alpha^2 c^2} = 1 + \frac{u}{1-u} = \frac{1}{\alpha^2} . \quad (322)$$

Consequently:

$$c^2 d\tau^2 = \alpha^2 c^2 dT^2 - \frac{dr^2}{\alpha^2} - r^2 d\Omega^2 . \quad (323)$$

The static spatial displacement is therefore $d\ell^2 = dr^2/\alpha^2 + r^2 d\Omega^2$. Substituting the spherical burden gives the Schwarzschild exterior interval. The metric records the clock and propagation relations obtained from the specified capacity, source and coupling equations.

Both signs of Br give the same static exterior interval because only Br^2 remains. In this calculation Br represents the propagation coupling, without establishing a literal material flow. It is different from Appendix F.1's gradient current $\mathcal{J}$: Br has units of speed and $\mathcal{J}$ of acceleration. We can eliminate Br from the final equations while retaining the spatial response it supplied.

Appendix E Galaxy inputs and uncertainty calculations

Galaxy	D Mpc	i degrees	Projected axis ratio	Radius status	Sources
NGC4051	18.00 ± 2.50	49.0 ± 3.0	0.880 ± 0.025	End	VS01,SV98
NGC5371	39.70 ± 9.92	53.0 ± 2.0	0.782 ± 0.012	Interior 0.50	Sa96,Be87
UGC07690	8.11 ± 2.43	41.0 ± 5.0	0.812 ± 0.035	Interior 0.57	Sw09,Sw02
NGC3949	18.00 ± 2.50	55.0 ± 2.0	0.669 ± 0.015	End	VS01,SV98
NGC4013	18.00 ± 2.50	89.0 ± 1.0	0.405 ± 0.010	Interior 0.43	VS01,SV98
UGC06818	18.00 ± 2.50	75.0 ± 3.0	0.467 ± 0.048	Interior 0.75	VS01,SV98
NGC2683	9.81 ± 0.49	80.0 ± 5.0	0.351 ± 0.018	Interior 0.33	Sa96,Br92
NGC3198	13.80 ± 1.40	73.0 ± 3.0	0.398 ± 0.014	Interior 0.36	Da06,Be91,Be87
NGC3992	23.70 ± 2.30	56.0 ± 2.0	0.594 ± 0.017	Interior 0.60	VS01,SV98
UGC08837	7.21 ± 0.36	80.0 ± 5.0	0.277 ± 0.003	End	Sw09,Sw02
NGC5585	7.06 ± 2.12	51.0 ± 2.0	0.650 ± 0.028	Interior 0.56	Bl99,Sa96,Co91
NGC4010	18.00 ± 2.50	89.0 ± 1.0	0.198 ± 0.017	End	VS01,SV98
NGC4565	9.7 fixed	86.6 fixed	0.205 ± 0.017	End	Yim14,Salo15
NGC4183	18.00 ± 2.50	82.0 ± 2.0	0.239 ± 0.016	Interior 0.54	VS01,SV98
NGC5907	17.30 ± 0.90	88.0 ± 2.0	0.197 ± 0.010	Interior 0.55	Sa96,SA87
NGC0100	13.50 ± 4.05	89.0 ± 1.0	0.185 ± 0.028	Interior 0.93	dB02,Rh96
NGC4100	18.00 ± 2.50	73.0 ± 2.0	0.341 ± 0.010	Interior 0.61	VS01,SV98
NGC5033	15.70 ± 4.70	66.0 ± 1.0	0.431 ± 0.019	Interior 0.54	Sa96,Be87
UGC07089	18.00 ± 2.50	80.0 ± 3.0	0.228 ± 0.004	End	VS01,SV98
NGC1090	37.00 ± 9.25	64.0 ± 3.0	0.458 ± 0.042	Interior 0.79	Ge04

Table 11. Central inputs and comparison-radius status. Distances, inclinations and their listed errors are from the adopted catalogue except NGC 4565. Source codes follow SPARC Table 1 [5]; Yim14 and Salo15 refer to [7] and [8]. The projected axis-ratio errors are formal image-fit errors. “Interior” and “End” refer to the adopted rotation dataset; they do not define a unique physical galaxy edge.

Galaxy	R range kpc	Mbar range 10^{10} M⊙	Reported ratio range	Adds M/L sensitivity
NGC4051	10.50–13.88	3.782–6.672	0.834–1.248	0.750–1.390
NGC5371	17.34–28.90	10.361–29.058	0.844–1.178	0.760–1.311
UGC07690	1.65–3.07	0.046–0.162	0.741–1.675	0.706–1.772
NGC3949	6.09–8.05	1.733–3.068	0.848–1.456	0.775–1.601
NGC4013	11.60–15.34	3.439–6.066	0.917–1.149	0.826–1.278
UGC06818	4.51–5.97	0.163–0.293	0.768–1.169	0.740–1.223
NGC2683	10.96–12.12	3.885–4.770	0.988–1.183	0.885–1.322
NGC3198	14.44–17.70	2.704–4.089	0.965–1.199	0.907–1.284

Galaxy	R range kpc	Mbar range 10^{10} M⊙	Reported ratio range	Adds M/L sensitivity
NGC3992	24.85–30.19	11.017–16.363	0.850–1.163	0.774–1.282
UGC08837	3.99–4.41	0.060–0.075	0.882–1.150	0.848–1.205
NGC5585	4.31–8.01	0.181–0.630	0.760–1.377	0.728–1.447
NGC4010	9.02–11.92	0.910–1.616	0.948–1.199	0.877–1.303
NGC4565	Unreported	Unreported	0.868–1.081	0.786–1.197
NGC4183	9.77–12.93	0.742–1.317	0.887–1.197	0.837–1.278
NGC5907	26.24–29.12	10.366–12.838	1.027–1.127	0.944–1.233
NGC0100	6.23–11.57	0.207–0.726	0.709–1.205	0.681–1.263
NGC4100	12.00–15.88	2.496–4.408	0.907–1.200	0.822–1.330
NGC5033	16.82–31.20	3.687–12.756	0.774–1.202	0.708–1.321
UGC07089	7.89–10.43	0.249–0.448	0.788–1.184	0.744–1.263
NGC1090	17.92–29.86	2.661–7.516	0.807–1.294	0.742–1.415

Table 12. Distance-coupled radius ranges and mass ranges, plus predicted-to-observed speed-ratio envelopes. Mass ranges in this table use reported distance and luminosity variation at fixed adopted stellar M/L. The final column adds the separate ±0.1 dex stellar M/L sensitivity. Each range permits simultaneous variation of its listed inputs; it is not a statistical confidence interval. NGC 4565 has no adopted distance or luminosity uncertainty here, so its mass and radius ranges are unreported, not zero-uncertainty measurements.

The envelope evaluates a 601 by 601 grid in projected axis ratio and inclination, retaining only physical oblate deprojections. For each geometry it rescales the observed velocity using that same inclination. Distance, luminosity and quoted velocity errors enter through their monotonic endpoints. No mass or radius is changed in the central Table 3 calculation. The source hashes, central rows, envelope assumptions and reproduction code accompany the manuscript. The ±0.1 dex exercise rescales all stellar components together; it is a model sensitivity, not an additional measured error for each galaxy. Common group-distance and mass-to-light systematics are not independent trials across the sample.

The envelopes omit unreported HI-flux and decomposition systematics. NGC 4565 retains fixed distance and inclination because their uncertainties were not recovered. The separate stellar mass-to-light range is a model sensitivity. Agreement within an individual envelope does not establish a simultaneous solution for all galaxies under one shared set of systematic parameters.

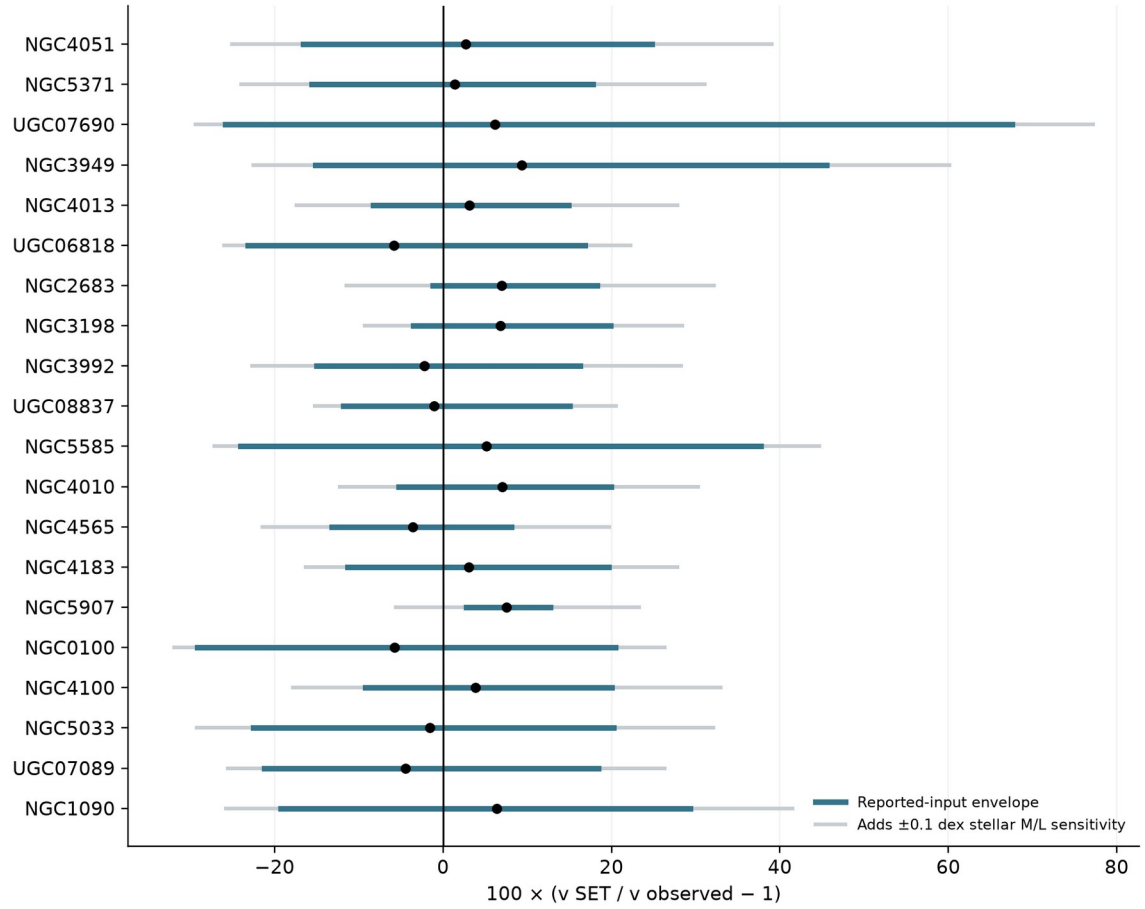


Figure 13. Central residuals and coupled-input envelopes for the selected examples. Blue lines use the reported-input ranges included in this calculation. Gray lines additionally vary the stellar mass-to-light normalization by ±0.1 dex. The two envelopes answer different questions. NGC 4565 retains fixed distance and inclination here. Unknown systematics and correlations between different galaxies are not assigned a probability model.

The projected stellar-aperture checks retain global HI and therefore are not complete enclosed-baryon reconstructions. For UGC 8837, the stellar correction changes the central discrepancy from −1.10% to approximately −1.55%; its close agreement survives that particular check. NGC 4565 has no equivalent enclosed-inventory check in this analysis. A comparison at an interior radius does not become a fully matched mass aperture merely because its central speed agrees.

Figure 5 uses a separate probabilistic input exercise, distinct from the endpoint envelopes in Table 12 and Figure 13. Its blue bars are the marginal 2.5th and 97.5th percentiles of 65,536 joint draws, with stellar M/L fixed at the adopted values. Quoted errors set Gaussian widths for projected axis ratio, inclination, distance, luminosity and speed; positive quantities and physical oblate deprojections are retained. Distance rescales mass and radius together, and the same inclination draw changes both the reconstructed area and the deprojected speed. The two plotted axes therefore have correlated errors.

The retained distance model includes a common 0.9 Mpc term for Ursa Major members and a common 7% term for Hubble-distance objects, with the remaining quoted variance assigned individually. These are assumptions of the sensitivity model. Gas flux and other unreported systematics are held fixed; NGC 4565 also retains fixed distance, inclination and luminosity. The archived draw code records the distributions and seed. These conditional percentiles do not include a complete mass-model covariance or selection adjustment, and the orange central predictions have no comparable full uncertainty distribution supplied here.

E.1 Additional galaxy source inputs and single-radius results

The following two table panels supply the 42 additions used in Section 8.7. Their source-group codes are P:PROBES, G:GHASP, S:SPARC and U:Ursa Major. Each row retains the global stellar-plus-HI/helium inventory, the actual adopted comparison radius and the image-derived area coefficient. The observed reading is the catalogue bin value for S and U and the common-radius two-side reading for P and G. All numerical propagation and comparisons use the archived full-precision inputs; table values are rounded for reading.

The displayed ± range is the quoted velocity error for NGC 3972 and UGC 06894. For P and G it is a descriptive combination of the source's branch error or dispersion and the half-difference of the two sides. These quantities are kept under their source definitions, rather than pooled as a single confidence level. A dagger marks a side half-difference greater than 20% of the two-side mean. The dagger is a property of the measured branches, not an adjustment of the calculated speed.

Table 13. Forty-two additional SET galaxy calculations, panel 1 of 2. Source codes and the meanings of the observational ranges are given in the text. The central source inputs are not retuned within their errors.

Galaxy	Source	R (kpc)	M_{bar} ($10^{10} M_{\odot}$)	f_{image}	v_{SET} (km/s)	$v_{obs} \pm e_v$ (km/s)	Difference
ESO013-016	P	5.946	0.4832	2.7069	87.37	81.51 ± 13.32	+7.18%
ESO285-048	P	8.459	1.2497	2.3852	133.68	124.21 ± 7.45	+7.63%
ESO291-024†	P	3.484	0.0974	2.1896	63.36	65.53 ± 39.90	-3.31%
ESO298-015†	P	7.166	0.3975	2.2679	86.15	86.43 ± 19.05	-0.32%
ESO346-001	P	7.655	0.7434	2.1604	119.66	110.02 ± 9.26	+8.76%
ESO362-011	P	10.392	1.3306	2.3539	126.10	130.67 ± 10.49	-3.50%
ESO407-009	P	3.700	0.1704	2.0580	86.49	86.33 ± 7.28	+0.19%
ESO418-008	P	2.289	0.1517	2.9784	71.71	78.51 ± 4.70	-8.66%
ESO482-035	P	6.163	0.7730	2.2927	128.14	121.39 ± 10.70	+5.56%
ESO482-046	P	10.341	0.6503	2.1387	97.27	94.08 ± 5.37	+3.39%
ESO508-011	P	13.714	1.2867	2.1345	119.04	116.65 ± 8.57	+2.05%
ESO533-004	P	8.988	1.4323	2.1770	152.12	148.70 ± 16.26	+2.30%
ESO540-016	P	6.747	0.2560	2.1272	75.97	83.58 ± 7.94	-9.11%
ESO544-027	P	6.566	0.6855	2.4395	109.87	116.01 ± 19.58	-5.29%
ESO548-032	P	8.001	0.3003	2.6717	60.16	64.06 ± 6.23	-6.10%
ESO549-018†	P	8.467	0.9067	2.1755	124.78	116.30 ± 26.89	+7.30%
ESO576-026	P	5.589	0.2141	2.1345	76.07	70.89 ± 4.30	+7.30%
ESO601-025	P	4.312	0.1699	2.5288	65.12	71.81 ± 11.44	-9.32%
IC5026	P	8.725	0.7289	2.1080	113.74	103.69 ± 7.71	+9.69%
IC5078	P	4.747	0.5875	2.3511	124.12	124.45 ± 11.85	-0.26%
NGC0672	G	6.465	0.4699	2.4783	90.24	96.85 ± 4.46	-6.82%

Table 14. Forty-two additional SET galaxy calculations, panel 2 of 2. Source codes and the meanings of the observational ranges are given in the text. The central source inputs are not retuned within their errors.

Galaxy	Source	R (kpc)	M_{bar} ($10^{10} M_{\odot}$)	f_{image}	v_{SET} (km/s)	$v_{obs} \pm e_v$ (km/s)	Difference
NGC0755	P	6.606	0.5502	2.3110	103.59	114.57 ± 15.89	-9.58%
NGC1145	P	8.780	1.7778	2.0795	179.51	179.67 ± 22.15	-0.09%
NGC1425	P	10.360	2.3180	2.3304	168.38	171.64 ± 7.49	-1.90%
NGC1438†	P	5.803	1.0954	2.4022	150.03	139.87 ± 32.71	+7.26%
NGC2543	G	13.301	3.6745	2.0492	212.77	214.72 ± 9.15	-0.91%
NGC3346	G	5.041	0.7124	2.9707	104.98	111.77 ± 3.21	-6.07%
NGC3403	P	2.747	0.4493	2.2066	152.04	146.99 ± 16.06	+3.44%
NGC3972	S	8.720	0.8791	2.1709	121.33	134.00 ± 4.90	-9.46%
NGC4062	G	6.008	0.9746	2.1313	156.77	156.03 ± 4.42	+0.47%
NGC4633†	P	6.187	0.3814	2.5214	81.69	90.06 ± 41.24	-9.29%
NGC4635	G	3.592	0.2302	2.2727	92.39	91.96 ± 4.87	+0.47%
NGC4651	G	10.487	4.4270	2.4039	224.21	207.27 ± 2.92	+8.17%
NGC5297	G	19.768	5.7025	2.2178	200.89	185.73 ± 11.20	+8.16%
NGC5774	G	11.610	1.4849	2.4612	120.54	116.04 ± 22.10	+3.87%
NGC5879	G	5.651	0.7654	2.2369	136.48	134.60 ± 4.88	+1.39%
NGC7218	P	7.040	1.1470	2.2718	147.40	141.16 ± 7.62	+4.42%
NGC7314	P	5.899	1.1347	2.1783	167.02	157.69 ± 5.23	+5.92%
PGC013716	P	4.429	0.6564	2.2281	143.33	158.20 ± 14.88	-9.40%
UGC05897	P	11.231	2.2939	2.1177	177.04	161.22 ± 29.65	+9.81%
UGC06894	U	3.607	0.1069	2.1373	66.81	63.00 ± 5.00	+6.05%
UGC10445	G	6.350	0.3200	2.7575	67.53	69.03 ± 10.47	-2.17%

The source archive resolves names and coordinates before combining surveys. The retained data priority is SPARC, then the source-complete Ursa Major measurements, GHASP and PROBES; additional observations of the same object remain cross-survey checks. The infrared shape is measured from the S4G fine free-ellipse product, with the existing coarse-product fallback used only when the fine product is absent. The image and angular comparison point are chosen by data coverage and quality, before the speed comparison. The ledger keeps all catalogue entries, including incomplete inputs and nonphysical oblate deprojections.

E.2 Coupled input propagation for the first 62 examples

For each trial, one distance factor rescales the stellar and atomic-gas masses by $(D / D_0)^2$ and the physical radius by D / D_0. Magnitude errors are propagated by drawing the magnitude and converting it to flux; the NGC 3972 luminosity and UGC 06894 HI flux use their quoted linear input errors. The projected axis ratio and inclination determine $\eta^2 = (p^2 - \cos^2 i) / \sin^2 i$ and hence f. Draws with nonpositive source quantities,

invalid projected shape or $\eta^2 \notin [0,1]$ are counted and excluded. They are never clipped to a thin disk or used to select a different radius.

The inclination transformation of the measured speed is applied only when it represents the same quantity varied in the shape calculation. This holds for the added SPARC and GHASP readings. PROBES has the same analytic dependence, but no error on its retained photometric inclination is supplied, so that input remains fixed. UGC 06894 has different global and local-ring inclinations, whose covariance is unreported. The fixed angular sampling rule remains unchanged throughout.

$$R^{(n)} = \frac{v_{SET}^{(n)}}{v_{obs}^{(n)}}, v_{obs}^{(n)} = v_{obs,0} \frac{\sin i_0}{\sin i^{(n)}} \text{ when the inclination is shared.} \quad (324)$$

Independent observational velocity noise is not included in these source-input draws. For the figure, the source-only percentiles of R are compared with the separate interval $[1 - e_v / v_{obs,0}, 1 + e_v / v_{obs,0}]$. These intervals retain the individual meanings of e_v. The 58/62 and 60/62 overlap counts are consequently descriptive geometric comparisons of displayed ranges. They are not a combined likelihood or a statement that all source and measurement systematics have been exhausted.

For the additions, the adopted GHASP distances and the Ursa Major 18.6 Mpc distance have no supplied uncertainty in the retained tables. All thirty PROBES inclinations have no supplied error. NGC 3972 has no per-object HI-flux error. The inventory marks each missing value explicitly. No default percentage is substituted. The .5/.7 stellar mass-to-light prescription, helium multiplier, bulge fraction, angular comparison location and unreported calibration/covariance terms remain fixed. The original-twenty source draws retain their archived distance model, including its stated common terms; the new PROBES distance errors are not assigned those older catalogue assumptions.

The full-precision rows, per-input provenance, attempted and accepted draw counts, seeds, percentile tables and file hashes accompany the calculation record. An independent implementation recomputed the 42 new ensembles through $Q / (\sqrt{2} A_{image})$ in SI units and reproduced the displayed percentiles to relative numerical precision better than 10^{-14}. This verifies the calculation and its bookkeeping; the observational interpretation follows the stated source and response prescription.

E.3 Twenty-one further galaxy comparisons

The complete new ledger covers 131 distinct identities: 84 calculations, 46 invalid central oblate geometries and one remaining missing permitted image product. Of the 84 calculations, 71 are eligible primary comparisons; 21 satisfy the fixed 10% central criterion and 50 lie outside it. Thirteen calculations remain separately labelled diagnostics, including the PHANGS inner-CO comparisons [44]. The 21 displayed additions are worked agreements from this retained screen. Their median absolute central discrepancy is 5.90%. Seventeen comparison points are interior to their adopted rotation datasets and four are endpoints; these labels locate the measurements and do not establish an enclosing source boundary.

Apertif supplies its published circular speeds, including the release's asymmetric-drift correction, and its adopted first CF4 distance; the catalogue HI flux in Jy Hz is converted with the authors' relation $M_{HI} / M_\odot = 49.7 (D / Mpc)^2 (F_{HI} / Jy\ Hz)$ [39]. WALLABY supplies the selected release's deprojected rotation bins, corrected HI mass and source distance; its kinematic and inclination contributions to the velocity range are combined in quadrature [40, 41]. UGC01551 uses the PROBES II archived observing curve with the original observing paper's distance and inclination [42, 43]. Already deprojected speeds receive no second inclination correction.

The additional rows use source codes S:SPARC with recovered S4G image aliases, A:Apertif, W:WALLABY and P2:PROBES II. E denotes the endpoint and I an interior point of the adopted rotation dataset. Four rows are endpoints and seventeen are interior. The table retains the global stellar-plus-HI/helium inventory and the actual common image/rotation radius. E and I do not designate an independently established enclosing boundary.

The displayed range is the recorded comparison-bin error for SPARC, Apertif and UGC01551. For WALLABY it combines the published kinematic-model and inclination contributions in quadrature. These values have different source definitions and do not form a shared confidence model. No source-input ensemble has yet been propagated for these 21 rows; the ensembles in Section 8.8 and Appendix E.2 remain those of the first 62 examples.

Table 15. Twenty-one further SET galaxy calculations. Source codes, radius labels and quality flags are defined in the text. Full-precision central source inputs are retained without retuning.

Galaxy	Source	R (kpc)	M_{bar} ($10^{10} M_{\odot}$)	f_{image}	v_{SET} (km/s)	$v_{obs} \pm e_v$ (km/s)	Difference
ESO576-050	W	11.357 I	1.2748	2.6630	104.36	98.10 ± 4.78	+6.38%
IC0758‡	A	5.574 E	0.3252	2.4874	80.55	87.80 ± 3.30	-8.26%
NGC3440‡	A	7.738 I	0.7591	2.5845	100.54	96.40 ± 3.30	+4.29%
NGC3835	A	12.861 I	3.0025	2.0213	198.29	203.60 ± 3.30	-2.61%
NGC4629	W	4.047 I	0.2818	3.6951	59.24	55.30 ± 1.53	+7.13%
NGC4902	W	17.693 I	9.9768	3.1968	194.86	200.00 ± 27.19	-2.57%
NGC5105	W	13.549 I	1.9671	2.6344	119.98	127.80 ± 3.80	-6.12%
NGC5963	A	5.160 I	0.5380	2.1786	122.95	116.10 ± 3.30	+5.90%
UGC01551§	P2	3.770 E	0.4204	3.5741	77.50	82.50 ± 10.30	-6.06%
UGC05721	S	1.710 I	0.1013	2.3580	85.62	82.20 ± 19.20	+4.16%
UGC07151	S	5.500 E	0.1961	2.2411	69.90	76.20 ± 3.17	-8.27%
UGC07153‡	A	10.766 I	0.8696	2.1695	108.67	108.30 ± 6.80	+0.34%
UGC07603	S	2.050 I	0.0531	2.4245	55.07	60.50 ± 2.89	-8.97%
UGC08053	W	2.615 I	0.0705	2.2963	59.31	62.30 ± 5.16	-4.80%
UGC08490	S	3.720 I	0.1466	2.2552	73.03	78.80 ± 3.69	-7.33%
UGC08550	S	2.920 I	0.0528	2.2429	49.71	51.60 ± 2.04	-3.66%
UGC08684	A	12.305 E	1.0366	2.2406	107.46	110.80 ± 3.30	-3.02%
UGC08699	S	13.390 I	3.3060	2.2767	181.05	177.00 ± 3.62	+2.29%
UGC08795‡	A	8.552 I	0.4611	2.1550	89.39	99.20 ± 3.30	-9.89%
UGC08851‡	A	4.763 I	0.1730	2.4200	65.33	67.50 ± 3.30	-3.22%
UGC09291	A	11.199 I	1.8313	2.3627	141.98	131.30 ± 3.30	+8.13%

The ‡ flags retain the published Apertif distinctions: NGC3440, UGC07153, UGC08795 and UGC08851 are A2 models, for which the release notes possible amplitude uncertainty of about 20%; IC0758 has inclination 34.2 degrees, below the release’s recommended 40-degree cut for amplitude analyses. These remain labelled

illustrations under the retained intake rule. The § flag records that the UGC01551 comparison is its fourteenth and final bin, outside the original paper’s five inner bar-affected points; its adopted distance and inclination are 20.2 Mpc and 63 degrees [39, 43].

The recovered SPARC/S4G identities are UGC05721/NGC3274, UGC07151/NGC4144, UGC07603/NGC4455, UGC08490/NGC5204, UGC08550/NGC5229 and UGC08699/NGC5289. All six previously lacked a cached permitted image product. Their original SPARC distances, inclinations, luminosities, HI masses and observed radial grids are retained; the name and coordinate audit recovers the image input for the first complete calculation. These objects are distinct from the earlier 62 examples.

The complete ledger records source files, input hashes, radius selection and every calculated outcome. The separate CO-aperture diagnostics are retained in that ledger and do not contribute to the 83-example figure.

Table 16. Adopted central source measurements for the 21 further examples. Masses use the displayed distance and the unchanged stellar mass-to-light and helium conventions. The image ratio is evaluated at the comparison radius listed in Table 15.

Galaxy	D (Mpc)	i (deg)	Image p	Stars (10^{10} M⊙)	HI+He (10^{10} M⊙)	S4G image
ESO576-050	28.2927	63.10	0.607952	0.816432	0.458378	ESO576-050
IC0758	21.9000	34.20	0.852875	0.215307	0.109866	IC0758
NGC3440	30.4000	83.50	0.430625	0.319286	0.439833	NGC3440
NGC3835	39.3000	69.40	0.355499	2.640423	0.362082	NGC3835
NGC4629	15.8422	67.60	0.901974	0.068391	0.213410	NGC4629
NGC4902	37.4683	34.20	0.912743	9.169972	0.806801	NGC4902
NGC5105	41.4451	50.00	0.726492	1.293949	0.673162	NGC5105
NGC5963	12.9000	50.90	0.648432	0.268503	0.269484	NGC5963
UGC01551	20.2000	63.00	0.873367	0.273699	0.146666	UGC01551
UGC05721	6.1800	61.00	0.552287	0.026550	0.074746	NGC3274
UGC07151	6.8700	90.00	0.234647	0.114200	0.081928	NGC4144
UGC07153	42.3000	89.00	0.188764	0.386590	0.482993	UGC07153
UGC07603	4.7000	78.00	0.390591	0.018800	0.034314	NGC4455
UGC08053	10.2423	56.90	0.590336	0.016031	0.054465	UGC08053
UGC08490	4.6500	50.00	0.669248	0.050850	0.095760	NGC5204
UGC08550	6.7000	90.00	0.235756	0.014450	0.038304	NGC5229
UGC08684	37.6000	83.50	0.258877	0.876084	0.160514	UGC08684
UGC08699	39.3000	73.00	0.381434	2.808810	0.497154	NGC5289
UGC08795	33.6000	75.20	0.307894	0.217194	0.243921	UGC08795
UGC08851	26.2000	68.00	0.487042	0.033882	0.139116	UGC08851
UGC09291	44.0000	65.40	0.500203	1.239012	0.592256	UGC09291

Appendix F Stationary source response and energy balance

F.1 Spherical charge and gradient flux

The source strength is $\mu Q = Q\Omega^2/(24\pi V\Omega) = GM$. In the isotropic weak-field representation, define $\Psi = S^2/2$ and the Euclidean diagnostic current $\mathcal{J} = -\nabla\Psi$. Its units are m/s^2 and its integrated flux has units m^3/s^2. This is the weak-field form of the exact physical work-current used in Section 3; the exact static source accounting also includes the stated clock and stress factors.

For the isotropic stationary branch, impose $\nabla\cdot\mathcal{J} = 4\pi G\rho$. Integration over a domain containing mass M gives $\oint\mathcal{J}\cdot dA = 4\pi GM = Q\Omega^2/(6V\Omega)$. In its spherical source-free exterior, symmetry therefore gives $\mathcal{J}r = GM/r^2$. Substituting $\mathcal{J}r = -d\Psi/dr = -\tfrac{1}{2}dS^2/dr$ immediately yields $-(r^2/2)dS^2/dr = GM$, equation (61). The boundary condition $\Psi \to 0$ at infinity gives $\Psi = GM/r$ and $S^2 = 2GM/r$.

The source identity fixes the integrated normalization, while $\mathcal{J} = -\nabla\Psi$ specifies the isotropic response. Here $\mathcal{J}$ is shorthand for the gradient and can be eliminated. The spherical areal radius is defined by $4\pi r^2$, no auxiliary radius calibration enters this calculation.

In the conditional collective extension, the corresponding representation is $\mathcal{J} = -C\nabla\Psi$. The tensor C belongs to that proposed response law, not to the source identity or to the invariant's remaining-capacity fraction. Appendix F states that stationary formulation. The stationary source equations determine the burden field for the specified response. The spherical clock and propagation relations follow from the physical-current and source-consistency construction in Section 3.

Earlier manuscripts developed this source and action language [1, 2]. The research supplement retains the previous auxiliary radial calibration for traceability. The source and current relations used here are given explicitly.

F.2 A stationary collective field

I now state a stationary source equation for a prescribed spatial response. Use Euclidean spatial derivatives in the weak field and a prescribed dimensionless symmetric positive tensor C:

$$-\nabla\cdot(C\nabla\Psi) = 4\pi G\rho_{\mathrm{bar}},\ \Psi = \frac{S^2}{2},\ a = \nabla\Psi. \qquad (325)$$

Equivalently, the source equation is $-\tfrac{1}{2}\nabla\cdot(C\nabla S^2) = 4\pi G\rho\mathrm{bar}$. Let fixed sources have normalized density profiles w_a, with $\int w_a\, d^3x = 1$, and define $\mu_a = Q_a^2/(24\pi V_a) = GM_a$. Its right side can then be written $4\pi\Sigma\mu_a w_a$. Moving sources changes their locations and thus the density; it need not change their individual M_a, V_a or Q_a. Splitting the same density into more numerical cells leaves this source unchanged.

The same source normalization can be written without abbreviating Q:

$$-\frac{1}{2}\nabla\cdot(C\nabla S^2) = \sum_a \frac{[4\pi\sqrt{2GM_aR_a^3}]^2}{6(4/3)\pi R_a^3} w_a(x) = 4\pi G\rho_{\mathrm{bar}}(x). \qquad (326)$$

For the specified constant response and isolated boundary condition, equation (327) gives $S^2 = 2\Psi$. Substituting that solution into equation (1), before simplifying the source coefficient, produces equation (218).

Define $\mathcal{J} = -C\nabla\Psi$. Any complete enclosing surface then has $\oint\mathcal{J}\cdot dA = 4\pi GM$. The source normalization is fixed while C determines how that gradient flux relates to the force $\nabla\Psi$. C contains response coefficients.

For constant positive C and an isolated boundary condition $\Psi \to 0$ at infinity, the solution is explicit. Write d = x − x′, C^{-1} for the matrix inverse and det C for its determinant:

$$\Psi(x) = \frac{G}{\sqrt{\det C}} \int \frac{\rho_{\mathrm{bar}}(x')}{\sqrt{d^T C^{-1} d}} d^3 x' . \quad (327)$$

The orientation at an enclosing boundary follows from this same solution. For a positive point source M_a at x_a, define $d_a = x - x_a$. Differentiating equation (327), with C constant and positive-definite, and using $\mathcal{J} = -C\nabla\Psi$ gives:

$$\mathcal{J}_a(x) = \frac{G M_a}{\sqrt{\det C}} \frac{d_a}{(d_a^T C^{-1} d_a)^{3/2}} . \quad (328)$$

Let n be the outward unit normal at x on a convex surface enclosing the sources, with x distinct from their locations. Geometry gives $d_a \cdot n \geq 0$. Since the coefficient multiplying d_a is positive:

$$d_a \cdot n \geq 0 \Rightarrow \mathcal{J}_a \cdot n \geq 0 , \mathcal{J} \cdot n = \sum_a \mathcal{J}_a \cdot n \geq 0. \quad (329)$$

The outward-normal contribution is strictly positive for a source strictly inside the surface. Summing positive source contributions preserves the sign; integrating over a nonnegative continuous density gives the same result. $\mathcal{J}$ retains its defined gradient-flux meaning. This orientation property does not establish QΣ = Qref; that boundary-loading normalization remains the collective prescription in Section 7.3.

The coordinate change $y = C^{-1/2}x$ converts the operator to an ordinary Laplacian and transforms the source volume. With C = I, equation (327) becomes the positive Newtonian potential. For specified constant C, equation (327) determines the stationary field of an arbitrary specified density.

Multiplying Ψ by a constant and dividing C by that constant preserves $\mathcal{J}$ and its source equation.

F.3 Work when the response changes

Energy provides a consistency test of the proposed response. Use a weak-field quasistatic family of equation (325), with finite smooth sources, Ψ fixed to zero at infinity and a symmetric positive C(x,T) prescribed independently of the source variations. Let matter obey mass continuity and the force $a = \nabla\Psi$. For the gravity-only case define:

$$E_{kin} = \frac{1}{2} \int \rho |v|^2 d^3 x , W = -\frac{1}{2} \int \rho \Psi \, d^3 x . \quad (330)$$

The factor ½ in the binding energy avoids counting each source interaction twice. Multiplying the source equation by Ψ and integrating by parts, with vanishing boundary terms, gives:

$$W = -\frac{1}{8\pi G} \int \nabla \Psi \cdot C \nabla \Psi \, d^3 x . \quad (331)$$

These expressions refer to a finite-energy source distribution. Unregularized point-source self-energies cannot be inserted into them. If pressure, internal energy, radiation or external matter forces are present, their corresponding energy and work terms must also be included.

The same source equation and force follow from a compatible quasistatic Lagrangian:

$$L_{qs} = E_{kin} + \int \rho \Psi \, d^3 x - \frac{1}{8\pi G} \int \nabla \Psi \cdot C \nabla \Psi \, d^3 x . \quad (332)$$

C is held fixed when varying Ψ and the source positions in this expression. To show the energy balance, define:

$$P_{response} = \frac{1}{8\pi G} \int \nabla \Psi \cdot \frac{\partial C}{\partial T} \nabla \Psi \, d^3 x . \quad (333)$$

Differentiate the source equation and use its self-adjoint spatial operator. Integration by parts gives $\int \rho (\partial \Psi / \partial T)\, d^3x = \int \Psi (\partial \rho / \partial T)\, d^3x - 2P_{response}$. The partial derivatives hold spatial coordinates fixed. Differentiate W in equation (330) and substitute this identity:

$$\frac{dW}{dT} = - \int \Psi \frac{\partial \rho}{\partial T} d^3 x + P_{response} . \quad (334)$$

Mass continuity, $\partial \rho / \partial T = -\nabla \cdot (\rho v)$, and the gravitational force give $dE_{kin}/dT = \int \rho v \cdot \nabla \Psi \, d^3x = \int \Psi (\partial \rho / \partial T)\, d^3x$ under the same boundary assumptions. Therefore:

$$\frac{d\left(E_{kin} + W \right)}{dT} = \frac{1}{8\pi G} \int \nabla \Psi \cdot \frac{\partial C}{\partial T} \nabla \Psi \, d^3 x . \quad (335)$$

The right side has units of power. Fixed C conserves Ekin + W. If C changes externally, it supplies or removes work through the stated term. As a sign check, take a uniform scalar C = kI at fixed density: $\Psi = \Psi_0 / k$ and $W = W_0 / k$. Since $W_0 < 0$, increasing k makes W less negative, as equation (335) requires.

The interpretation changes if C is determined by the sources themselves. At fixed boundary conditions the full variation is:

$$\delta W = - \int \Psi \, \delta \rho \, d^3 x + \frac{1}{8\pi G} \int \nabla \Psi \cdot \delta C \nabla \Psi \, d^3 x . \quad (336)$$

If a source coordinate Xa changes C, this second term contributes an additional generalized force:

$$F_{a,response} = - \frac{1}{8\pi G} \int \nabla \Psi \cdot \frac{\partial C}{\partial X_a} \nabla \Psi \, d^3 x . \quad (337)$$

Its work cancels the part of Presponse caused by that internal coordinate change. Updating C from a rearranging source distribution while retaining only the force ∇Ψ would omit that interaction. Alternatively, C can be assigned independent dynamics and its own energy, but those equations must then be supplied. Equation (335) alone is not a closed evolution law for a galaxy.

Appendix G A finite flat speed profile and its exterior

To illustrate the relation between potential depth and orbital speed, prescribe a weak-field circular speed v_0 from an inner radius Rin > 0 to a transition radius Rt. Outside Rt, prescribe $vorb^2(R) = v_0^2 Rt/R$. Both the speed and the force are continuous at Rt, and the inverse-radius exterior allows S^2 to approach zero at infinity.

Integrating equation (109) first in the exterior gives:

$$S^2(R) = \frac{2 v_0^2 R_t}{R}, R \geq R_t. \quad (338)$$

Inside the flat-speed interval, split the same integral at Rt:

$$S^2(R) = 2 \int_R^{R_t} \frac{v_0^2}{r} dr + 2 v_0^2. \quad (339)$$

$$S^2(R) = 2 v_0^2 \left[1 + \ln\left(\frac{R_t}{R} \right) \right], R_{\text{in}} \leq R \leq R_t. \quad (340)$$

Differentiation gives $-(R/2)dS^2/dR = v_0^2$ throughout that interval. At Rt, both sides give $S^2 = 2v_0^2$ and $dS^2/dR = -2v_0^2/Rt$. Thus the boundary conversion $S/\sqrt{2} = v_0$ is consistent at the transition, even though it does not hold at every interior point. For example, at R = Rt/2, $S/\sqrt{2} = v_0\sqrt{(1 + \ln 2)} \approx 1.3012 v_0$ while the circular speed remains v_0.

Below Rin the model needs an inner solution. Its exterior gravitational parameter is $\mu ext = v_0^2 Rt$, or $\mu ext = (4/f)^2 GMbar$ under the boundary calibration at Rt. Thus a choice $f \neq 4$ retains an enhanced exterior monopole. This construction joins a flat-speed segment to the edge conversion; the source law would still have to determine Rt, f and the profile.

Appendix H Quadratic gradient normalization

The current conversion also supplies a check for any expression quadratic in the burden gradient. Squaring equation (90) gives

$$\left(\frac{dS^2}{d\chi}\right)^2=\alpha^8\left(\frac{ds_\star^2}{d\chi}\right)^2. \quad (341)$$

Thus, for any specified weight $W(\chi)$ and the same integration measure,

$$I_S=\int W(\chi)\left(\frac{dS^2}{d\chi}\right)^2 d\chi, \quad (342)$$

$$I_S=\int W(\chi)\alpha^8\left(\frac{ds_\star^2}{d\chi}\right)^2 d\chi. \quad (343)$$

If W depends on the field, its dependence must also be expressed in the new variable. The equations above only convert an already specified expression. Its units, physical role, integration domain and weight remain to be supplied by that expression; it is not automatically a field energy. Omitting α^8 would generally give a different functional rather than a different notation for the same one.

The finite-interval counterpart follows directly from equation (89):

$$\mathcal{J}_{12}^2=\alpha_1^4\alpha_2^4\left(\frac{s_{\star1}^2-s_{\star2}^2}{2\Delta\chi}\right)^2. \quad (344)$$

These exact identities keep a change from capacity burden to receiving-clock units from silently changing the field functional. In this revision they complement the complete-energy construction of Section 3 and Appendix K. The normalization identities themselves do not select that functional.

The main derivation needs the first-gradient conversion in Sections 4.2 and 4.3 and the transformed motion equation in Section 5.1. Those checks, together with this quadratic identity, keep the physical work-current, the auxiliary orbital force and a possible quadratic field expression distinct. The orbit and Mercury benchmark remain those derived in Sections 5.3 and 5.6.

Appendix I Numerical and algebraic verification

The coordinate transformation was checked for both signs of the radial drift at six radii and multiple non-null displacement increments. The largest scaled residual was 1.6e-15. The analytic oblate surface coefficient was also compared with a direct numerical surface integral at six axis ratios; the maximum coefficient difference was 4.9e-15.

A separate trajectory integration advances the spatial motion law, rather than the energy first integral, in units c = Rs = m = 1. For p = 8Rs and eccentricity 0.3, fourth-order Runge–Kutta steps of 0.4 and 0.2 over proper duration 2400 both give nine perihelion passages. Maximum relative energy drift falls from 1.57e-11 to 8.8e-13 as the step is halved. The finer trajectory gives mean perihelion advance 1.6698605690 radians, compared with 1.6698605710 from the independent anomaly integral.

Mercury's weak-field integral uses a cancellation-resistant form of $(1 - z)^{-1/2} - 1$, evaluated by Gaussian quadrature. It gives 42.98048057 arcsec per Julian century. The Earth normalization and radial signal delay were recalculated from the stated inputs. The checks in this appendix verify algebra and implementation; observational comparisons are reported separately.

The source-to-boundary identities were checked over multiple masses, radii and area coefficients, including spherical and thin-area limits. Their largest relative numerical discrepancy was 6.7×10^{-16}. The exact spherical escape condition $\alpha\,\gamma_{escape} = 1$ was checked at five occupancies, with a largest energy-normalization discrepancy of 1.2×10^{-15}. Numerical differentiation of Appendix G's matched profile recovered its stipulated circular-speed function to within 8×10^{-11} in normalized squared-speed units.

The response-work identity was independently checked with finite source distributions and symmetric positive tensors, including changing eigenvalues, rotating axes and spatially varying response. A three-dimensional finite-element differentiation check had maximum normalized disagreement below 8.2×10^{-11}. A separate evolving-sphere calculation checked integrated mechanical-energy change against response work for constant, increasing, decreasing and cyclic prescribed coefficients. Halving the integration step reduced the fourth-order integration error by approximately sixteen. The sign and normalization of the work term passed those tests. Their assumptions, inputs and scripts accompany the research record.

For Mercury, the archived JPL-based example uses $GM_\odot = 1.3271244004127942\times10^{20}\ m^3/s^2$, $a = 5.790922654\times10^{10}$ m, e = 0.20563593 and period 87.96946593 days [9]. The result is about 0.1035 arcsec per orbit, or 42.98 arcsec per Julian century. Treating these inputs as exact turning-point parameters, the exact integral differs from the leading term by approximately 5.16×10^{-6} arcsec per century. That is a mathematical comparison, without converting fitted ephemeris elements to second post-Newtonian accuracy. The model uses an isolated spherical Sun and a test probe. Planetary and other contributions must be added when comparing with the total observed precession [4].

Appendix J Cosmological distance and domain checks

The grouped brightness result concerns the redshift dependence of relative flux. The checks below retain the same source inputs and distinguish calculated quantities from the observations and inference procedures used to assess them.

The absolute rate and nearby distances

The source and clock prescription gives $H_0 = 30.54$ $\mathrm{km\,s^{-1}Mpc^{-1}}$. The cited megamaser analysis infers 73.9 ± 3.0 $\mathrm{km\,s^{-1}Mpc^{-1}}$ from geometric distances and galaxy redshifts [19]. This is not a ΛCDM prediction. Its comparison with SET depends on how the model connects its calculated rate to those distances and redshifts.

Under the propagation relation retained in Section 11.1, the leading small-redshift distance is $D_L \simeq cz / H_0$. The fixed inputs therefore predict a nearby distance–redshift slope that does not reproduce the megamaser inference under that mapping. The relative supernova contrasts cancel the common absolute distance scale and do not test this normalization.

The BAO geometry comparison

The separate comparison with six released BAO geometry quantities gives $\chi^2 = 48.70$ [20]. This check uses the dimensionless ratio of transverse to radial distance, for which the common acoustic-ruler scale cancels. The released constraints still depend on the survey analysis and its standard-ruler interpretation. Under the retained flat propagation coupling, the statistic indicates poor agreement for this realization. It tests additional distance geometry rather than equality with ΛCDM parameters, and is separate from the supernova brightness statistics.

The domain of the retained history

The physical domain of this history ends at $z = 2.7244$. This is a limit of the retained source and front evolution, not an observed edge of the universe. The supernova groups reported here lie within that domain. Extrapolation beyond it is not supplied by this branch.

The source-radius formulation explains this endpoint. At $R_h = 0$ and finite T_Q, (243c) gives $d\,R_h / d\,T_Q = c$, so the regular front starts growing immediately. The right-hand side of (244) is not differentiable at $u = 0$ and also admits waiting branches; those do not satisfy the regular radius equation.

For the retained varying branch, the primitive in Section J.1 gives $a_i / a_0 = \left(1 + z_{max}\right)^{-1} = 0.2684965101$ and $T_{Qi} / T_{Q0} = \left(a_i / a_0\right)^{3/2} = 0.1391258939$. The zero-radius front is therefore reached at finite source time and finite density, with a nonzero material scale factor. It is the start of this front history, not the formal zero-volume origin chosen for the material clock.

The fixed trajectory $\eta_h = 3 / \sqrt{7}$ is the one solution of (243d) that extends over every $T_Q > 0$ with nonnegative radius and positive clock factor. Lower trajectories reach zero radius at finite T_Q when followed backward; upper trajectories reach $u = 1$ at finite T_Q instead. The fixed trajectory has constant α, so its normalized distance history has the constant-clock shape in (266).

J.1 Additional supernova identities and covariance

All 580 Union2.1 identities were compared with the actual Pantheon+ and DES-Dovekie Hubble-diagram inputs using normalized identifiers, survey aliases and primary-source coordinates. The 278 overlaps comprise 228 Pantheon+-only objects, 46 present in both prior compilations and four DES-only objects. Candidate positional matches within 3 arcsec and remaining near-neighbors out to 30 arcsec were checked. The final strict set also excludes SN 2001go because the earlier spectral-aging calculation used it. The identity auditor had not inspected SET residuals; membership was determined from source identity, prior use and declared data-quality rules.

The three primary redshift-quality exclusions are SN 2005gj, 04D4dw and n404, whose released redshifts differ by more than 0.05 from the later coordinate-provider catalogue. Their original released redshifts are retained unchanged in the sensitivity calculation. Lesser recorded redshift revisions are also left unchanged. The primary set contains 298 objects and the sensitivity set 301; both contain the same 23 nearby references and 15 objects outside the fixed redshift windows.

The official 580×580 statistical-plus-systematic covariance is subset on the retained original row indices. The statistical diagonal agrees with the published squared distance-modulus errors; the full covariance is positive definite. The separate CosmoMC product is a projected precision matrix and is not the covariance used here. The standardized distances retain the published stretch, color and host corrections and the catalogue nuisance estimation. Consequently, the quoted Gaussian probabilities are conditional on these released quantities and their covariance.

For an individual target let $P_i = e_i - w_a$, where w_a contains the fixed inverse-marginal-variance nearby-reference weights. Both P_i and each grouped row L_i sum to zero, cancelling the common additive distance zero point. With C the retained magnitude covariance, the individual covariance is $PC\,P^T$, which includes the variance and cross-covariances of the shared reference.

The grouped and individual contrast matrices have ranks 7 and 260, respectively. No further fitted intercept is subtracted from those ranks. The anchor variance is 0.00195881292 mag², and group correlations reach 0.76014. The displayed brightness intervals are the asymmetric transforms of marginal magnitude intervals; the formal statistics remain in magnitude space. The relative contrasts test distance shape and cancel the common absolute scale.

The history was independently integrated through $F(y) = [\ln(1+y^2) + 2\sqrt{6}\arctan y - 2\ln(\sqrt{6} - y)]/7$, with $\ln(1+z) = F(y_0) - F(y)$. Inversion and quadrature in y recover the saved per-object corrections without importing the production model or its contrast functions. The branch ends at $z = 2.72444319594$; every retained Union2.1 object is within it. The interpretation and additional geometry checks in Appendix J remain applicable to this unchanged history.

The same fixed flat $\Omega_m = 0.3$ comparison used in the earlier research gives $\chi^2 = 2.3703$ for the seven groups and 253.5987 for the 260 individual contrasts. Both fixed histories are compatible with these released errors. The additional test therefore supports the SET brightness relation across distinct objects, while leaving the physical interpretation to further observable comparisons.

Appendix K Exact complete-energy selection and its boundary terms

This appendix supplies the coordinate and endpoint details behind (52). All candidates have the same fixed physical source radius, supported source state and complete source energy. Their isotropic coordinate radius at that surface may differ; a coordinate relabeling does not change the physical source.

K.1 Radial reduction of the off-solution energy

For $q_{ij}=\psi^4\delta_{ij}$, the angular part fixes $r=\rho\psi^2$. The radial parts then give $f_B=B^{-1/2}=1+2\rho\psi_\rho/\psi$ and $dr=\psi^2 f_B\,d\rho$. Starting before any field solution is used, the angular integral in (33) is

$$W_{ext}=-\frac{2c^4}{G}\int_{\rho_0}^{\rho_c}\rho^2\psi_\rho(\alpha\psi)_\rho\,d\rho \qquad (345)$$

Use $\psi_\rho=\psi(f_B-1)/(2\rho)$ and $\alpha_\rho=\alpha_r\psi^2 f_B$. Substitution produces (46) with a finite upper radius. The identity $\alpha_r f_B=b/r^2$ then gives (47). This verifies the sign and coefficient of the full energy without evaluating it on the selected field.

K.2 The integration-by-parts identity

Compare candidates at the same upper capacity $a_c<1$. Their coordinate radii need not be the same, which is why both the logarithmic term and the boundary term in (47) must be included. With the definitions in (49), expand the transformed functional:

$$F[y]-F[y_*]=2b\int_{A_0}^{a_c}\left[a\eta_a^2+2a\,y_*'\eta_a+e^{-2y_*}\left(e^{-2\eta}-1\right)\right]da \qquad (346)$$

Integrating the cross term uses $(a\,y_*')'=-e^{-2y_*}$ and gives

$$\begin{aligned} F[y]-F[y_*]&=2b\,I_c[\eta]+\left[4ba\,y_*'\eta\right]_{A_0}^{a_c}\\ I_c[\eta]&=\int_{A_0}^{a_c}\left[a\eta_a^2+\frac{e^{-2\eta}-1+2\eta}{(1-a)^2}\right]da \end{aligned} \qquad (347)$$

The difference of the integrated $2b/r$ term is $-2b\eta$, and the difference of the final radial boundary term is $-2\sqrt{a}(r-r_*)$. The complete endpoint is therefore

$$\begin{aligned} B(a)&=4ba\,y_*'\eta-2b\eta-2\sqrt{a}(r-r_*)\\ B(a)&=-\frac{2b}{1-a}\left[(1-\sqrt{a})^2\eta+2\sqrt{a}\left(e^{-\eta}-1+\eta\right)\right] \end{aligned} \qquad (348)$$

At the actual source, $\eta(A_0)=0$ makes $B(A_0)=0$. At infinity, the stated spatial energy normalization and current equation give $\eta=O((1-a)^2)$. Since $e^{-\eta}-1+\eta=O(\eta^2)$, the combined expression also tends to zero at $a=1$. These statements concern the full endpoint; they do not separately discard divergent integrals.

$$W_{ext,c}-W_{*,c}=-\frac{c^4}{2G}\left[2b\,I_c+B(a_c)-B(A_0)\right] \qquad (349)$$

Take $a_c\to 1$, add the unchanged material contribution $\alpha_0 E_m$, and use $W_*=E_s-\alpha_0 E_m$. The result is (52). Its normalization is exact because $(c^4/(2G))(2b)=E_s$.

K.3 A current-conserving control

A control makes the role of complete accounting explicit. In diagnostic units $b=c=G=1$ with $R_0=4$, consider

$$A=1-\frac{2}{r}+\lambda\frac{(r-4)^2}{r^5}$$
$$\kappa_r=1+\lambda\frac{(r-4)(20-3r)}{2r^4},\ B=\frac{\kappa_r^2}{A} \qquad (350)$$

For the regular cases compared here these profiles keep the source boundary, the current $J_W=1/r^2$ and both leading isolated charge readings. The shell has $E_s=1$ and $\alpha_0 E_m=0.828427124746190$. Direct integration yields

$$\lambda=0: W_{total}=1.000000000000000$$
$$\lambda=1: W_{total}=0.999934780923883 \qquad (351)$$
$$\lambda=-1: W_{total}=0.999935009917085$$

The two deformed profiles miss the complete source budget despite passing the current and boundary checks. The small numerical deficit is sufficient for this exact mathematical test. These values are consistency controls, not observational fits or empirical uncertainty bars.

K.4 Static variation, stress and source charge

The lapse variation in (54) and the retained conformal spatial variation give the pair

$$\Delta\psi=-\frac{2\pi G}{c^4}\psi^5\varepsilon$$
$$\Delta(\alpha\psi)=\frac{2\pi G}{c^4}\alpha\psi^5(\varepsilon+2\,\mathrm{tr}\,\sigma) \qquad (352)$$

For an isotropic fluid $\mathrm{tr}\,\sigma=3p$. The physical spatial Laplacian of the lapse is

$$D^2\alpha=\psi^{-6}\left[\psi\,\Delta(\alpha\psi)-\alpha\psi\,\Delta\psi\right]$$
$$D^2\alpha=\frac{4\pi G}{c^4}\alpha(\varepsilon+\mathrm{tr}\,\sigma) \qquad (353)$$

Multiplication by c^2 recovers (34). Integrating over the complete stationary source, including its supports, gives (35). This places the spherical current law and the source-energy normalization in the same retained energy construction.

Appendix L Source burden and curvature invariants

L.1 General spherical contraction

Before imposing the reciprocal radial relation, write $d\ell = \sqrt{B}\, dr$. Four independent orthonormal sectional coefficients for the static spherical interval are

$$X = \frac{A''}{2AB} - \frac{(A')^2}{4A^2B} - \frac{A'B'}{4AB^2}, Y = \frac{A'}{2ABr},$$
$$Z = \frac{B'}{2B^2 r}, W = \frac{1 - B^{-1}}{r^2}, K = 4\left(X^2 + 2Y^2 + 2Z^2 + W^2\right). \qquad (354)$$

Setting $B = A^{-1}$ yields the reduced expression in Section 6.4. Substituting the selected A then gives its source and tidal forms. The squared-length factors in each sectional coefficient show why K has units of inverse length to the fourth power. The contraction is a tidal/interval diagnostic, not a local material density.

L.2 Field energy in the selected spherical solution

The off-solution field functional in Section 3 assigns a definite energy decomposition. Evaluated on its selected positive-mass spherical solution and expressed per physical spatial volume, its exterior contribution is

$$u_W = \frac{\mathcal{J}_W^2}{2\pi G(1+\alpha)^2} = \frac{c^4 r^2 K}{96\pi G(1+\alpha)^2},$$
$$u_W = \frac{c^4(1-\alpha)^2}{8\pi G r^2}, W_{>r} = M c^2 \frac{1-\alpha}{1+\alpha}. \qquad (355)$$

The density follows by converting the retained functional to the physical measure; the tail follows by integration from r to infinity. It is the positive field contribution accompanying clock-weighted material energy. Expressing the same complete energy as proper material energy plus binding energy gives a different decomposition. The invariant K rewrites the selected result; it does not determine the full functional away from that solution.

For example, in isotropic coordinates the selected family satisfies $F = \alpha - 2/\psi + 1 = 0$. Adding a multiple of $\int |\nabla F|^2 d^3x$ would leave the energy and its first variation on that family unchanged while altering other configurations. This algebraic check explains why the field functional is stated before the selection calculation.

L.3 Rotation and the scope of the spherical diagnostic

The normal lapse and shift of Section 6.5 give $K_{r\varphi} = q_{\varphi\varphi}\partial_r v^\varphi/(2N)$ and $K_{\theta\varphi} = q_{\varphi\varphi}\partial_\theta v^\varphi/(2N)$. Substitution into $K_{ij}K^{ij}$ and into the spatial Laplacian of N gives the same function. At $b = c = 1$, $a_J = 4/5$, $r = 3$, $\theta = \pi/2$, both $c^2\Delta_q N/N$ and $c^2 K_{ij}K^{ij}$ are $15279392/1154300625 = 0.013236926039\ldots$. The horizon is at $8/5$, so this is a regular exterior check.

The spherical positive contraction must be distinguished from its general rotating value. For the retained Kerr interval, with $u_\theta = \cos\theta$, The invariant expression agrees with the Kerr contraction in [37].

$$K = 48\,b^2\,\frac{r^6 - 15\,r^4\,a_J^2\,u_\theta^2 + 15\,r^2\,a_J^4\,u_\theta^4 - a_J^6\,u_\theta^6}{\left(r^2 + a_J^2\,u_\theta^2\right)^6}. \qquad (356)$$

It can be negative: $b = 1$, $a_J = 4/5$, $r = 2$, $u_\theta = 3/4$ gives $K = -0.102511322304\ldots$ outside the horizon. The Lorentzian contraction is not a positive norm in every geometry. The square-root energy inverse and positive tidal reading used in Section 6.4 therefore remain explicitly spherical.

Appendix M Causal evolution and radiation normalization

M.1 Causality of the complete local tidal response

Let $V = v_\gamma^2 / c^2$ be the tensor propagation-speed ratio, and A_r, C_v, C_t the scalar-rate, vector and tensor coefficients in the linear completion normalized by κ_f. Its complete local tidal response contains four independent inverse-spatial-Laplacian residues. Their cancellation requires [35]

$$\begin{gathered} -2/V + C_t/4 = 0,\, C_v/(2V) - C_t/2 = 0, \\ -2/V + 2A_r + C_v = 0,\, -(2A_r + C_v)/V + C_t/4 = 0. \end{gathered} \qquad (357)$$

The first three give $C_t = 8/V$, $C_v = 8$, and $A_r = 1/V - 4$. The fourth then requires $V = 1$ for positive finite propagation speed. Thus $A_r = -3$ and $C_v = C_t = 8$. The causal quantity is the complete local tidal observable, including its constrained sectors; the spatially projected tensor alone is not that complete observable.

M.2 Common evolution and local constraints

The local completion uses one positive spatial ruler tensor and its conjugate momentum, common lapse and shift multipliers, spatial covariance, a Hamiltonian even and quadratic in momentum, no momentum derivatives, at most two spatial derivatives, constant coefficients and no extra propagating clock field. Flat unloaded calibration fixes the zero-source state. In this subsection set $c = 1$; the source law elsewhere retains its displayed SI factors.

For a trial scalar constraint with coefficients a_f, b_f, c_f, require that two changes of normal slicing describe the same physical history. The bracket must be the corresponding spatial change:

$$\begin{gathered} \mathcal{C}_f = \frac{a_f\,\pi^{ij}\,\pi_{ij} + b_f\,\pi^2}{\sqrt{q}} + c_f\,\sqrt{q}\,R^{(3)}, \\ \{\mathcal{C}[n], \mathcal{C}[m]\} = D[q^{ij}(n\,\partial_j m - m\,\partial_j n)]. \end{gathered} \qquad (358)$$

Direct evaluation in the stated class gives $a_f\,c_f = -1$ and $a_f + 2\,b_f = 0$. Matching the static SET energy fixes $c_f = -\kappa_f/8$, $a_f = 8/\kappa_f$ and $b_f = -4/\kappa_f$. These are the coefficients of the common field system presented in Section 3. The normal-composition requirement is a retained evolution principle stronger than agreement of field and matter clock rates alone. Constraint preservation then applies at finite wavelength, not only after averaging a wave. The normal-deformation algebra has the established canonical precedent of Hojman, Kuchar and Teitelboim [36].

M.3 Energy flux and quadrupole normalization

Variation of the quadratic tensor density yields the wave equation in Section 6.6. Its free canonical energy and flux follow from the same density. For an outgoing far-zone wave, the radial derivative is $-\dot{\gamma}/c$ to leading order in $1/R$, which gives the displayed positive flux. No independent flux coefficient is assigned.

Complete local source conservation is $\dot{\varepsilon}+\partial_i j_i=0$ and $\dot{j}_i+c^2\partial_j\sigma_{ij}=0$. Two integrations by parts for a compact source give $\ddot{I}_{ij}=2\int\sigma_{ij}d^3x$. The retarded Green function then supplies the $1/R$ amplitude. For a trace-free symmetric tensor A_{ij}, the angular identity

$$\int|\Lambda A|^2 d\Omega=\frac{8\pi}{5}A_{ij}A_{ij} \quad (359)$$

converts the wave flux into $G\langle \ddot{I}_{ij}^{STF}\ddot{I}_{ij}^{STF}\rangle/(5c^5)$. For a circular two-body orbit the ordinary quadrupole evaluation gives $32G^4m_1^2m_2^2(m_1+m_2)/(5c^5d^5)$. Applying the previously calibrated source identity to each mass gives

$$P=\frac{Q_1^4Q_2^4(Q_1^2/R_1^3+Q_2^2/R_2^3)}{5\,3\,2^4\pi^{10}Gc^5d^5R_1^6R_2^6}. \quad (360)$$

The source reference radii remain distinct from orbital separation and observer distance. This result is obtained within the weak, slow-source, outgoing tensor completion. The compact-source conservation step includes stresses that drive or support the source; a changing density without its companion momentum and stresses would not satisfy the derivation.

M.4 Source exchange and field history

The pulse identity in Section 6.7 follows by integrating the exact emitter balance over the retarded interval: $E_c(T-r/c)-E_c(T)=\int_{T-r/c}^{T}P(u)du$. Changing variables $u=T-s/c$ identifies that difference with the exported energy still inside the sphere. The local causal tidal equations, including the changing spatial response, then supply the pulse tides. Applying only a stationary Poisson equation to the moving pulse would omit those time-dependent terms.

The relevant conservation statement concerns the complete state. Two different histories may leave different field configurations; they need not assign the same energy to those different states. What must be consistent is the energy assigned to the same complete source, support and field state. This is the sense in which conserved accounting and field memory coexist in the present calculation.

Appendix N Cosmological domain energy and clock matching

N.1 The material-domain constraint and its clock assignment

The source law also enters the common field constraint directly. For a material domain of irrotational pressureless matter, use the matter's physical clock t, its volume V_D, and the volume average $\langle\cdot\rangle_D$. Let θ be the local volume-expansion rate and let σ_{kin}^2 denote one half of the squared shear tensor. Define the motion-fluctuation term

$$B_D=\frac{2}{3}\left(\langle\theta^2\rangle_D-\langle\theta\rangle_D^2\right)-2\langle\sigma_{kin}^2\rangle_D. \quad (361)$$

Writing the averaged field constraint with $Q_D^2=24\pi G M_b V_D$ gives

$$\left(\frac{dV_D}{dt}\right)^2=Q_D^2-\frac{3}{2}V_D^2\left[B_D+c^2\langle R^{(3)}\rangle_D\right]. \quad (362)$$

Here $R^{(3)}$ is the scalar of the spatial reading in the common field system; B_D has units of inverse time squared and is distinct from the source quantity Q_D. For a flat homogeneous dust state both averaged terms vanish. The positive-volume branch then gives $dV_D/dt=Q_D$. The tested cosmological realization instead assigns $dV_D/dT=Q_D$, giving $dV_D/dt=Q_D/\alpha$. These are different physical volume histories when $\alpha\neq1$, rather than two labels for the same constraint. The source, clock and photon relations are common to both; the retained front and volume prescriptions specify the history compared with the catalogue here.

This distinction makes the complete-energy connection calculable. Set $x=\alpha^{-2}-1=y^2$ and let a prime denote $d/d\ln a$. To realize the retained history as a material-domain solution of the averaged field equations, the required motion and spatial moments would be

$$\frac{B_D}{H_b^2}=\frac{3}{2}(x'-x),\frac{c^2\langle R^{(3)}\rangle_D}{H_b^2}=-\frac{3}{2}(x'+3x). \quad (363)$$

The domain response energy represented by their contribution to (362) is

$$E_{response}\equiv-\frac{c^2V_D}{16\pi G}\left[B_D+c^2\langle R^{(3)}\rangle_D\right]=M_bc^2x. \quad (364)$$

Equations (363) and (364) identify the magnitude that a complete interacting source state would have to supply. They are compatibility targets obtained from the retained history. A local source state realizing those moments must also determine its own photon and beam propagation; an averaged volume history alone does not impose the flat homogeneous optical map. The isolated spherical energy-selection proof establishes its spherical result within its stated class, while these equations locate the corresponding cosmological field-to-history question.

N.2 Source clocks and expansion energy

For an expanding, smooth material domain in the same irrotational pressureless class, before caustics, define its source-volume clock from its actual volume and fixed baryonic mass. The clock origin is the one specified in Section 10.1:

$$T_D=\sqrt{\frac{V_D}{6\pi G M_b}},H_D=\frac{1}{3V_D}\frac{dV_D}{dt},H_{b,D}^2=\frac{8\pi G M_b}{3V_D}. \quad (365)$$

Let $E_b=M_b c^2$. Here $E_{response}$ means the first expression in (364), evaluated on the actual source state; its target value $M_b c^2 x$ is not imposed. Dividing (362) by $9V_D^2$ and multiplying by $3c^2V_D/(8\pi G)$ gives the expansion-side energy accounting

$$E_{\exp}=E_b+E_{response}=\frac{3c^2V_D H_D^2}{8\pi G}. \quad (366)$$

Differentiating T_D then gives an exact identity for this state, without assuming uniform density:

$$\alpha_D\equiv\frac{dt}{dT_D}=\frac{H_{b,D}}{H_D}=\sqrt{\frac{E_b}{E_{\exp}}},U_D\equiv 1-\alpha_D^2. \quad (367)$$

This accounting is not an isolated boundary energy or the full constrained Hamiltonian. The further correspondence needed by the retained luminosity history is

$$U_D=u_h \Leftrightarrow (1-u_h)E_{\exp}=E_b. \quad (368)$$

At the frozen reception benchmark this requires $E_{\exp}/E_b=(1-u_0)^{-1}=4.16479720005$. For these source tests, $u_h=(2G/c^2)\int_0^{r_h} dM_b/R$ is the equal-time central shell reading inside the advancing front, using proper baryonic shell mass and areal radius. The actual source and its front must preserve this equality as they evolve. The source-clock/expansion-energy identity remains exact, while the expanding homogeneous pressureless class and the nonuniform controls examined do not preserve the further equality with that shell burden. These results locate the remaining source-evolution requirement; they do not change the conditional radius history or the catalogue calculation.

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